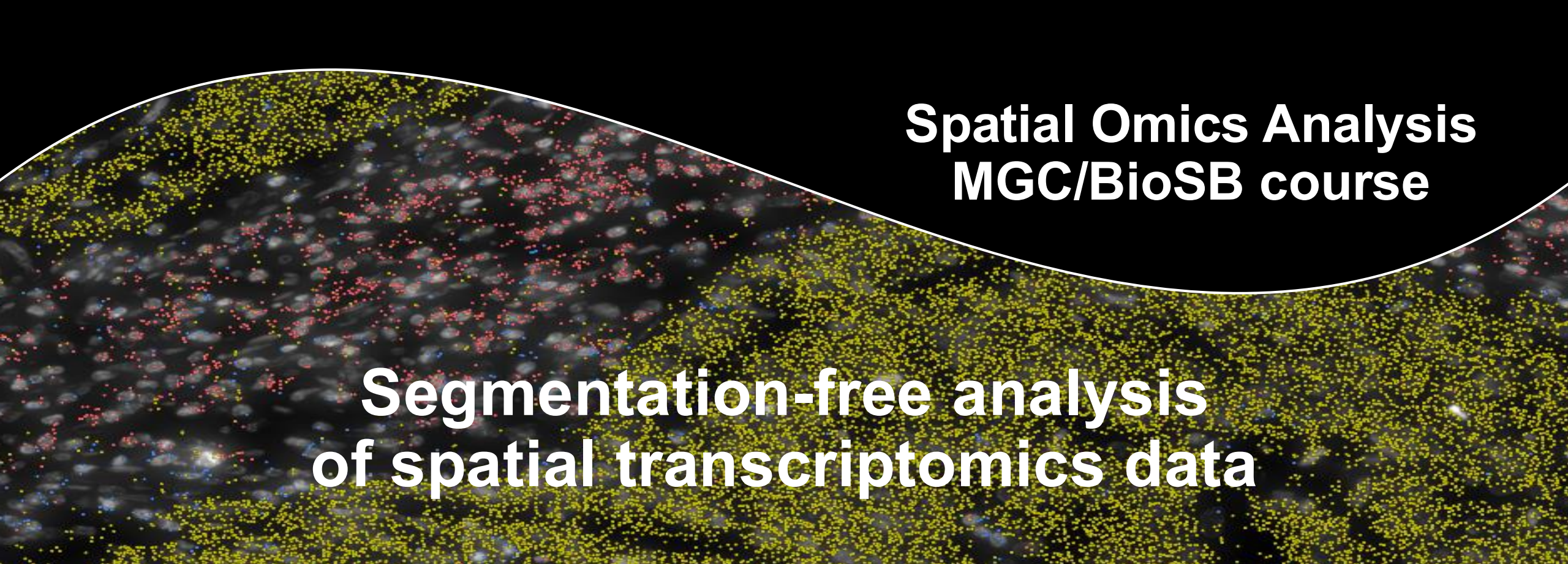




Spatial Omics Analysis MGC/BioSB course

Segmentation-free analysis of spatial transcriptomics data

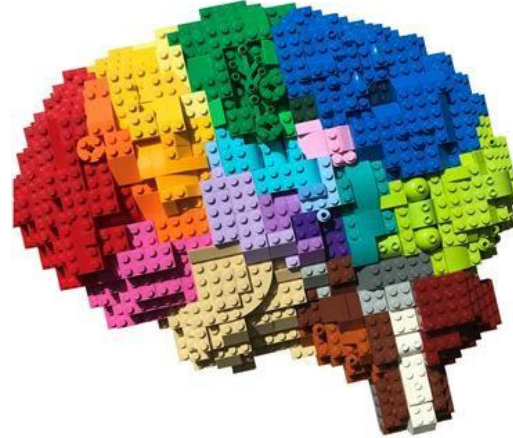
Scott Engels
Pathology department



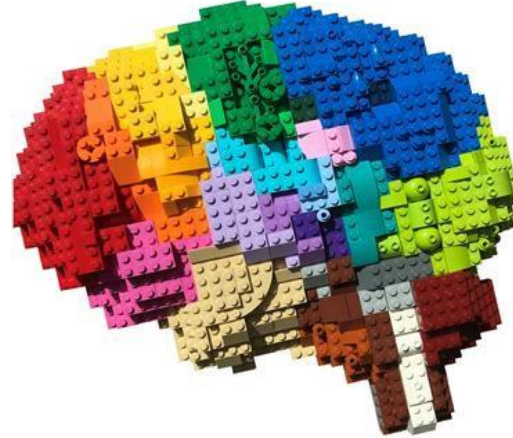
11-11-2025



Resolution revolution in transcriptomics:



Resolution revolution in transcriptomics: Bulk

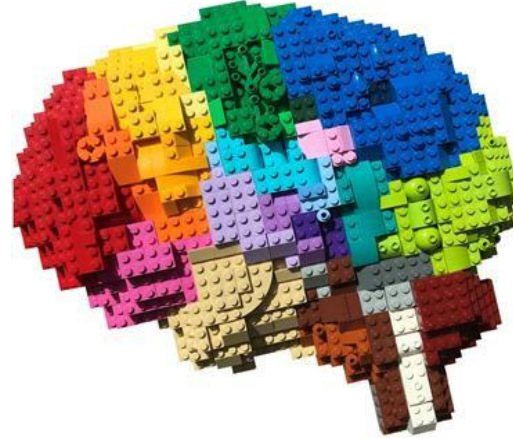


Bulk

Resolution revolution in transcriptomics: Single-cell

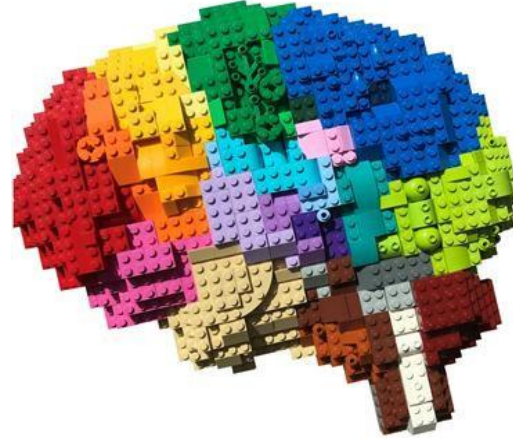


Bulk



Single-cell

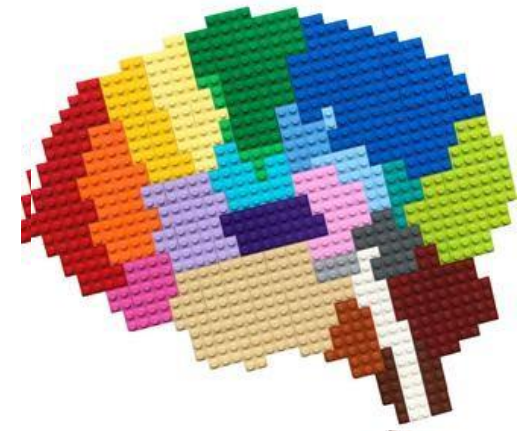
Resolution revolution in transcriptomics: Spatial



Bulk



Single-cell



Spatial

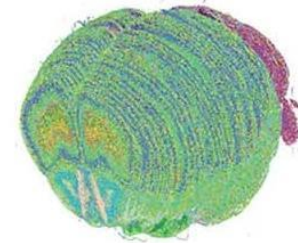
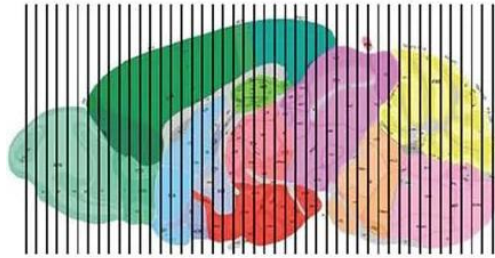
Size and resolution



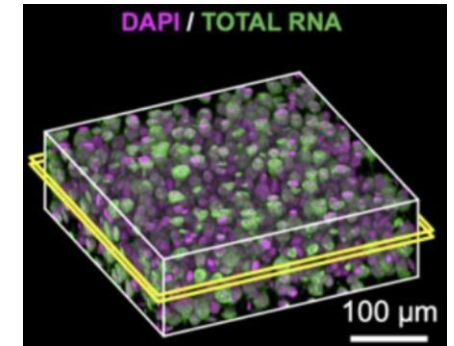
Size and resolution



Pseudo 3D



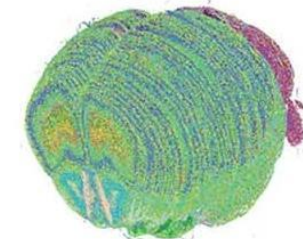
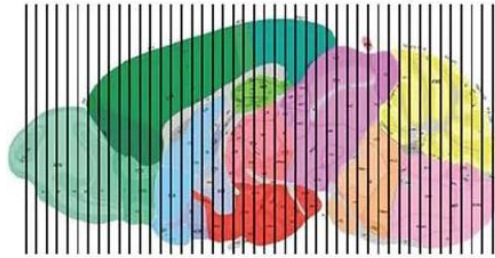
Real 3D



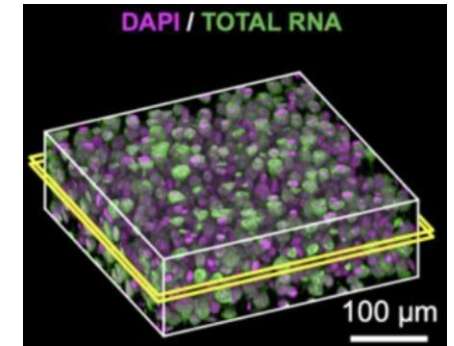
Size and resolution



Pseudo 3D

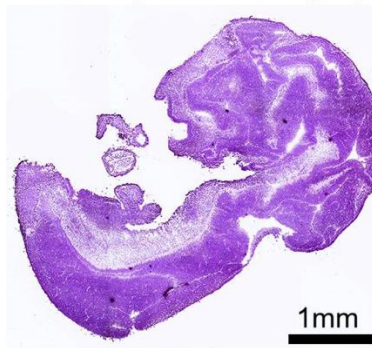


Real 3D

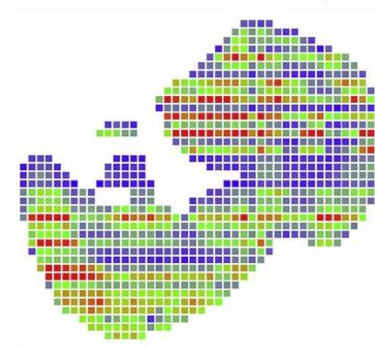


Multi-omics

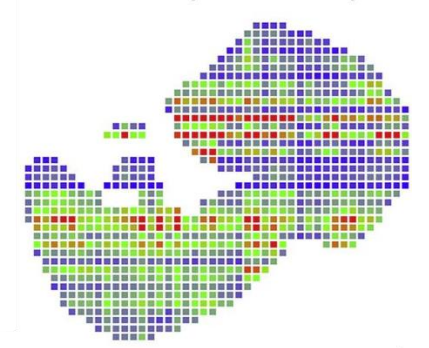
H&E (E.10 Embryo)



Pan-mRNA map



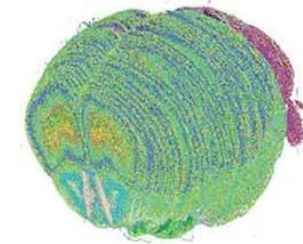
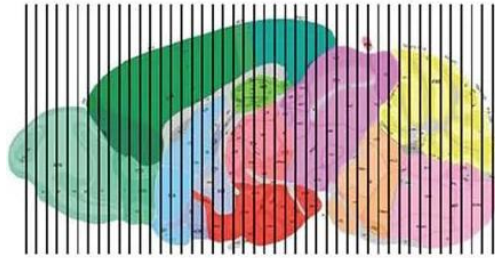
Pan-protein map



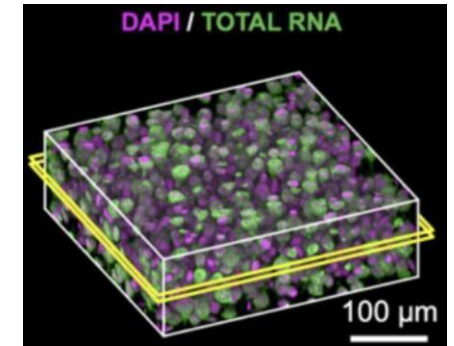
Size and resolution



Pseudo 3D

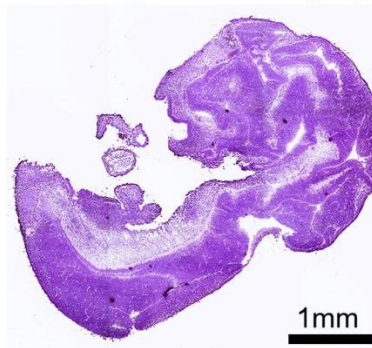


Real 3D

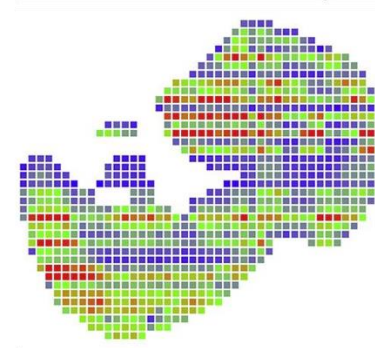


Multi-omics

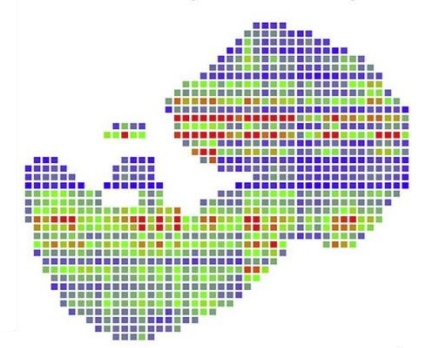
H&E (E.10 Embryo)

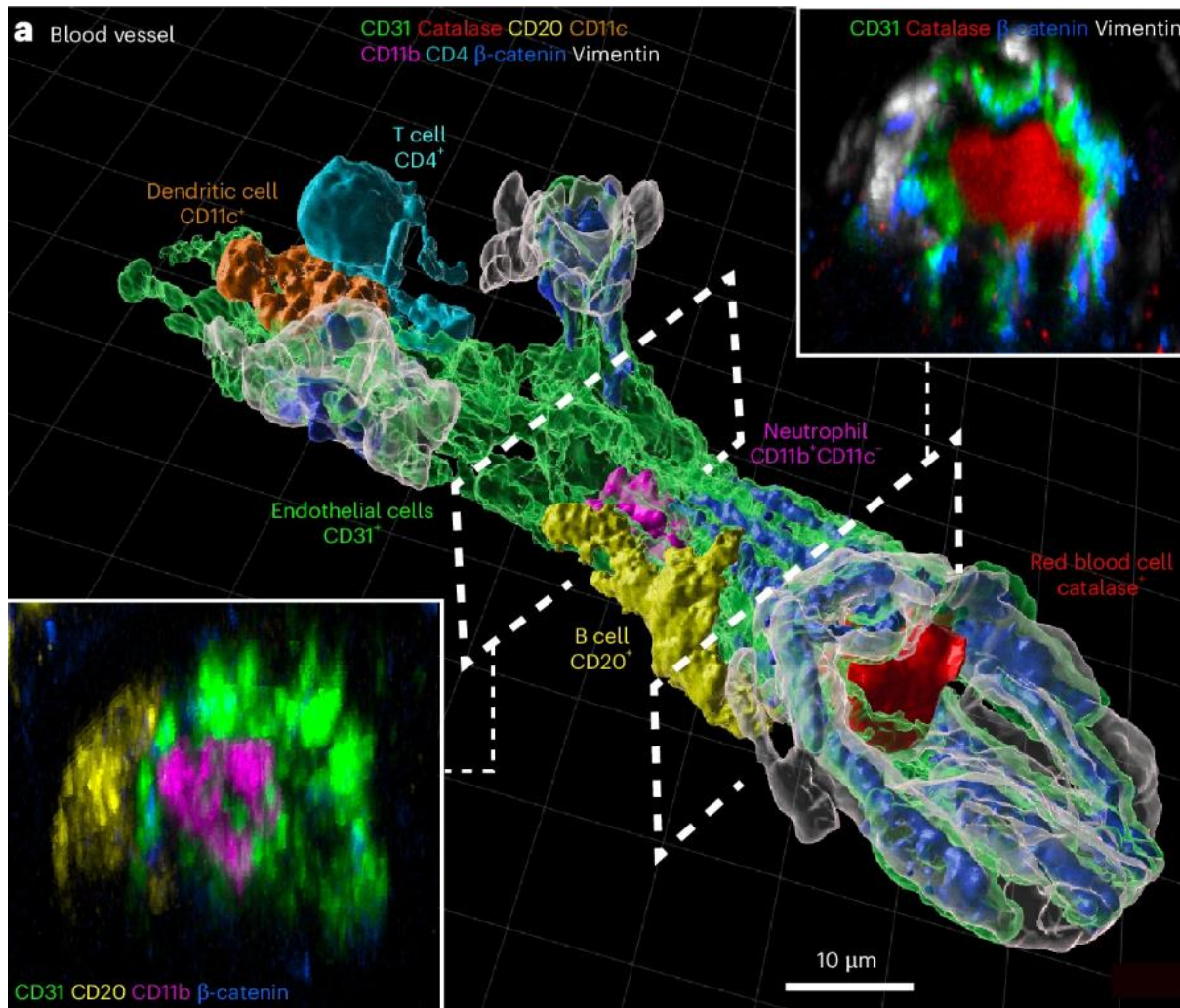


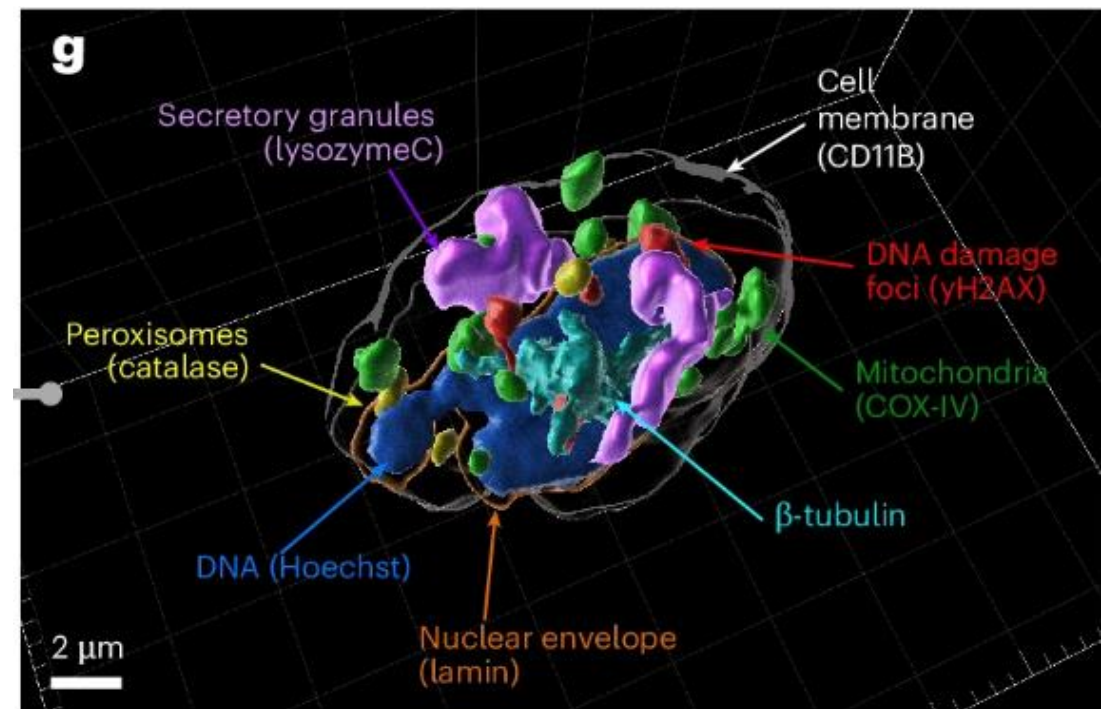
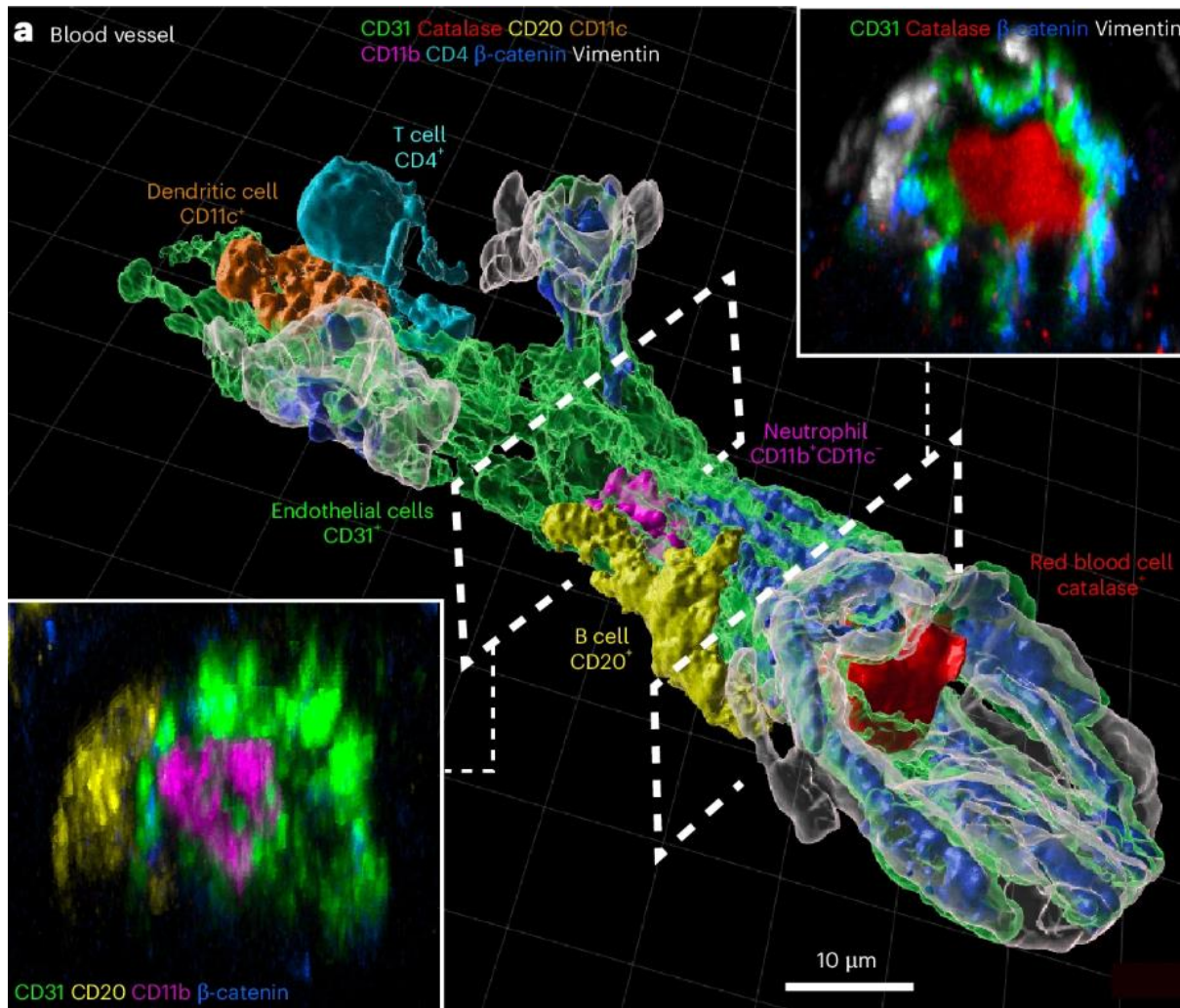
Pan-mRNA map

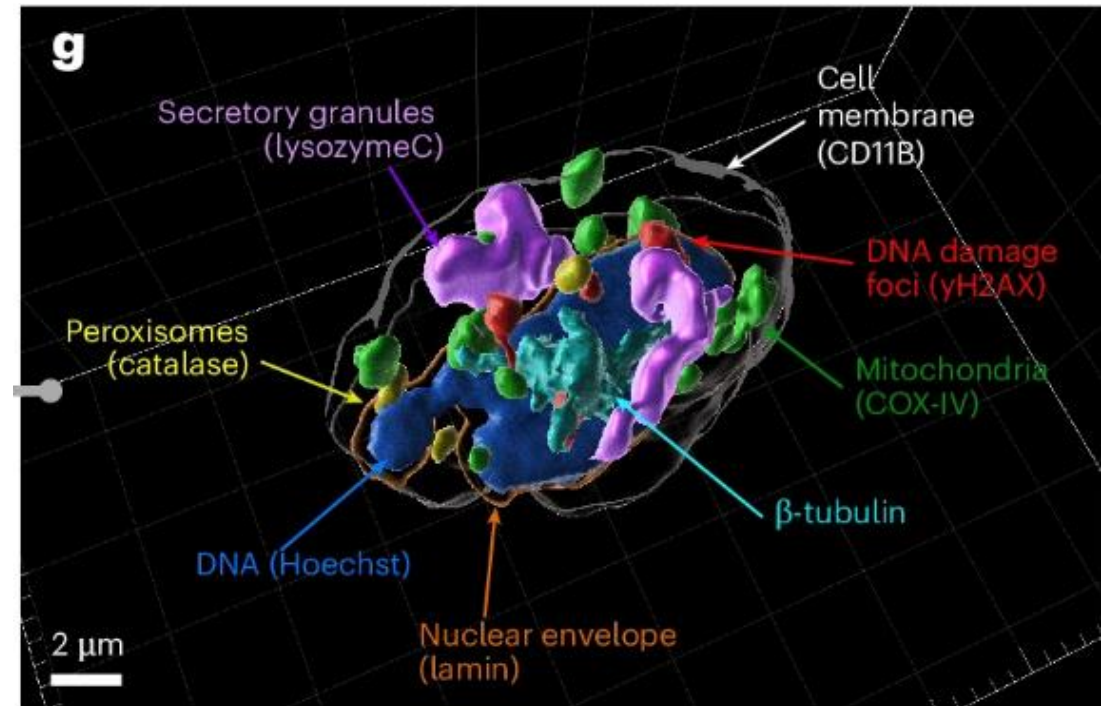
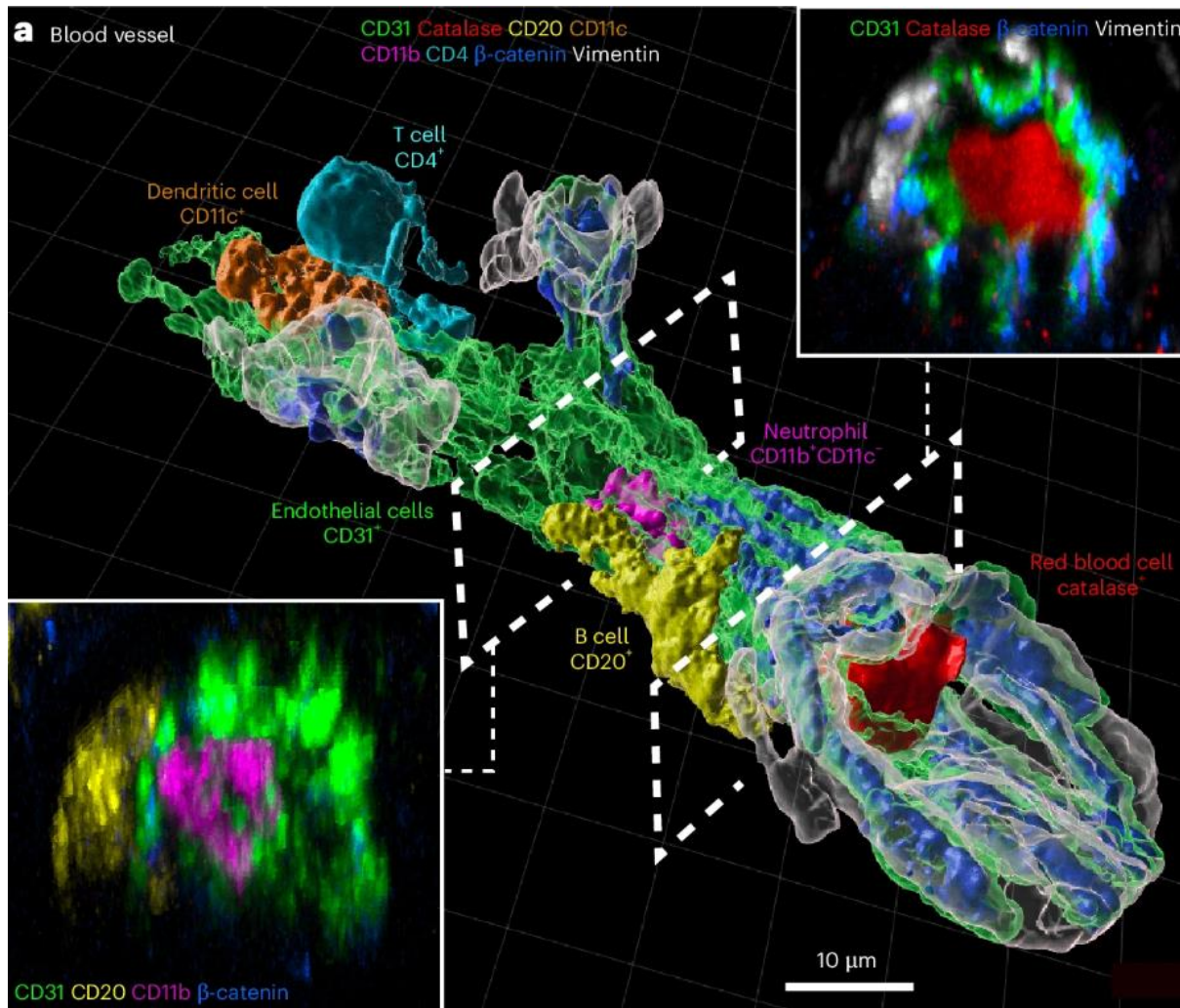


Pan-protein map

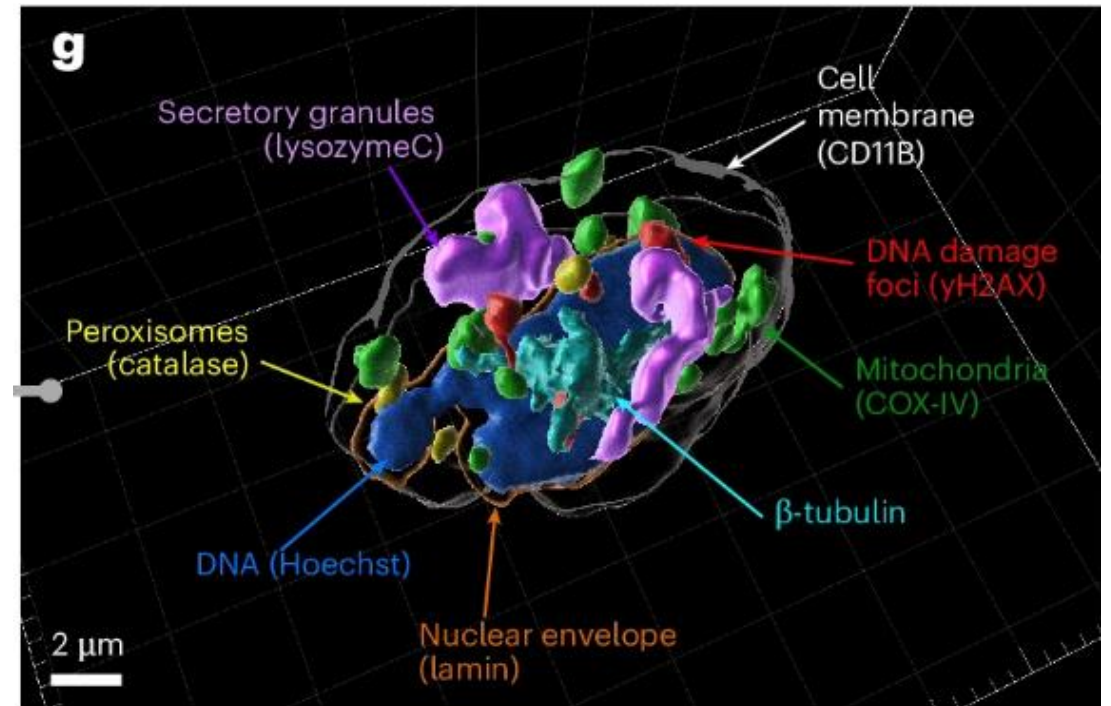
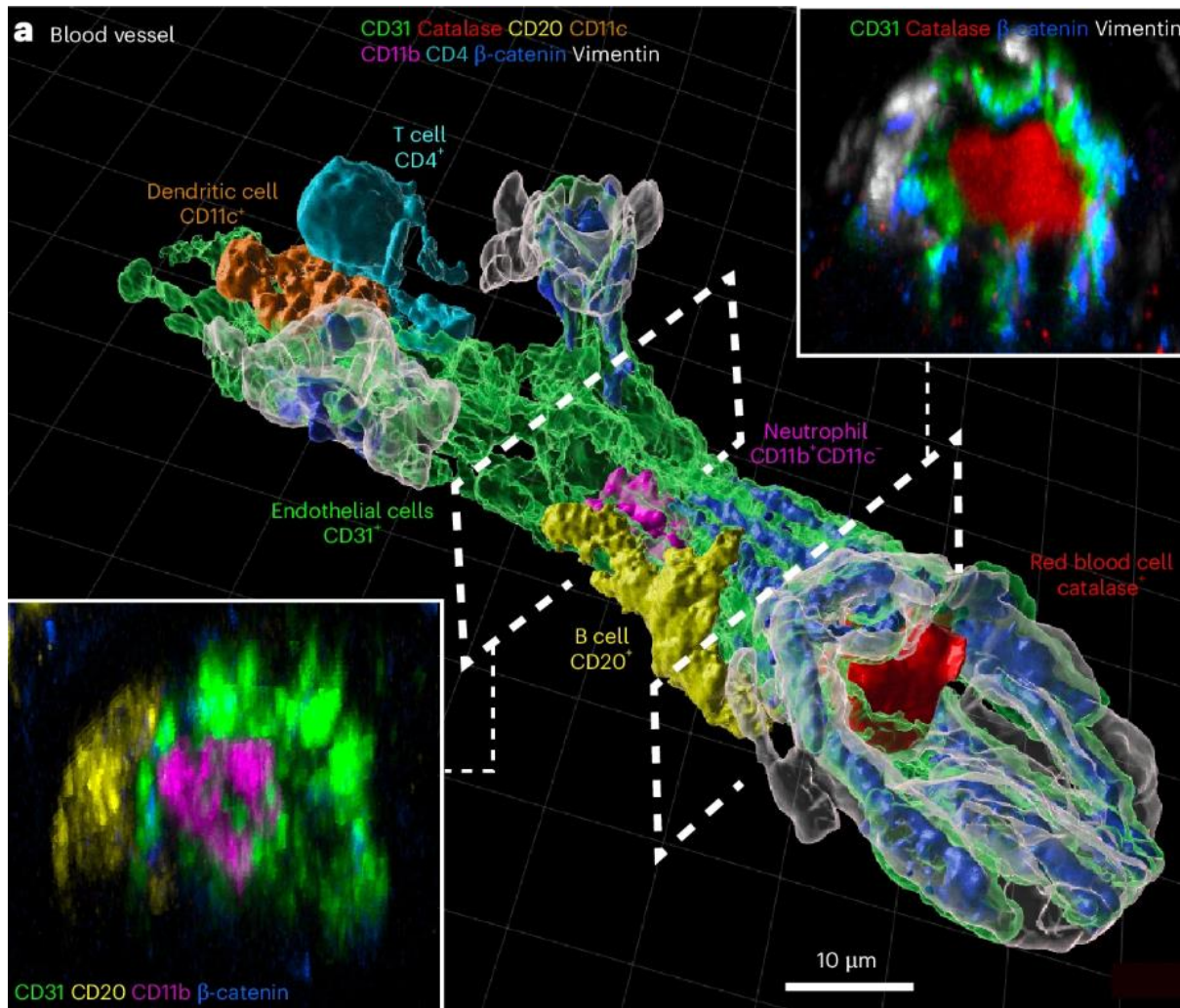








**Current limitation:
3D technologies not scalable yet**



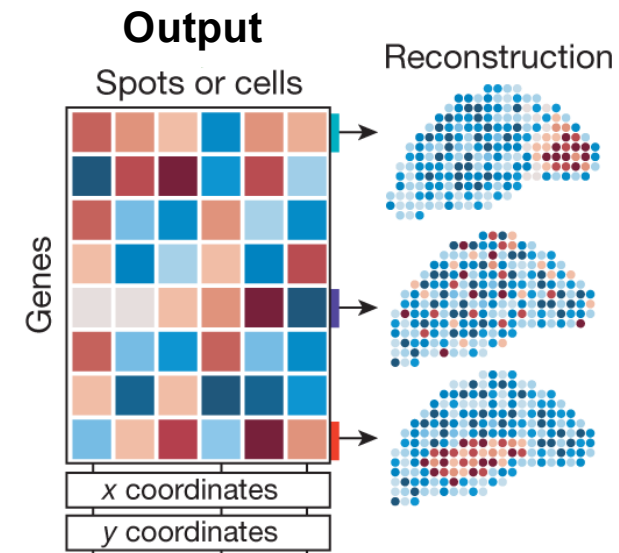
Current limitation:
3D technologies not scalable yet

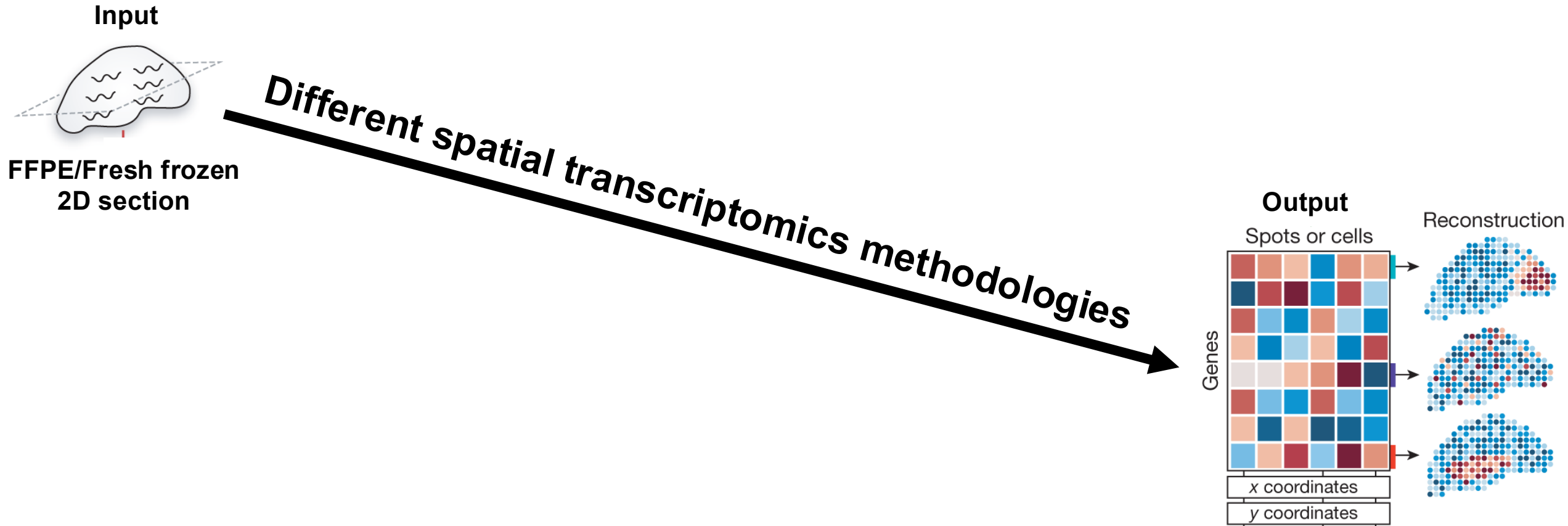
2D spatial transcriptomics

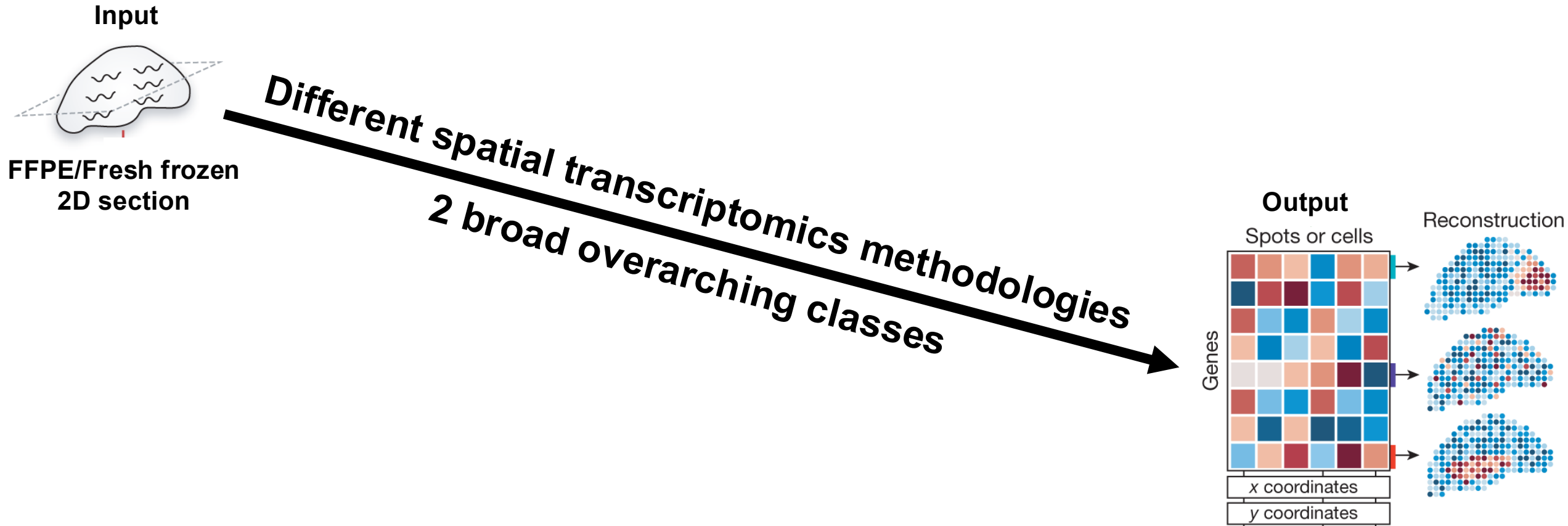
Input



FFPE/Fresh frozen
2D section

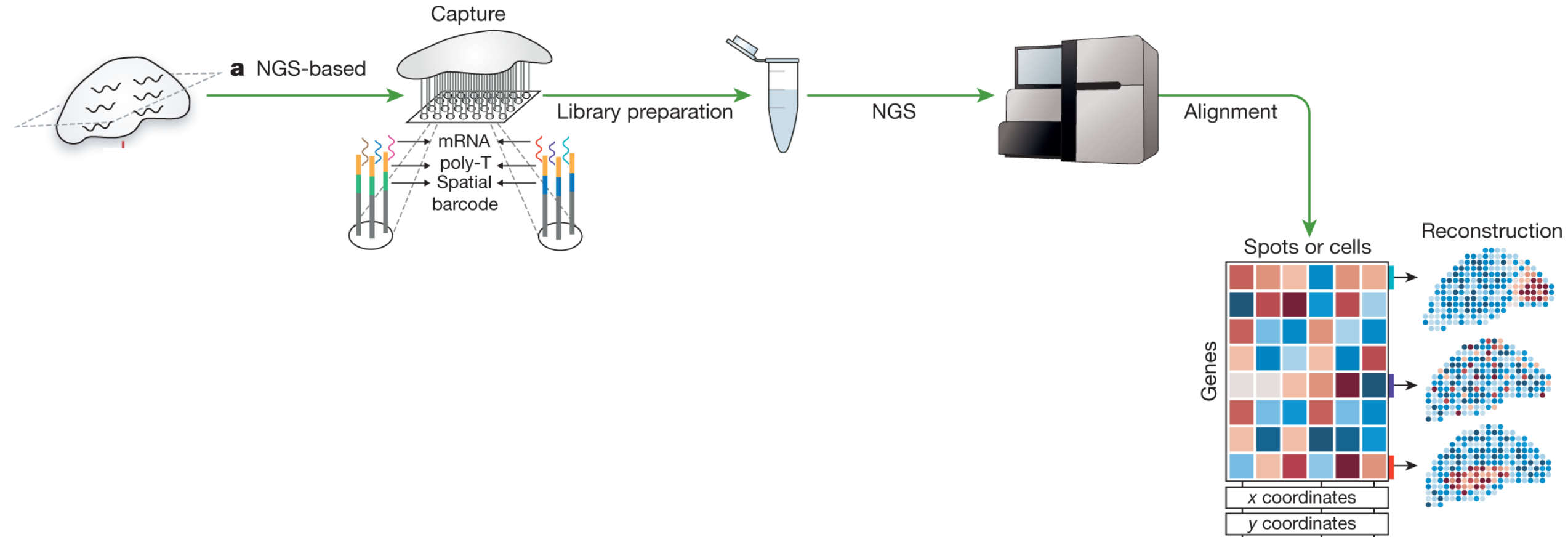






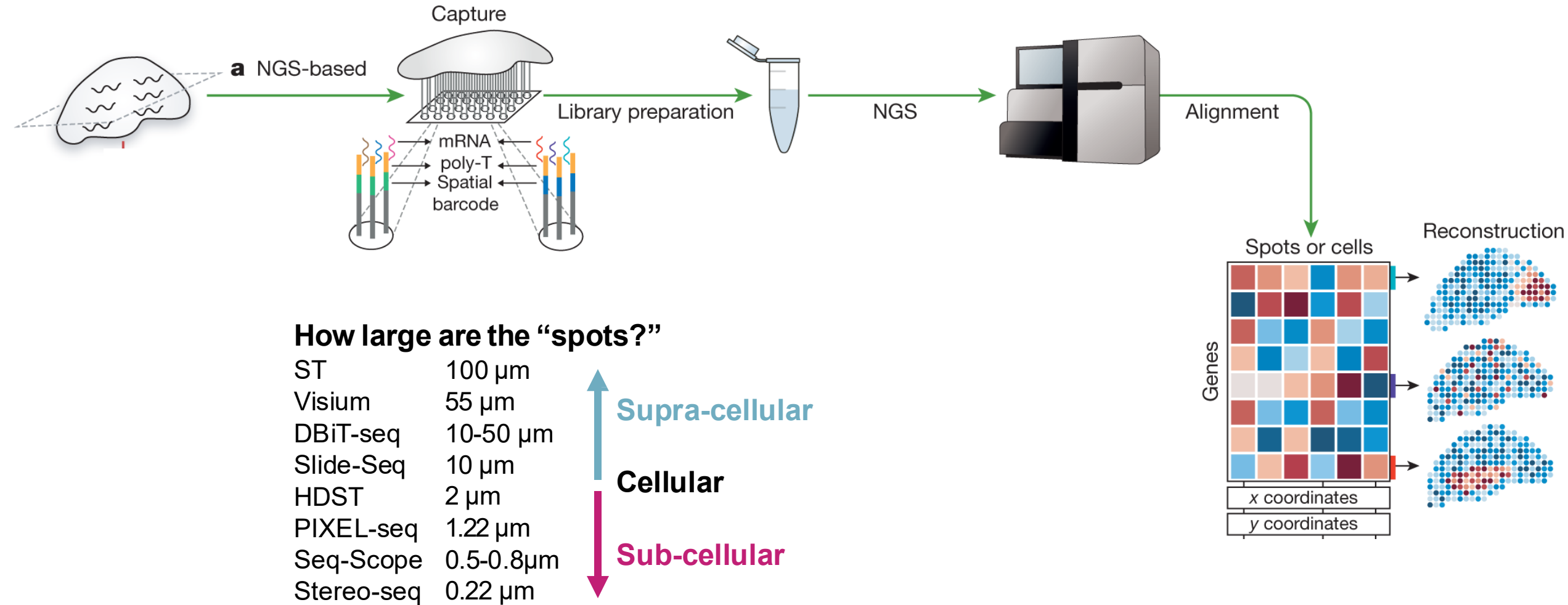
Spatial transcriptomics methodologies

1. Next-generation sequencing-based



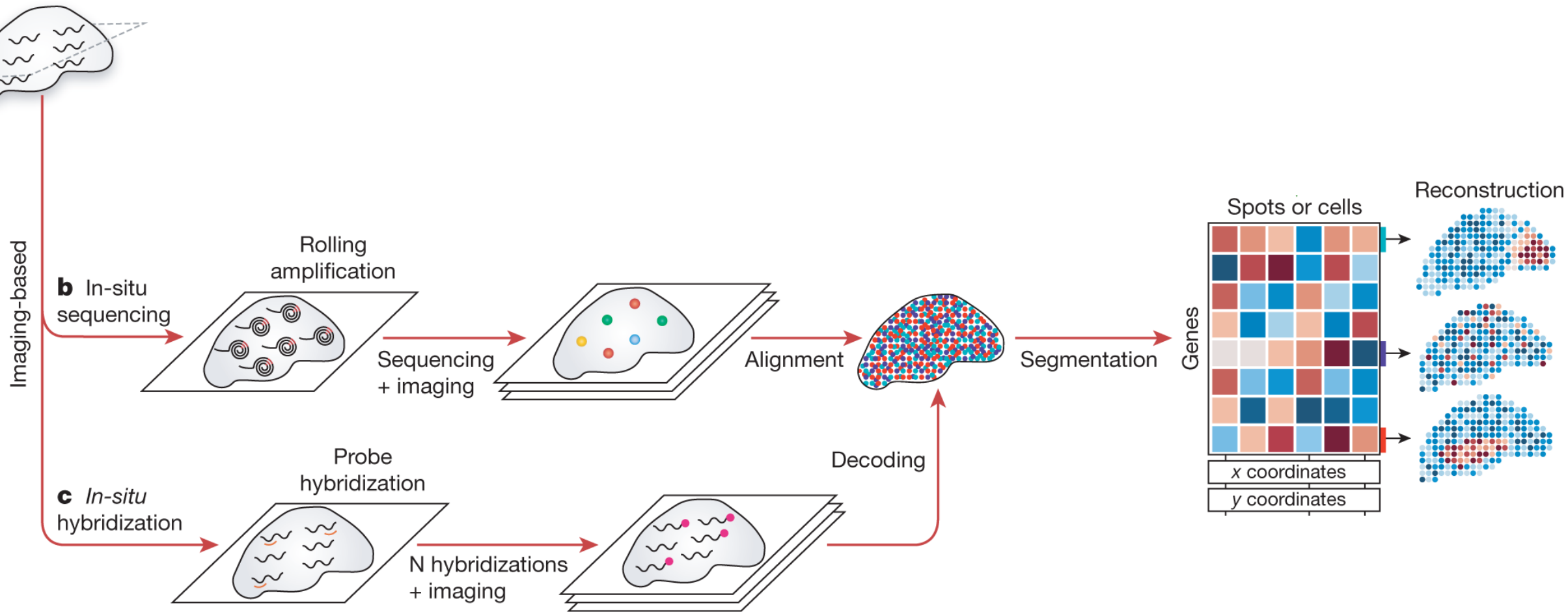
Spatial transcriptomics methodologies

1. Next-generation sequencing-based



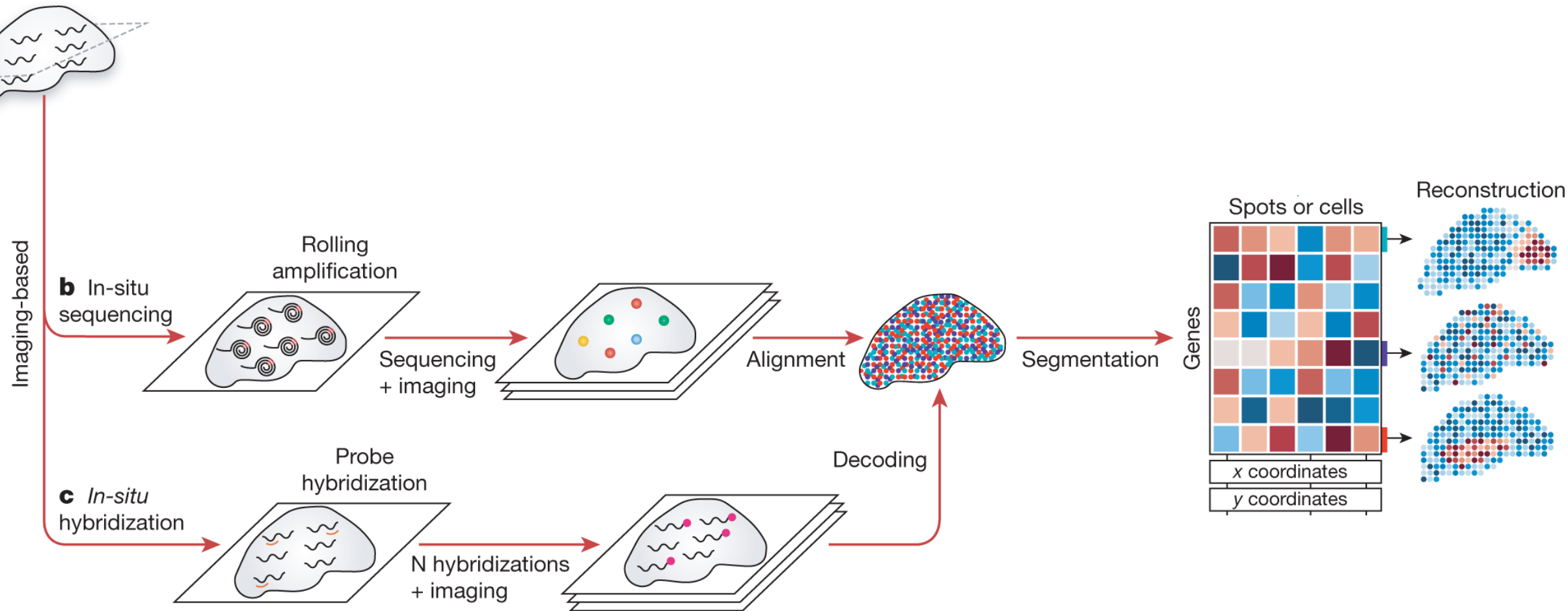
Spatial transcriptomics methodologies

2. Imaging-based

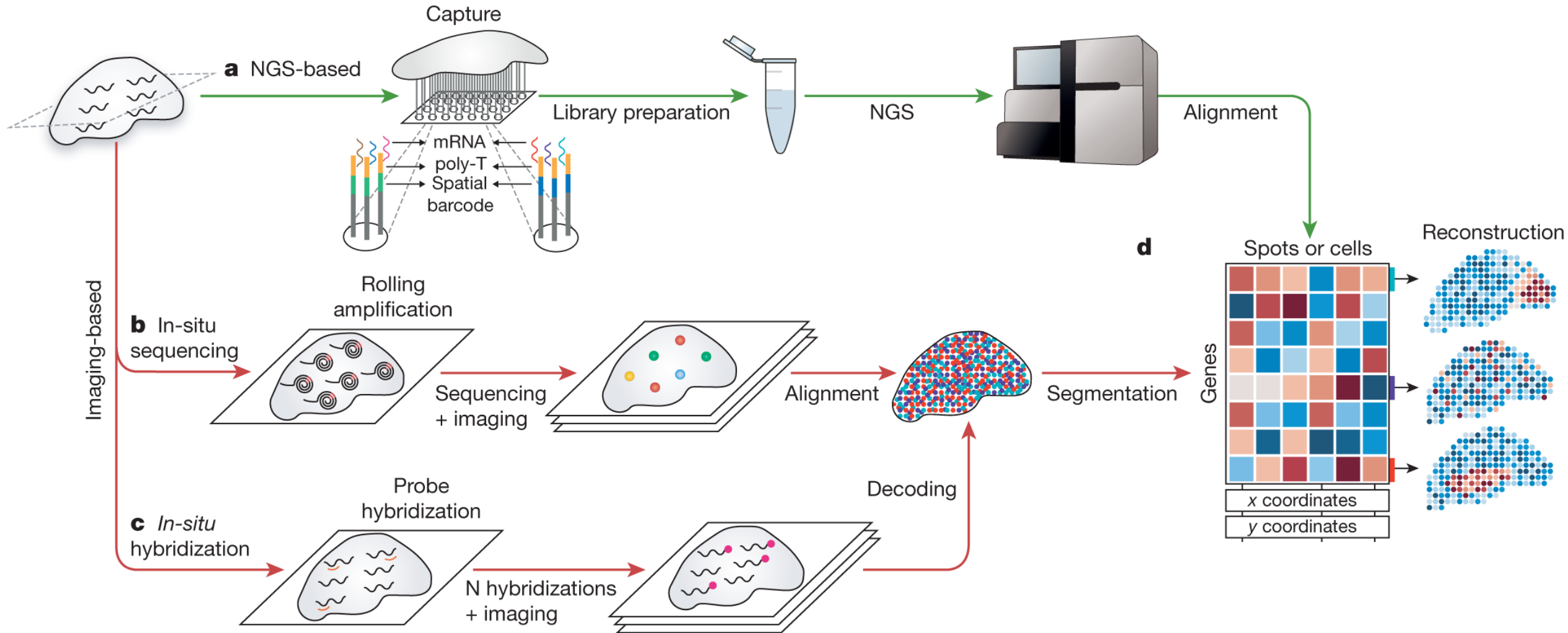


Spatial transcriptomics methodologies

2. Imaging-based

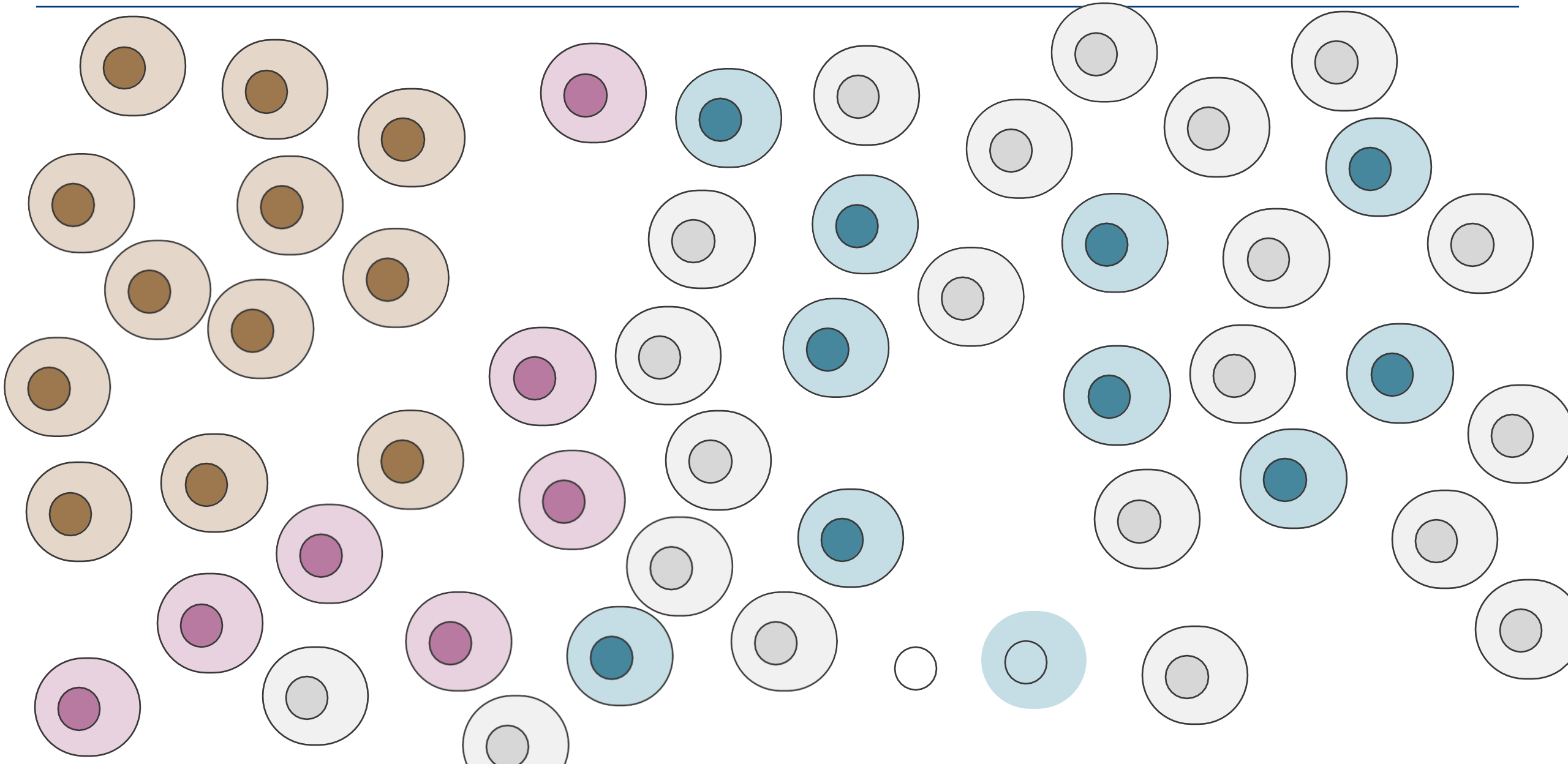


**MERFISH/VizGen, ISS/Xenium, Molecular Cartography,
 CosMx, Barista-seq, seqFISH, STARmap, EEL-FISH, ...**

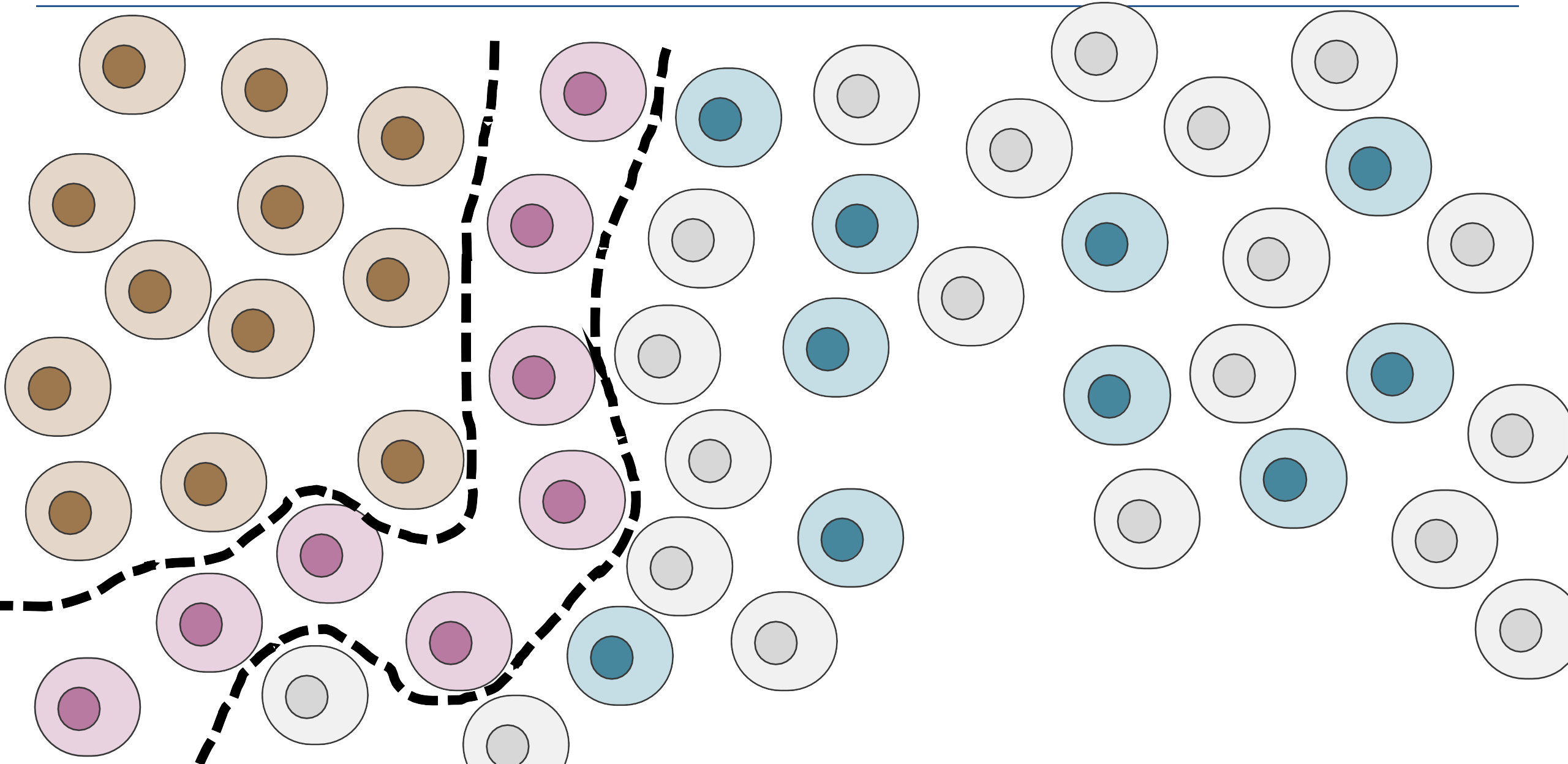


Huge variation in spatial resolution between these two categories, and also within categories

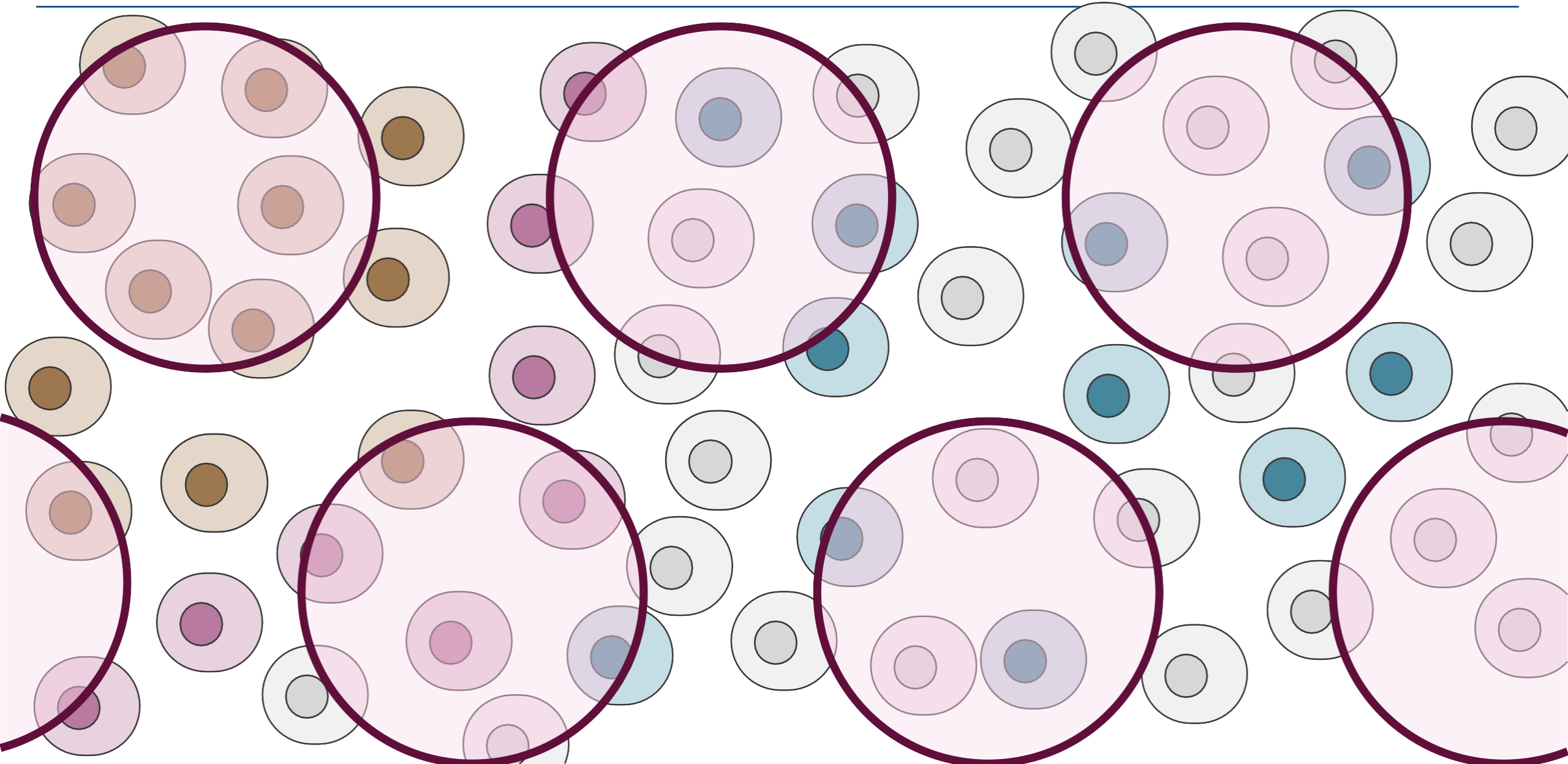
Spatial resolution



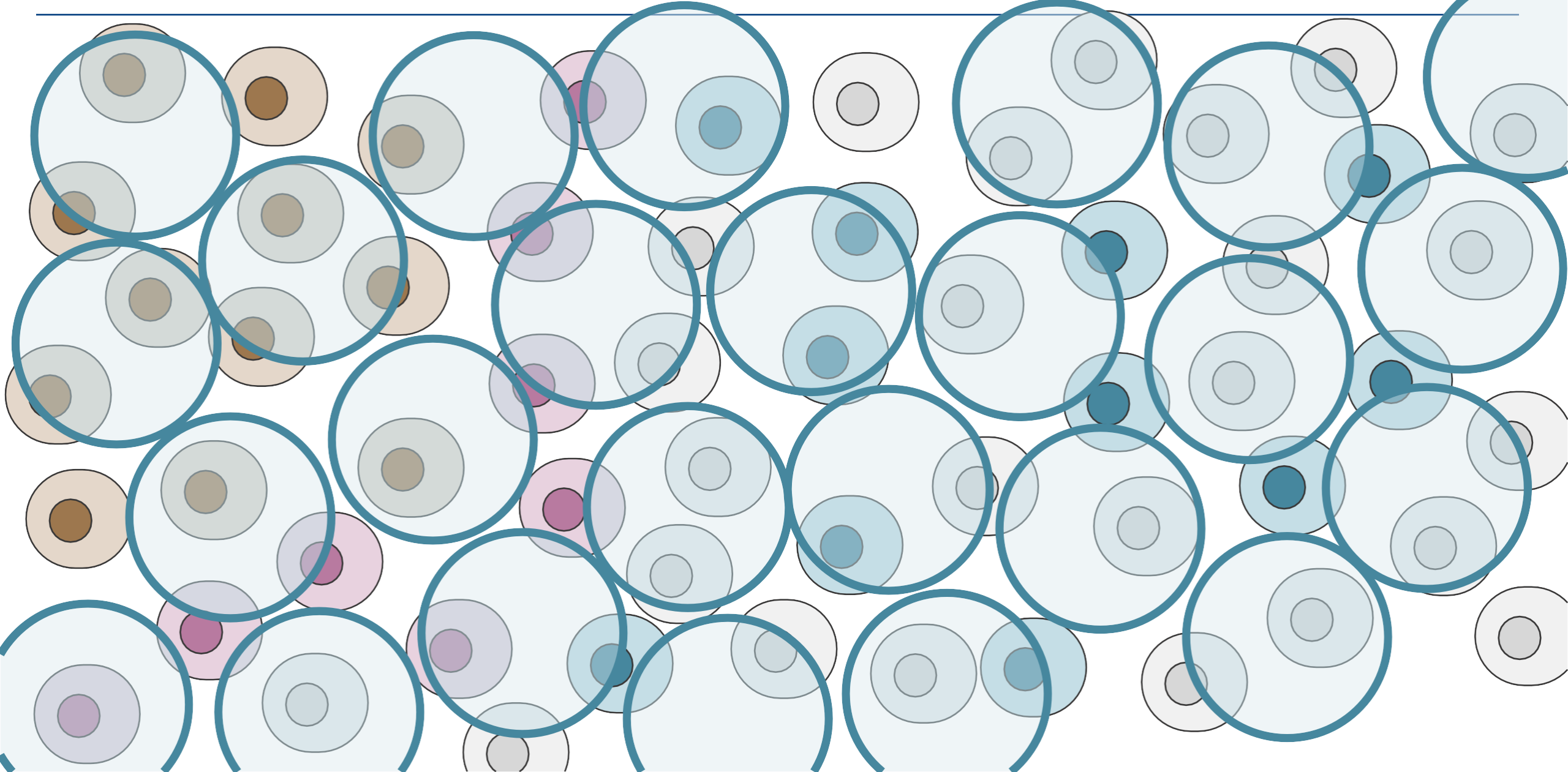
Spatial resolution (NGS-based) Microdissection, e.g. TIVA, Geo-seq, etc



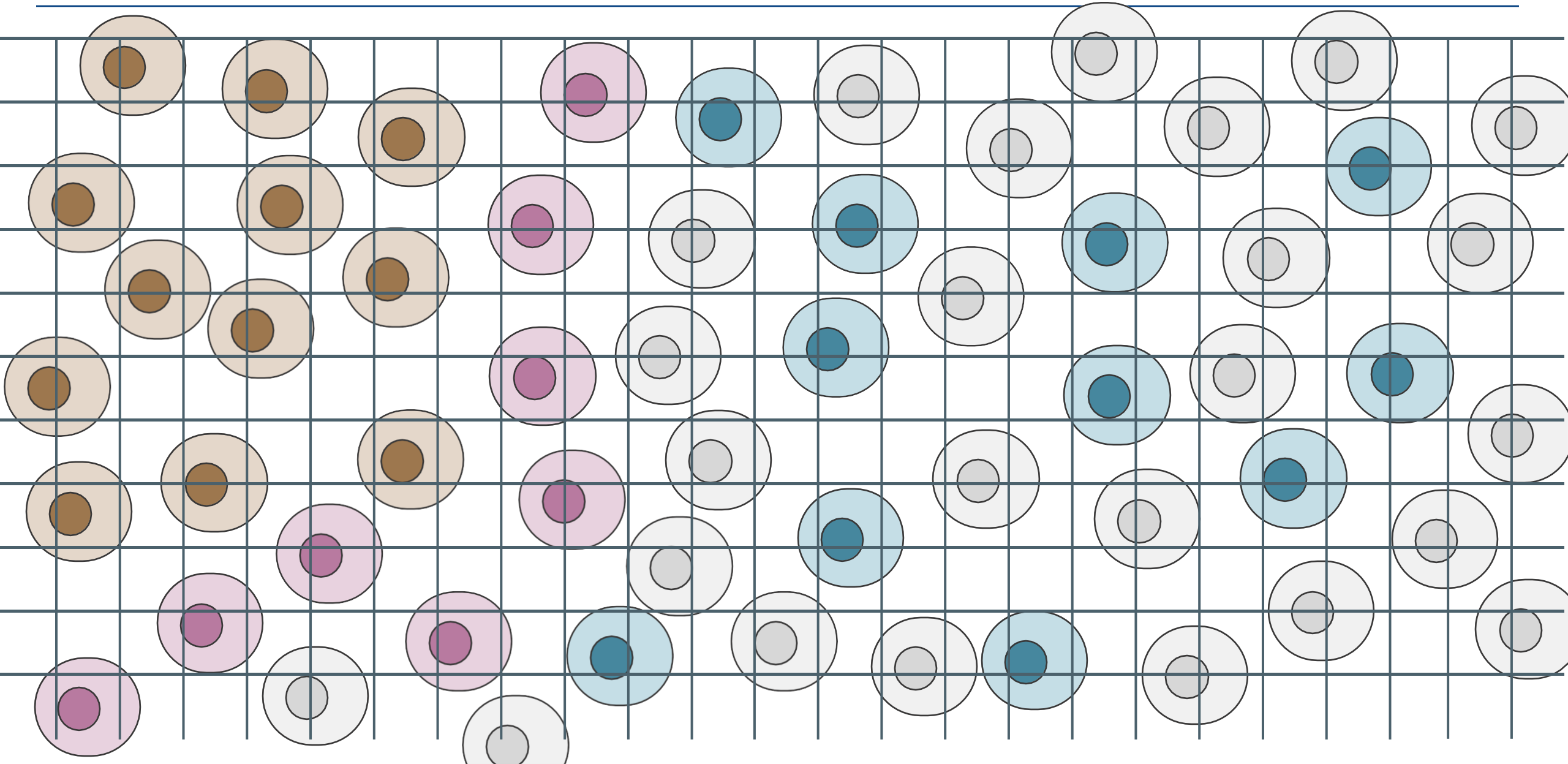
Spatial resolution (NGS-based) Supra cellular grids, e.g. Visium



Spatial resolution (NGS-based) ~1-3 cell profiling, e.g. Slide-Seq/Slide-tags

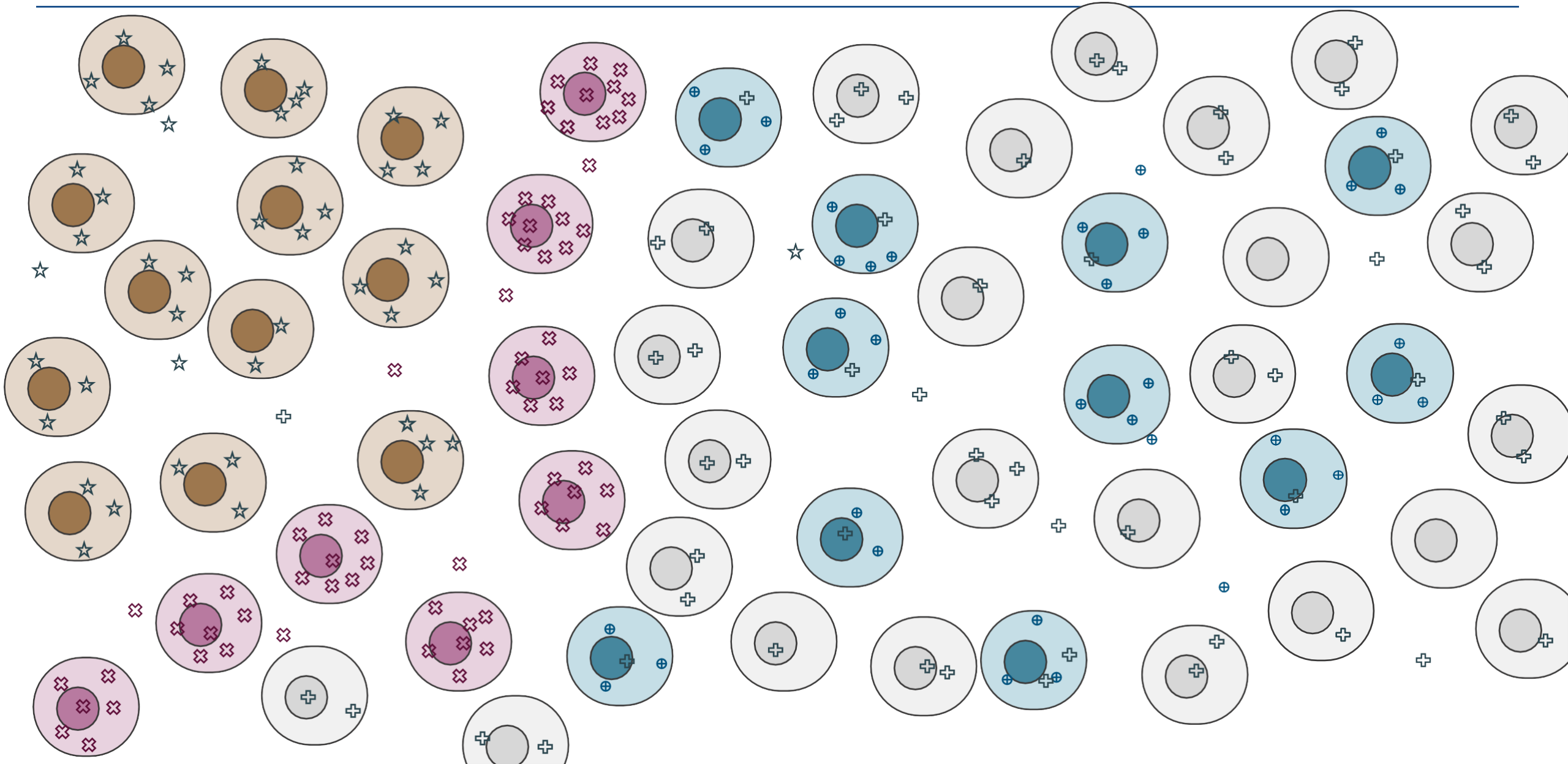


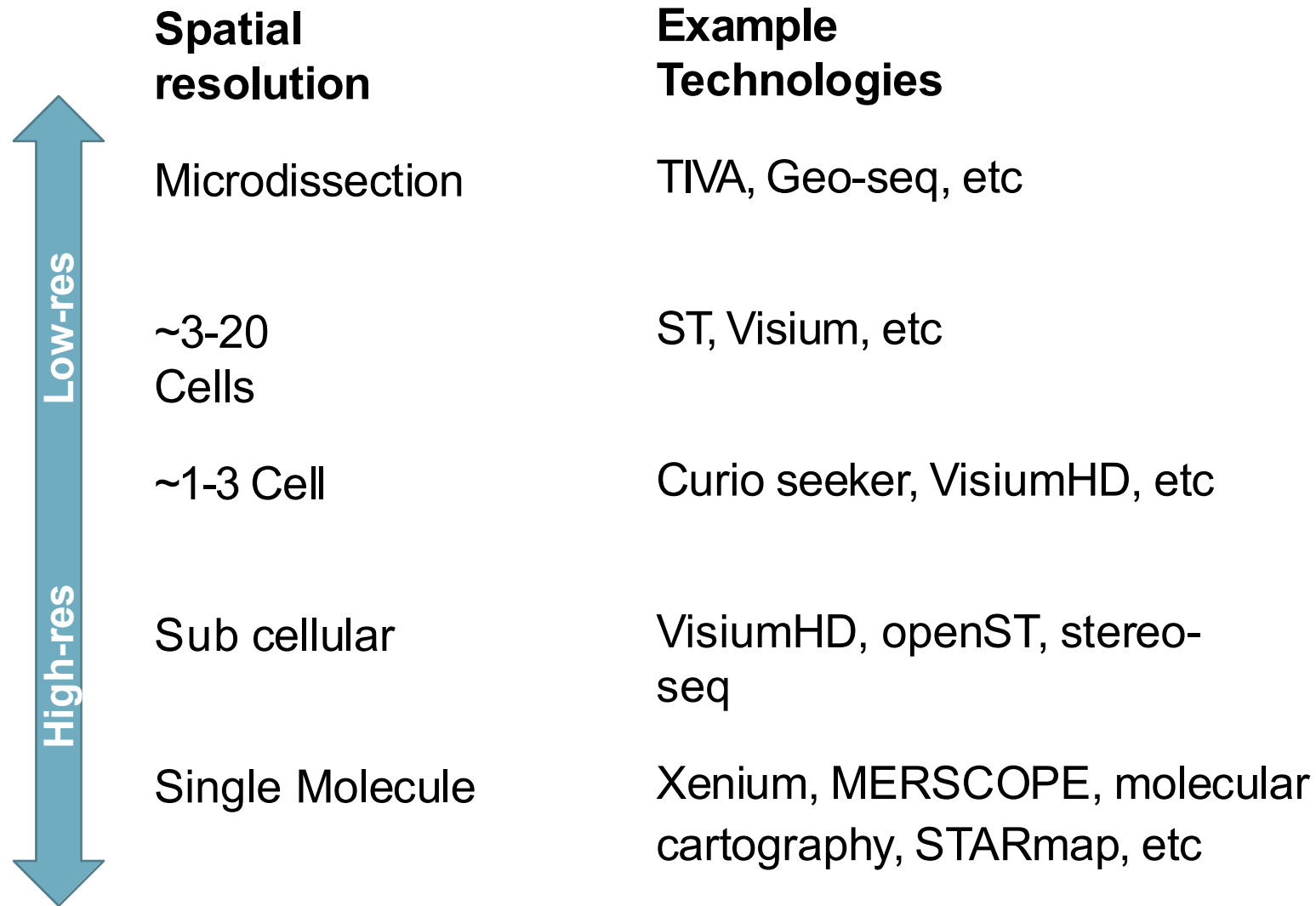
Spatial resolution (NGS-based) Sub-cellular, e.g. VisiumHD

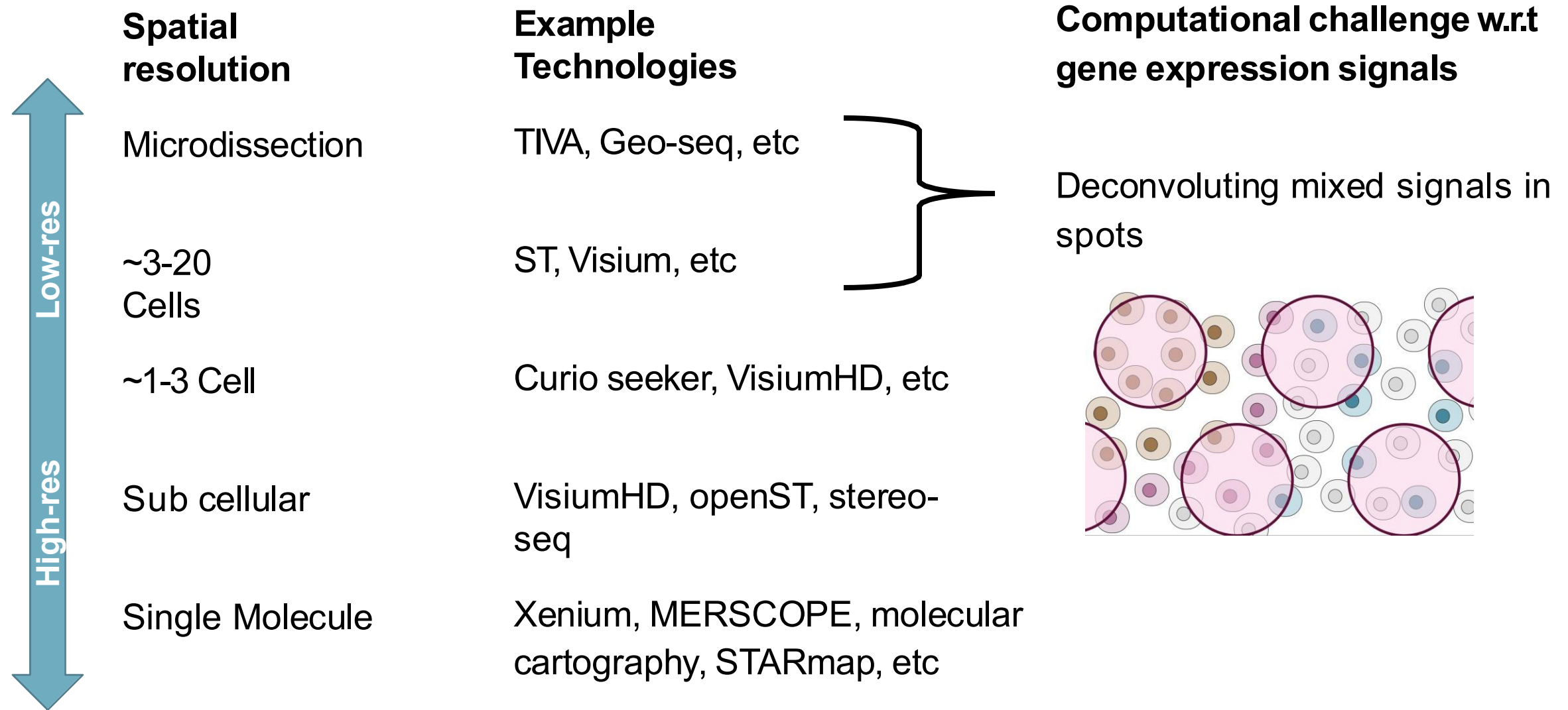


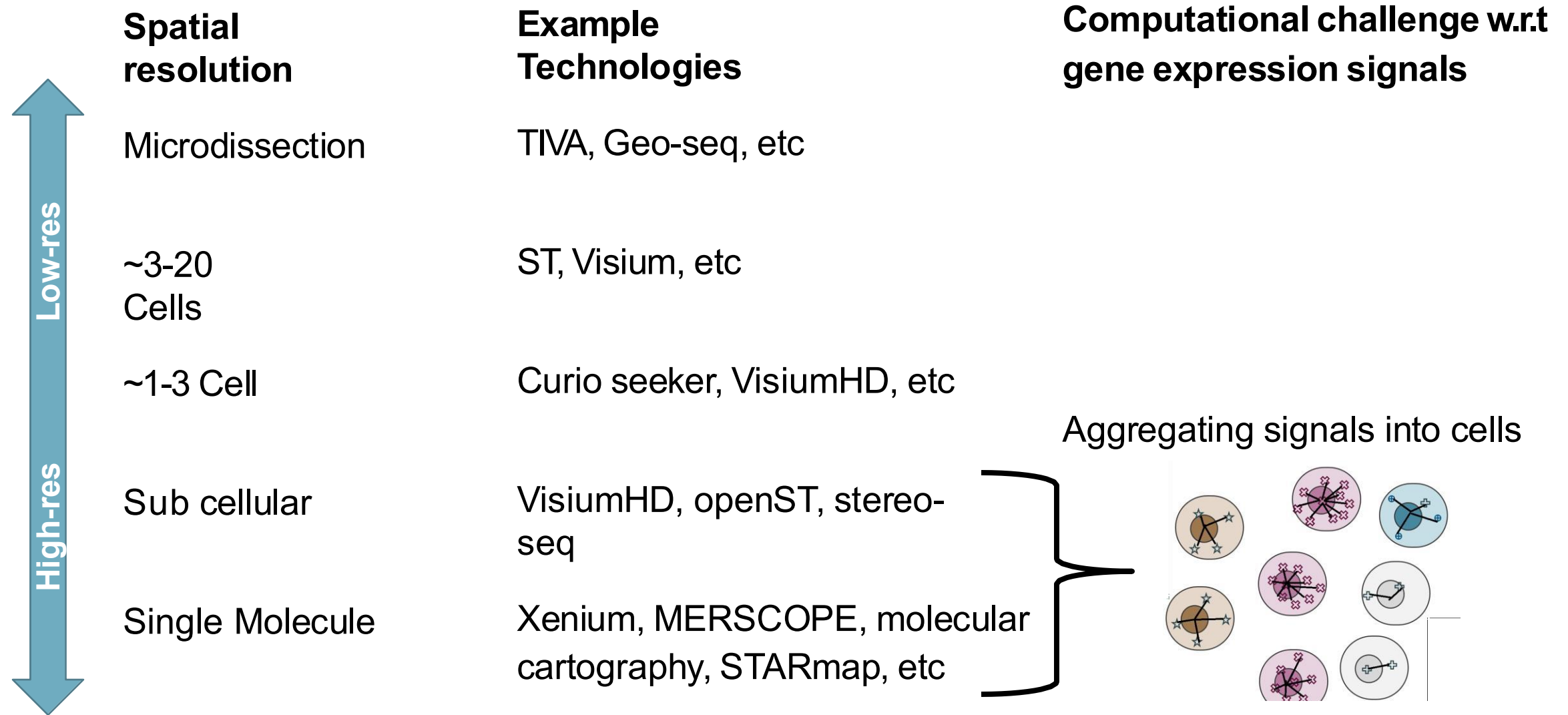
Spatial resolution (Imaging-based)

Single-molecule, e.g. cosMX, Xenium









	Spatial resolution	Example Technologies	Computational challenge w.r.t gene expression signals
	Microdissection	TIVA, Geo-seq, etc	
	~3-20 Cells	ST, Visium, etc	
	~1-3 Cell	Curio seeker, VisiumHD, etc	
	Sub cellular	VisiumHD, openST, stereo-seq	
	Single Molecule	Xenium, MERSCOPE, molecular cartography, STARmap, etc	Single-molecule spatial transcriptomics offers high spatial resolution, but is limited by panel size (typically 500–5000 genes), meaning not all transcripts can be captured and transcript coverage decreases as panel size increases

	Spatial resolution	Example Technologies	Computational challenge w.r.t gene expression signals
	Microdissection	TIVA, Geo-seq, etc	
	~3-20 Cells	ST, Visium, etc	
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	Single Molecule	Xenium, MERSCOPE, molecular cartography, STARmap, etc	

!Choice of technology depends on your biological question!

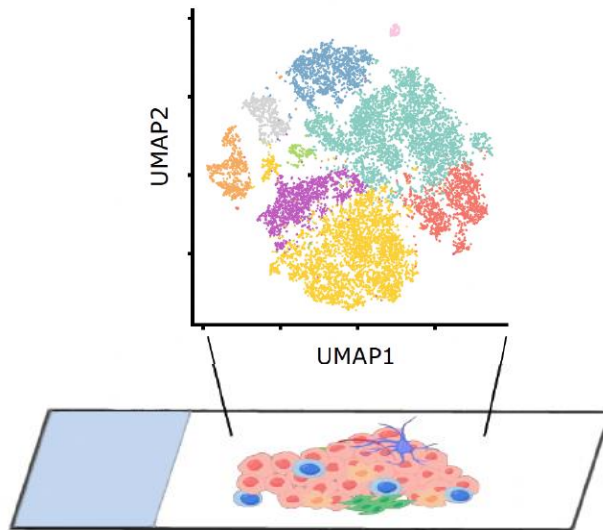
	Spatial resolution	Example Technologies	Computational challenge w.r.t gene expression signals
 Low-res High-res	Microdissection	TIVA, Geo-seq, etc	
	~3-20 Cells	ST, Visium, etc	
	~1-3 Cell	Curio seeker, VisiumHD, etc	
	Sub cellular	VisiumHD, openST, stereo-seq	
	Single Molecule	Xenium, MERSCOPE, molecular cartography, STARmap, etc	

Imaging-based spatial transcriptomics e.g. Xenium (10x Genomics)



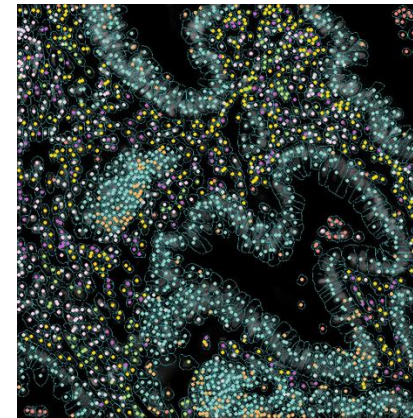
~500 genes

Spatial transcriptomics- Xenium



Transcriptome:
Cell phenotypes
and functional states

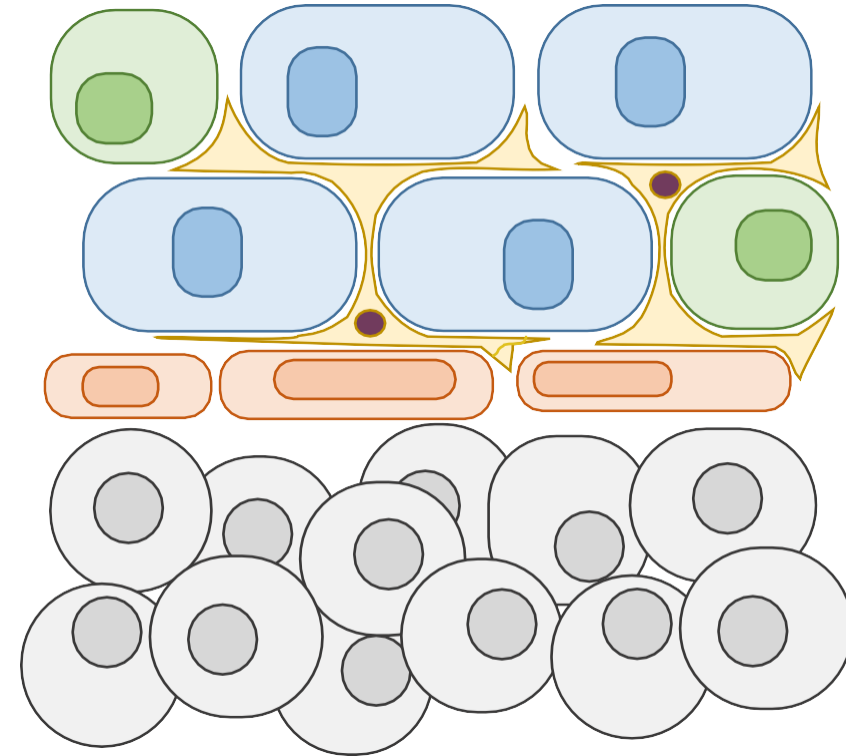
X,Y coordinates:
Mapping cells to
location



Imaging-based spatial transcriptomics e.g. Xenium (10x Genomics)



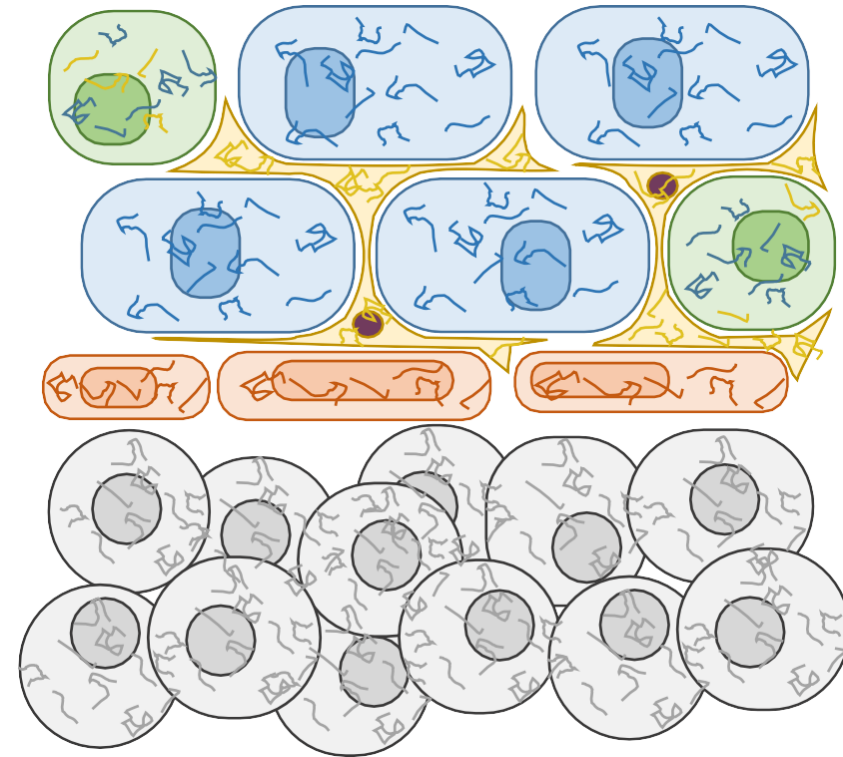
- Cells in tissue



Imaging-based spatial transcriptomics e.g. Xenium (10x Genomics)



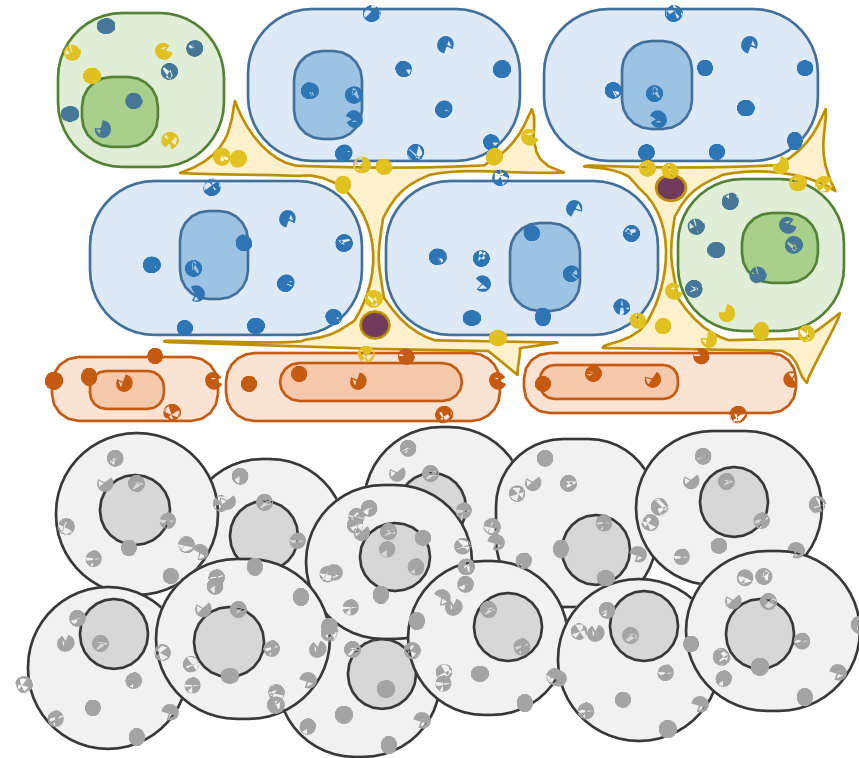
- Cells in tissue
- Transcripts locations



Imaging-based spatial transcriptomics e.g. Xenium (10x Genomics)



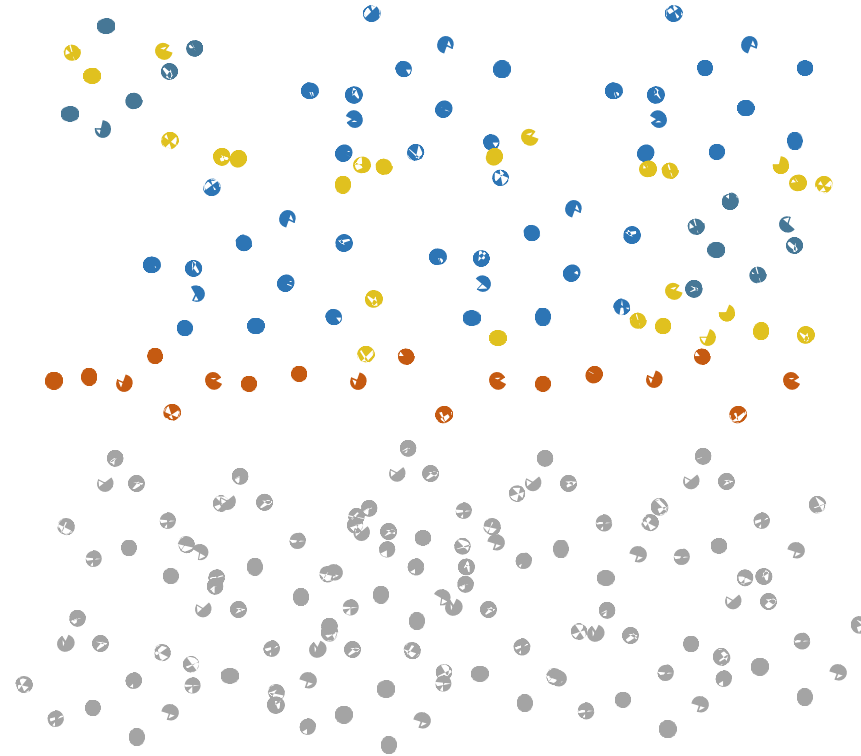
- Cells in tissue
- Transcripts locations
- Image transcript locations



Imaging-based spatial transcriptomics e.g. Xenium (10x Genomics)



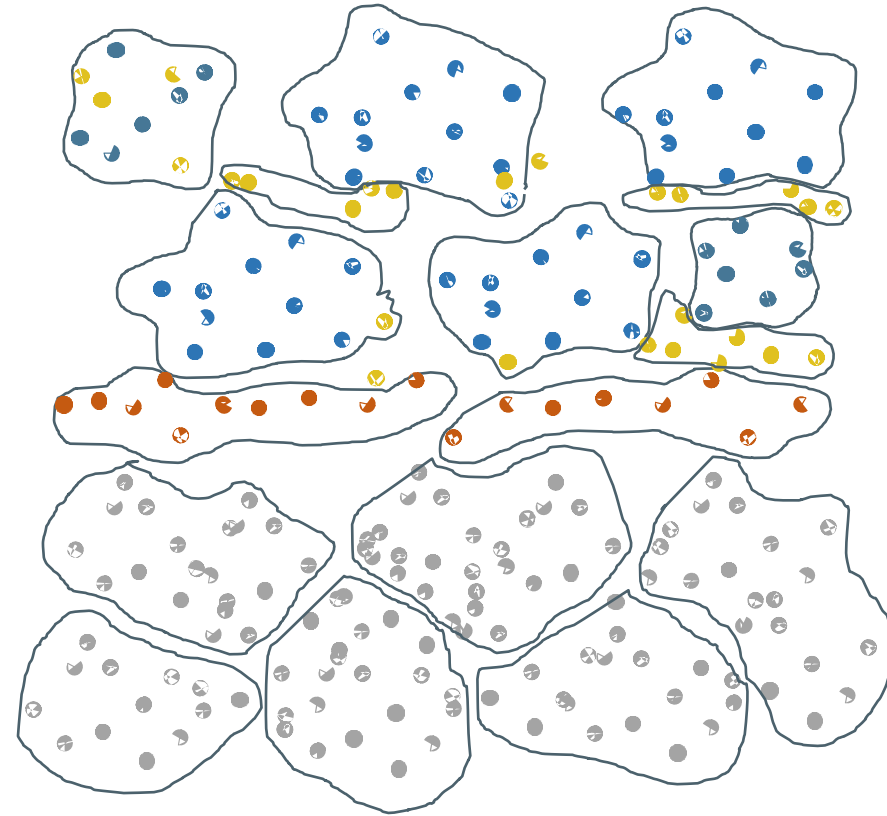
- Cells in tissue
- Transcripts locations
- Image transcript locations
- Report transcript locations



Imaging-based spatial transcriptomics e.g. Xenium (10x Genomics)



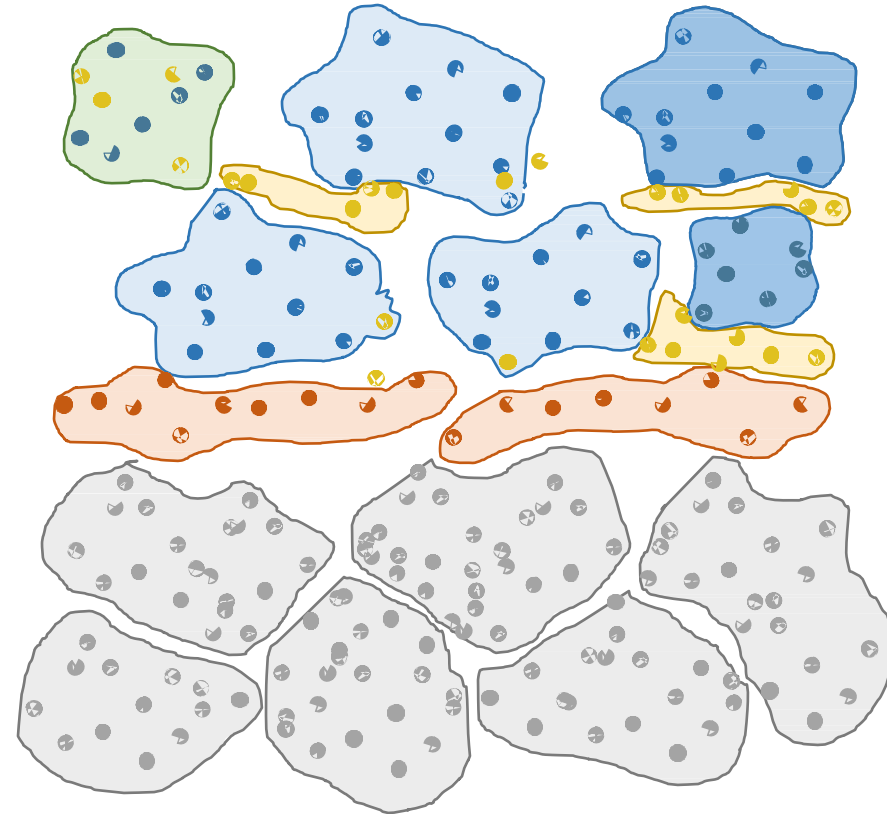
- Cells in tissue
- Transcripts locations
- Image transcript locations
- Report transcript locations
- Identify cells (segmentation)



Imaging-based spatial transcriptomics e.g. Xenium (10x Genomics)



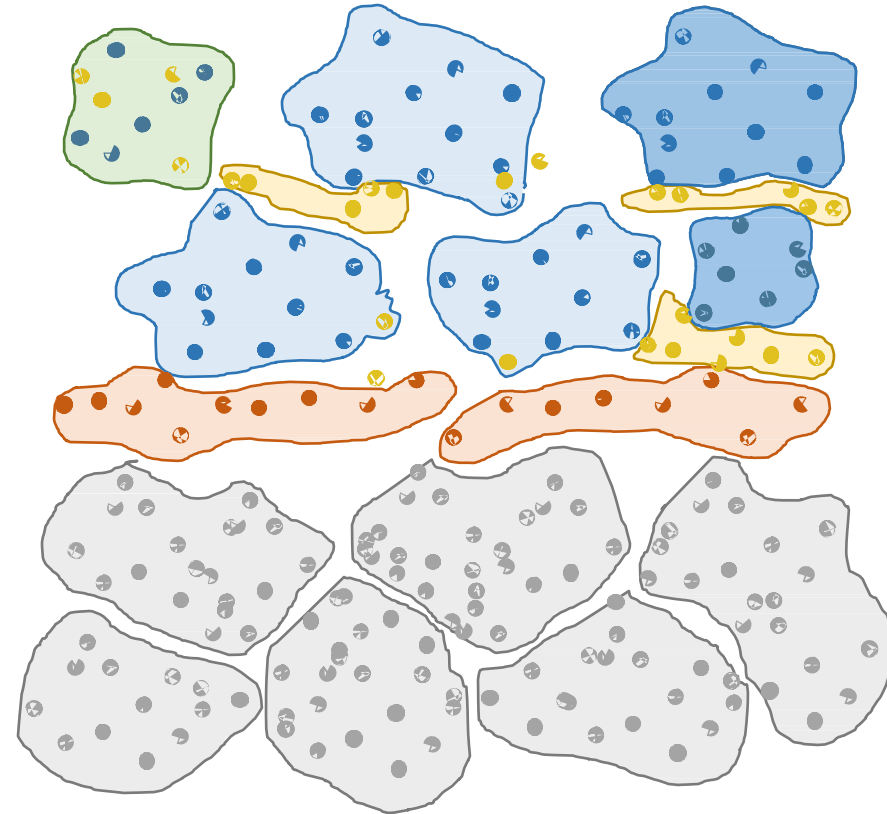
- Cells in tissue
- Transcripts locations
- Image transcript locations
- Report transcript locations
- Identify cells (segmentation)
- Analyse cells

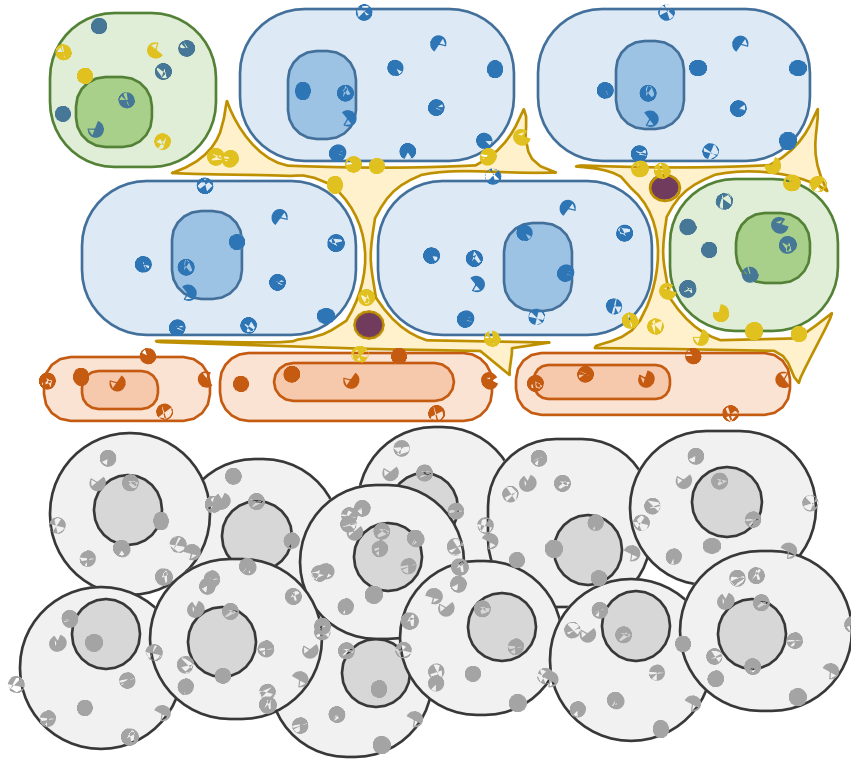


Imaging-based spatial transcriptomics e.g. Xenium (10x Genomics)

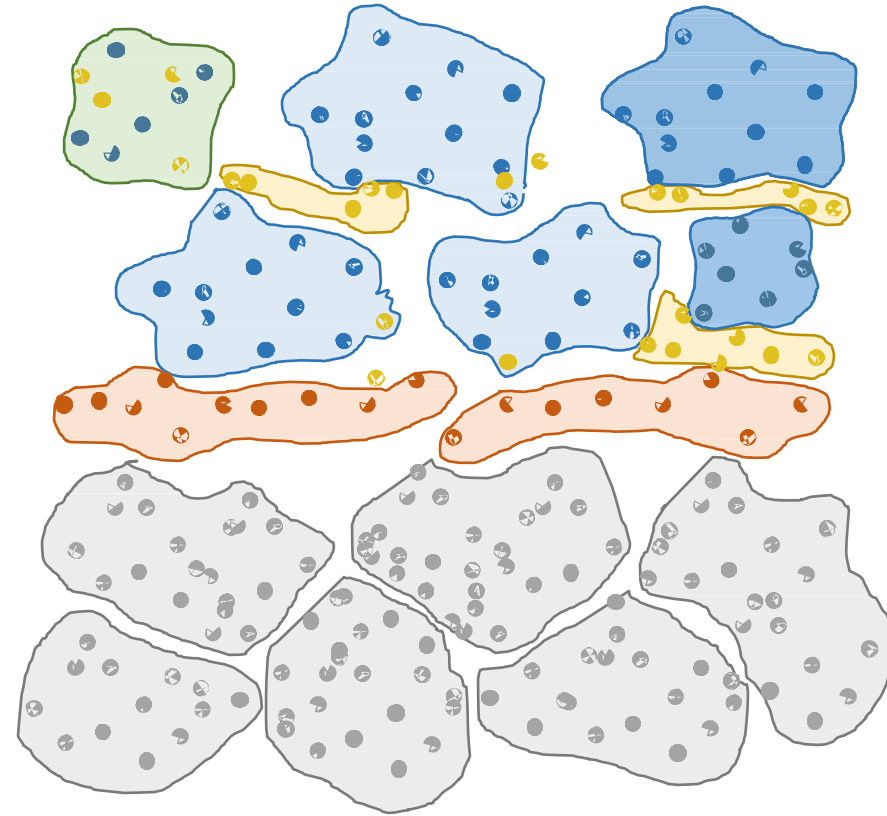


- Cells in tissue
- Transcripts locations
- Image transcript locations
- Report transcript locations
- **Identify cells (segmentation)**
- Analyse cells





Spatial orientation in the tissue

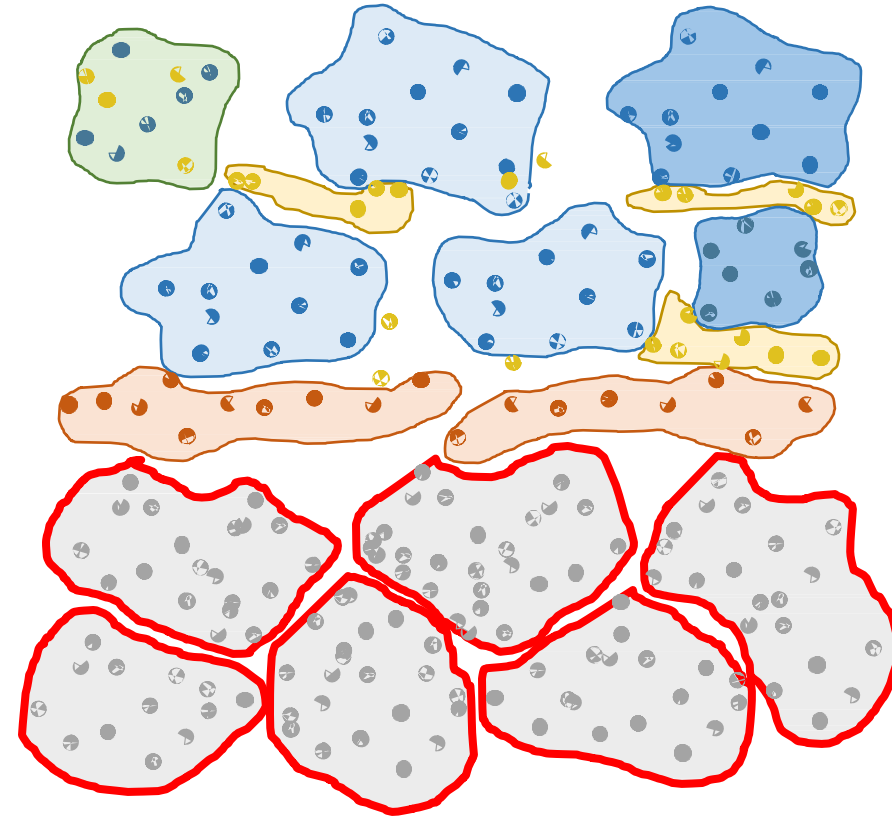
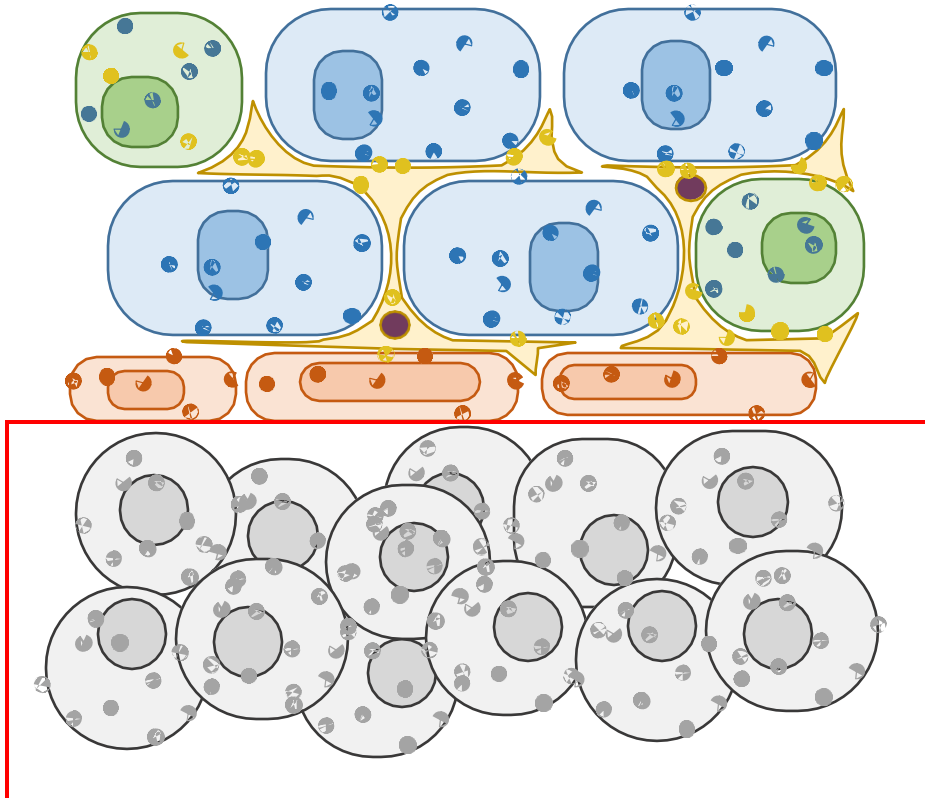


What Xenium maps

Evaluation – cell proportions

Overlapping cells

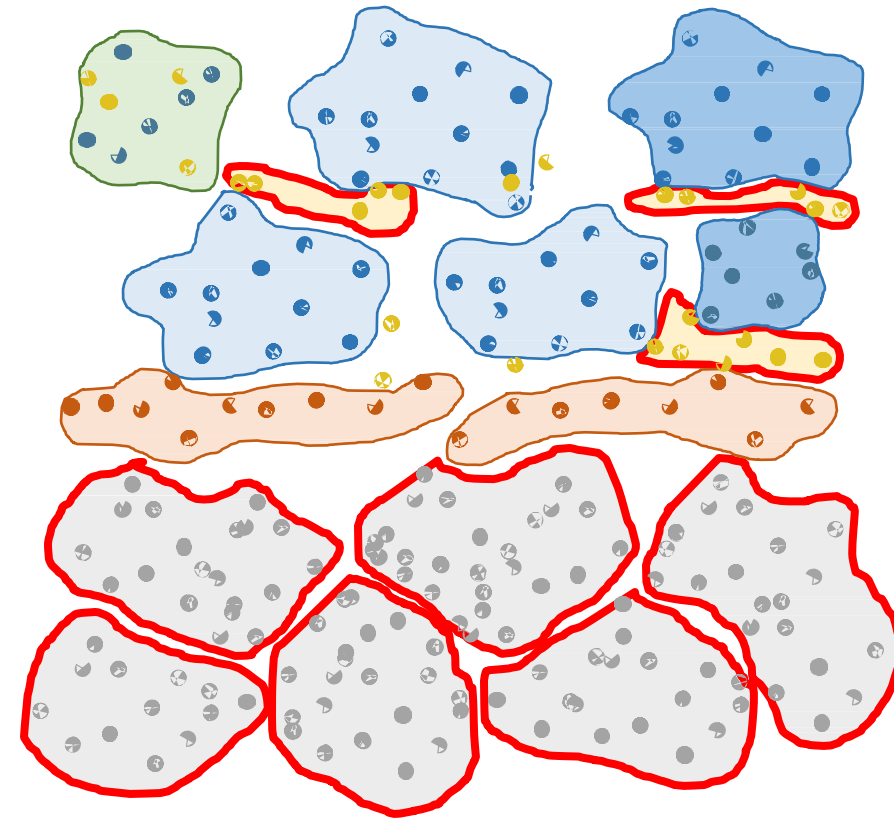
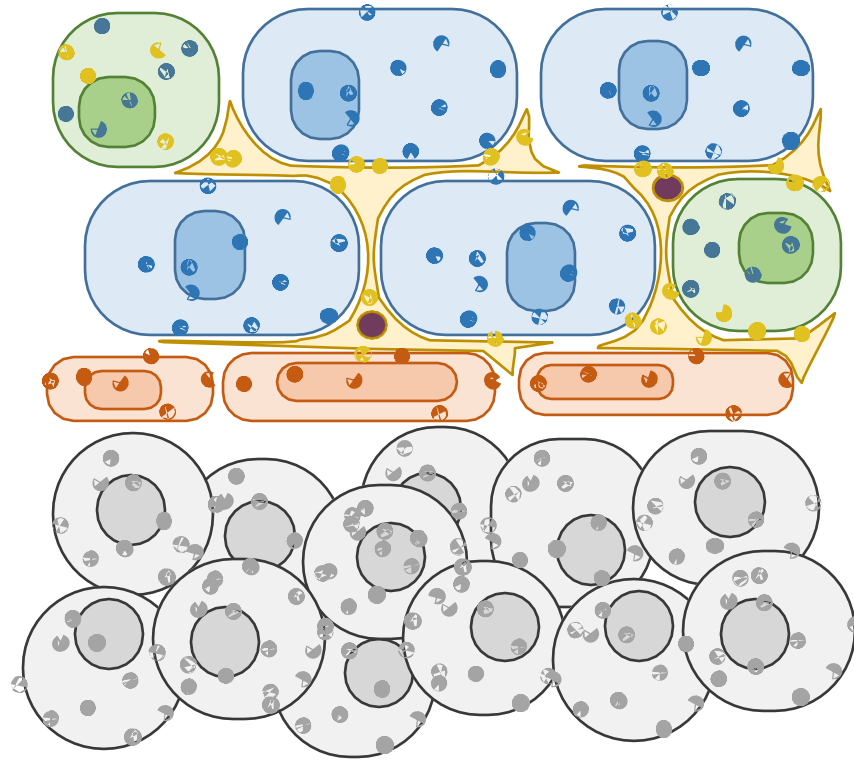
Green, blue, orange, yellow cells all +/- 1



Highly dense grey cells are hard to separate

Evaluation – cell size

Unique cell types that are not “classically round”

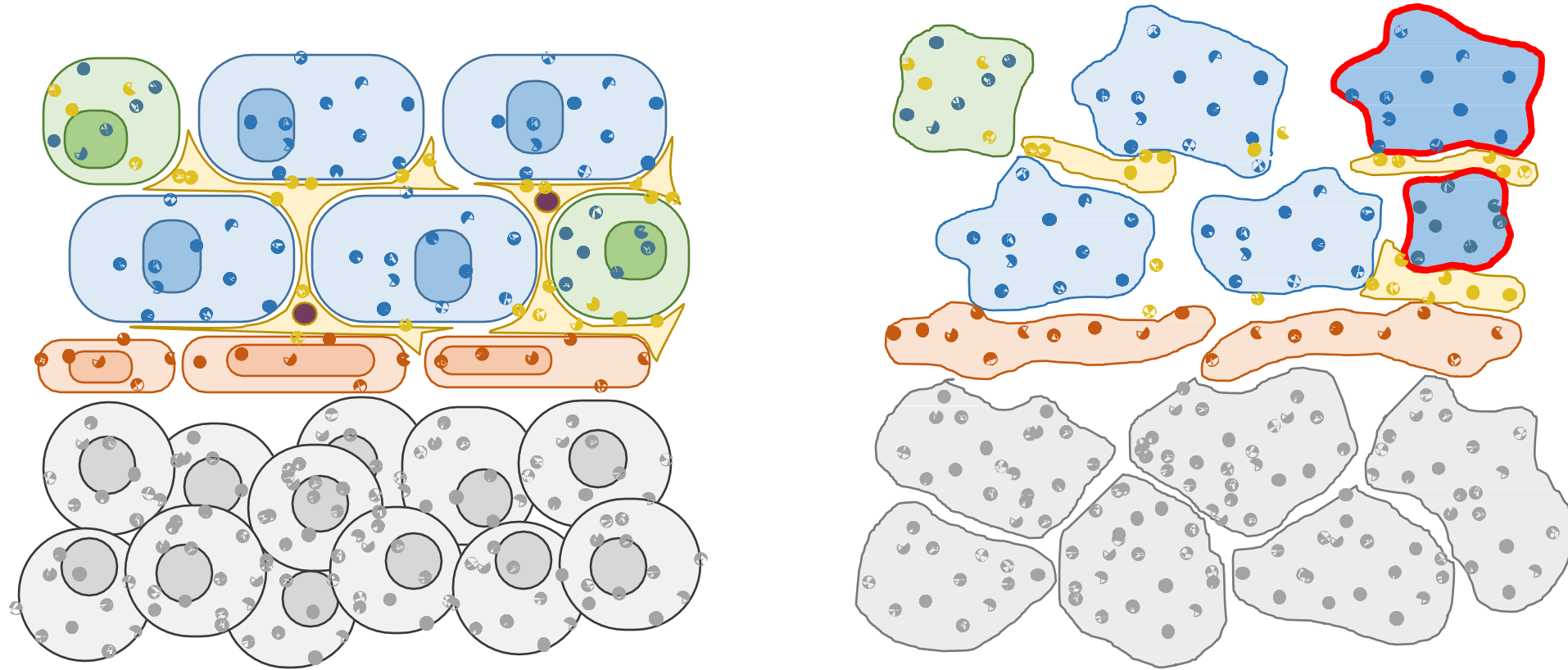


yellow
cells are
split or
missed

Highly dense grey cells are too
large

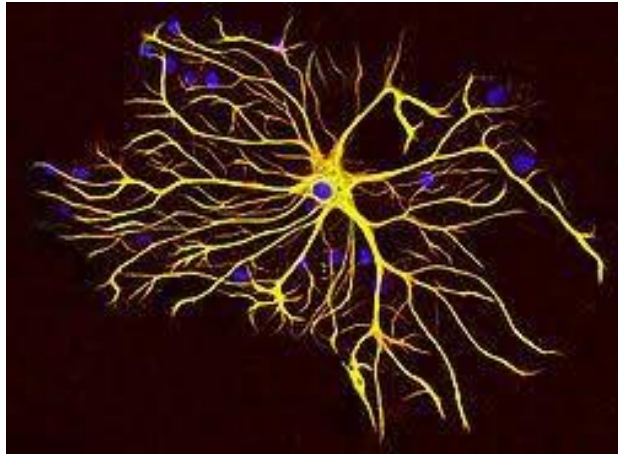
Evaluation – cell type annotation

Spatial co-localization signatures, contamination

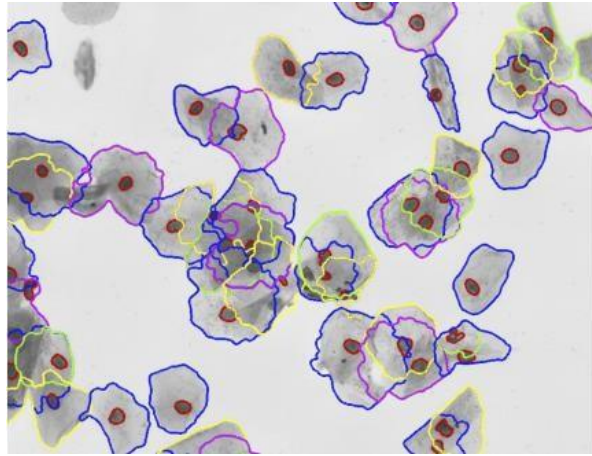


Cell segmentation is not always easy and never perfect

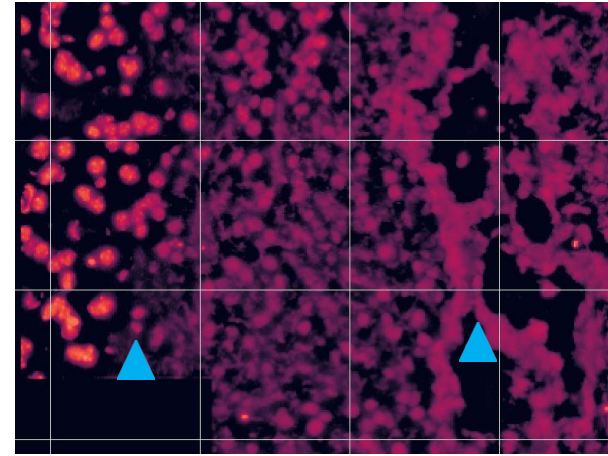
Transcript density-based cell segmentation (e.g. Baysor)



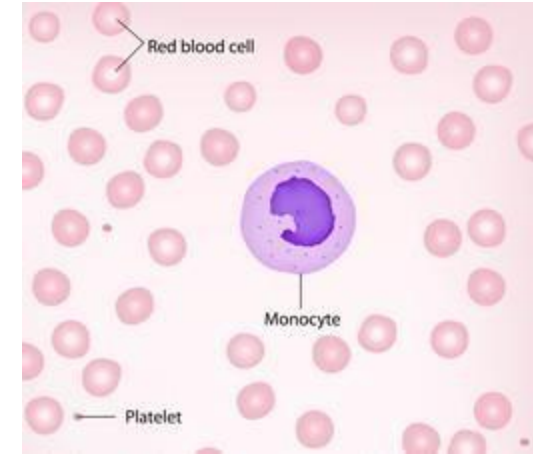
Complex shapes



Overlapping cells



Cell stains might have issues

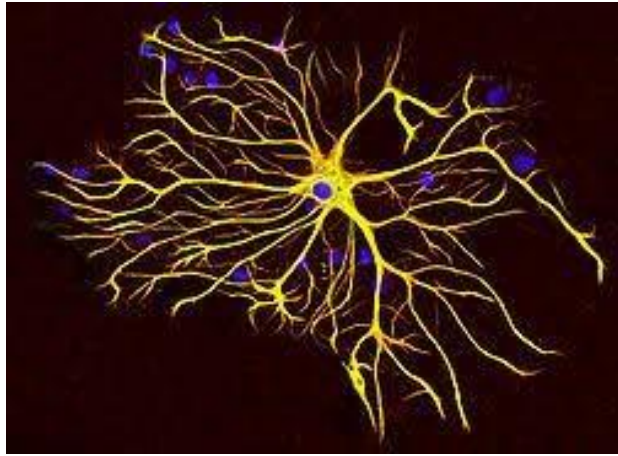


Cells might not be stained correctly (e.g. red blood cells)

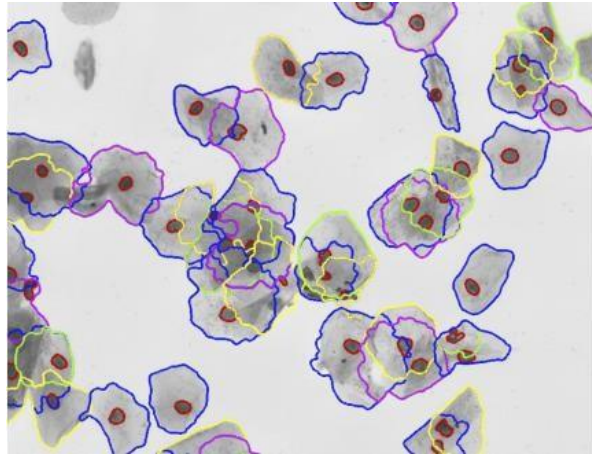
Cell segmentation is not always easy and never perfect



Transcript density-based cell segmentation (e.g. Baysor)

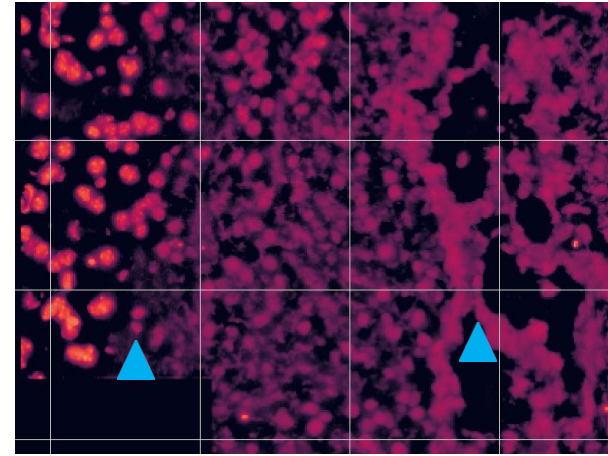


Complex shapes

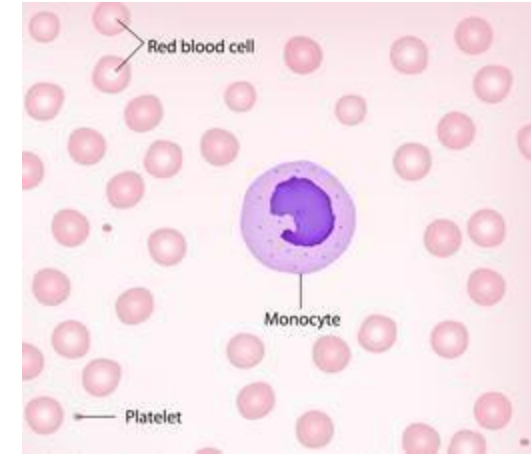


Overlapping cells

Cell staining kit-based cell segmentation



**Cell stains might have
issues**



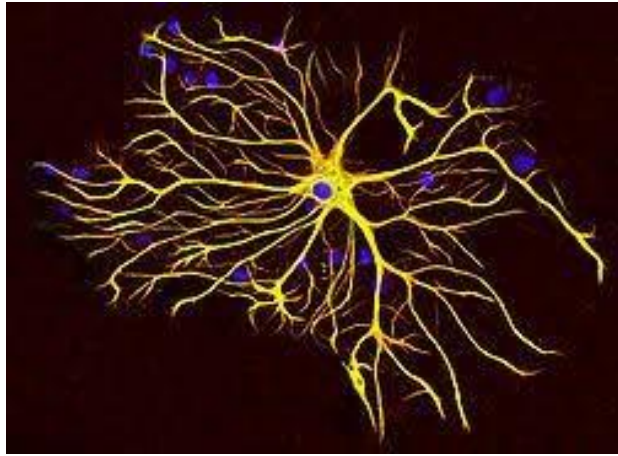
**Cells might not be
stained correctly (e.g.
red blood cells)**

Are there analyses methods that we can use in parallel to uncover biology?

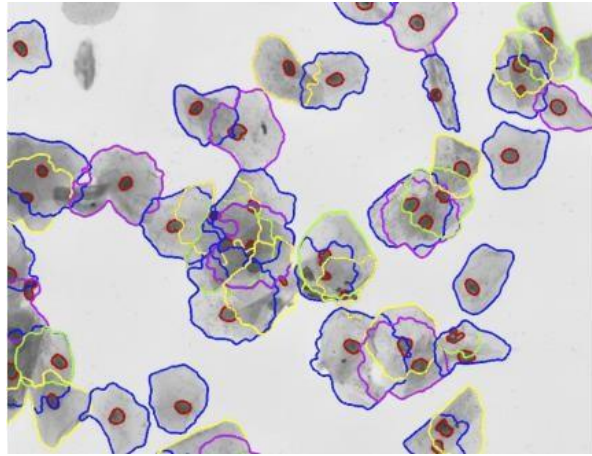
Cell segmentation is not always easy and never perfect



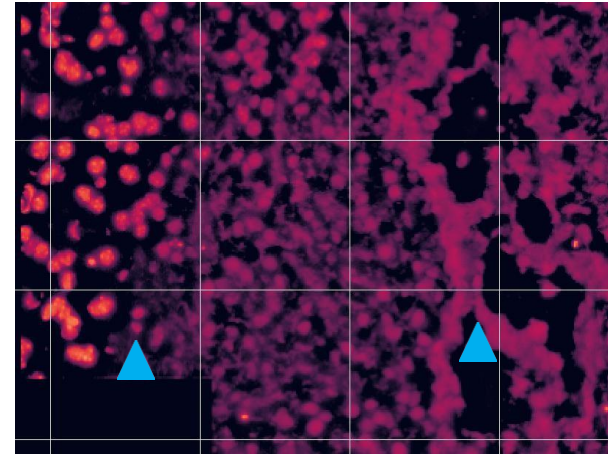
Transcript density-based cell segmentation (e.g. Baysor)



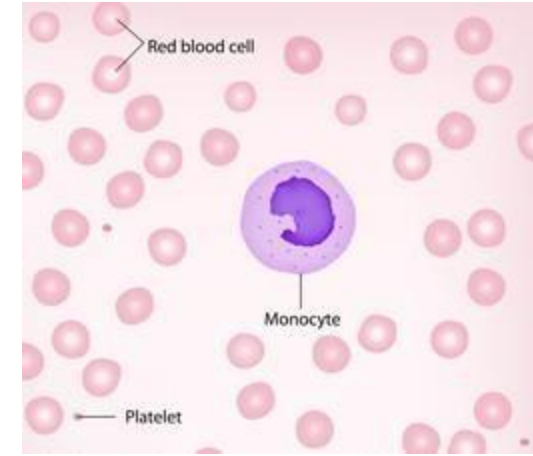
Complex shapes



Overlapping cells



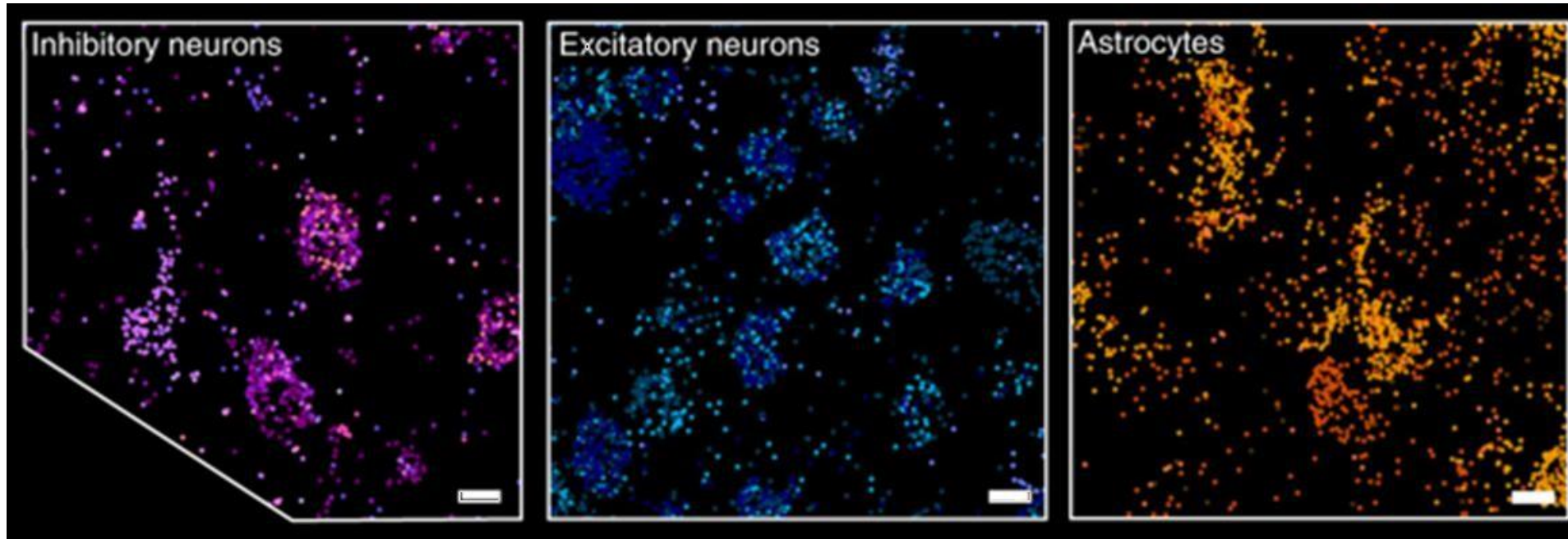
Cell stains might have
issues



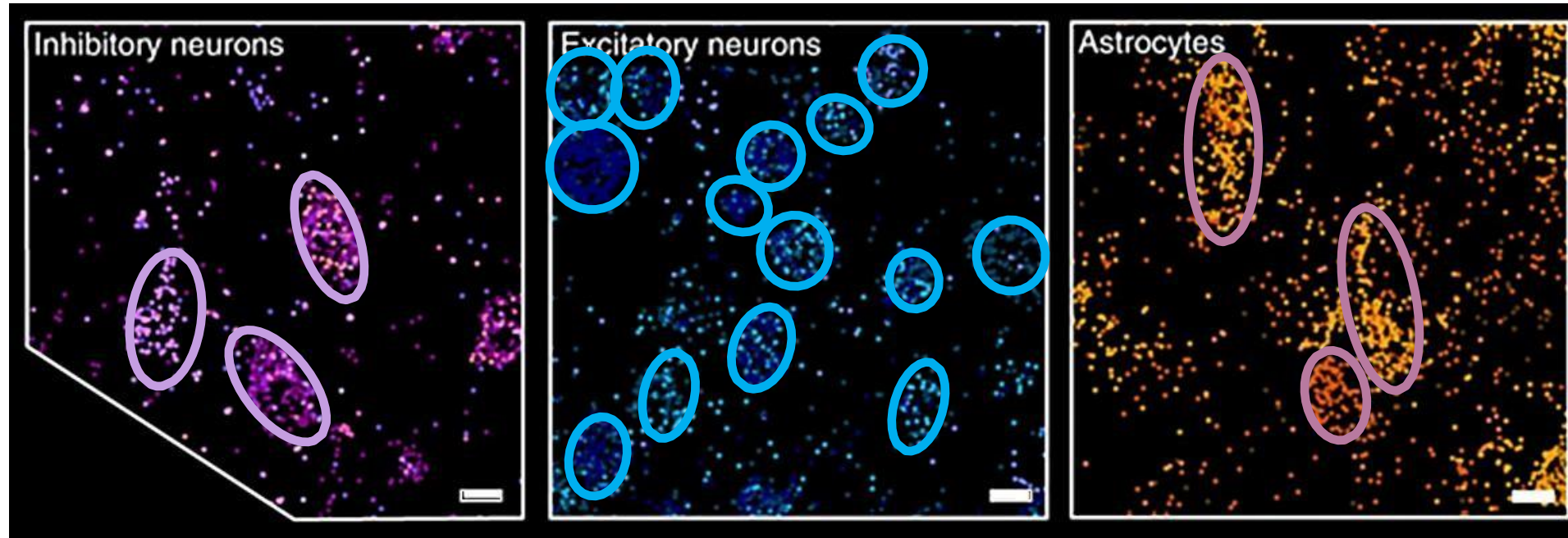
Cells might not be
stained correctly (e.g.
red blood cells)

Are there analyses methods that we can use in parallel to uncover biology?
Segmentation-free analysis methods

mRNA molecule organization patterns are not random



mRNA molecule organization patterns are not random



GRIDGENE: Guided Region Identification based on Density of GENEs – a transcript density-based approach to characterize tissues by spatial transcriptomics



New Results

 Follow

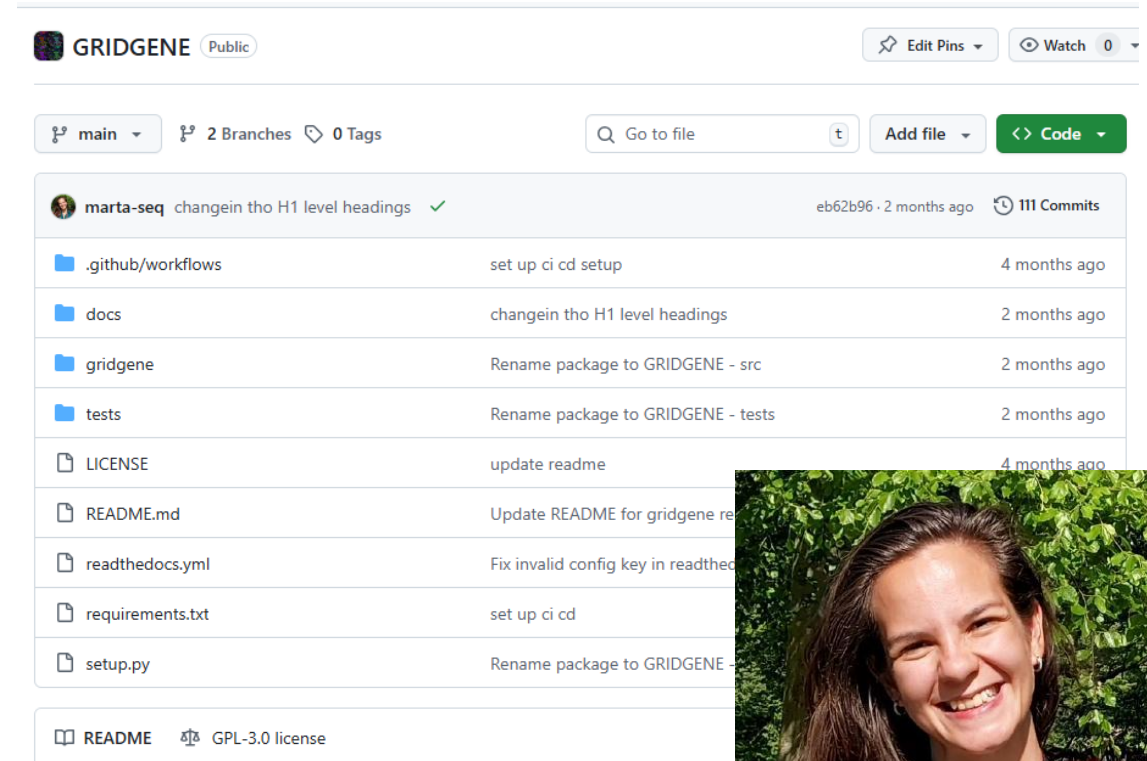
GRIDGENE: Guided Region Identification based on Density of GENEs – a transcript density-based approach to characterize tissues by spatial transcriptomics

 A. M. Sequeira,
  M. E. Ijsselsteijn,
  M. Rocha,
  J. Roelands,
  Noel F.C.C. de Miranda

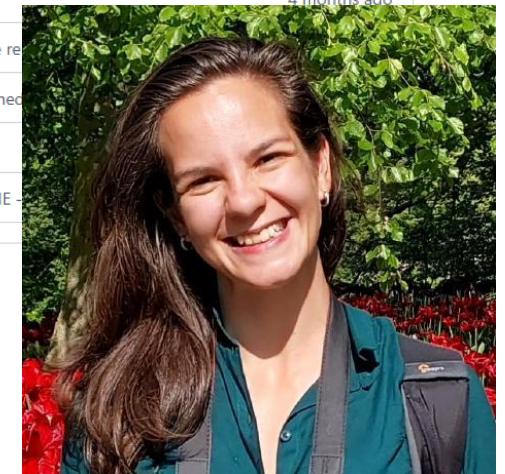
doi: <https://doi.org/10.1101/2025.08.14.670318>

This article is a preprint and has not been certified by peer review [what does this mean?].

Paper available on bioRxiv



Code available on GitHub



Ana Marta Sequeira

Defining a characteristic gene set for your region of interest

GRIDGENE identifies regions of interest by identifying areas that have high expression density for a user-defined gene set signature.

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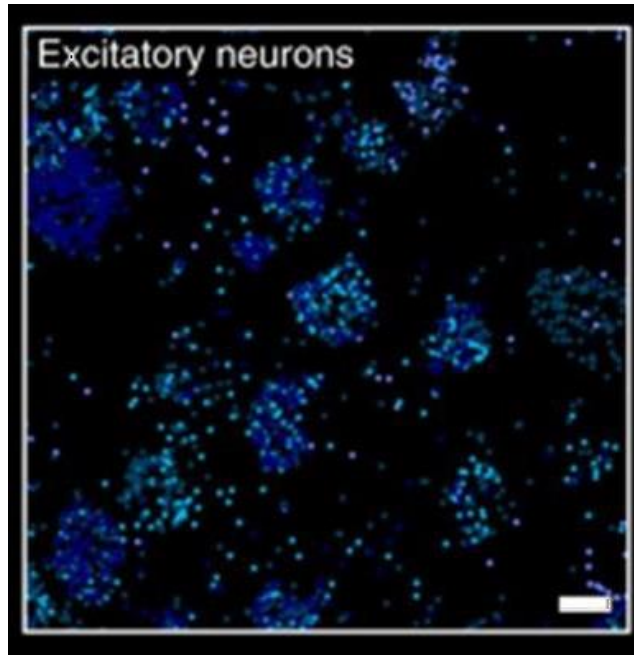
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Cell types



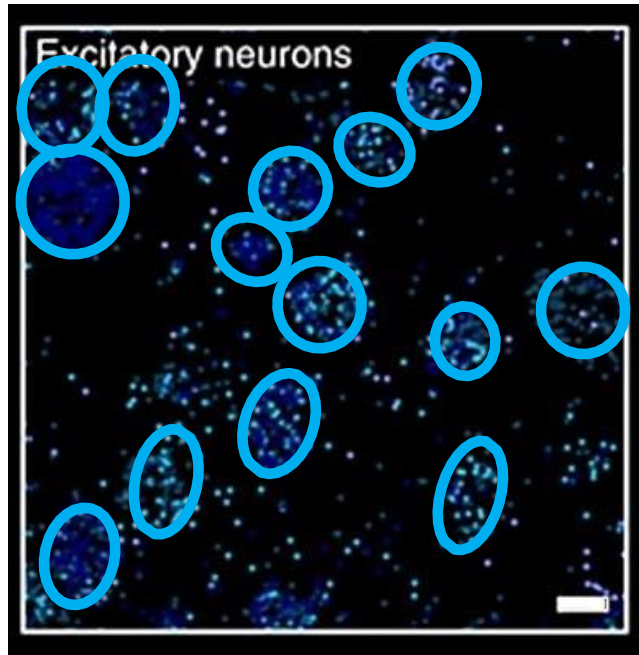
Expression of
genes associated
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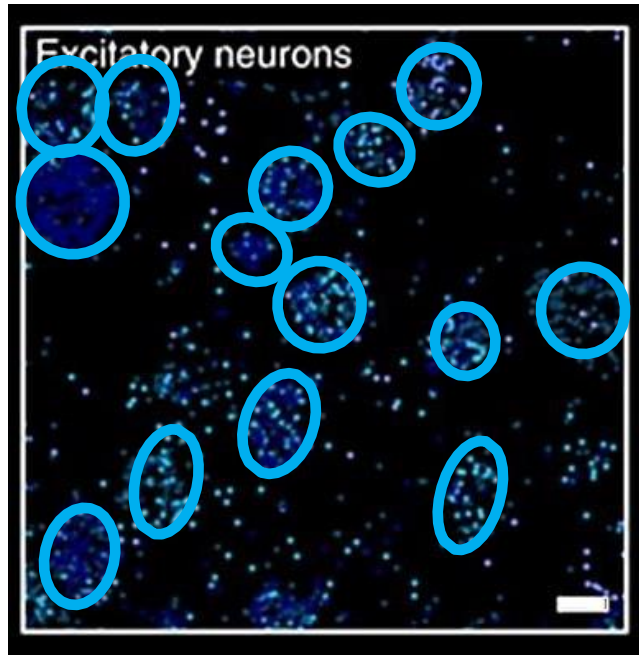
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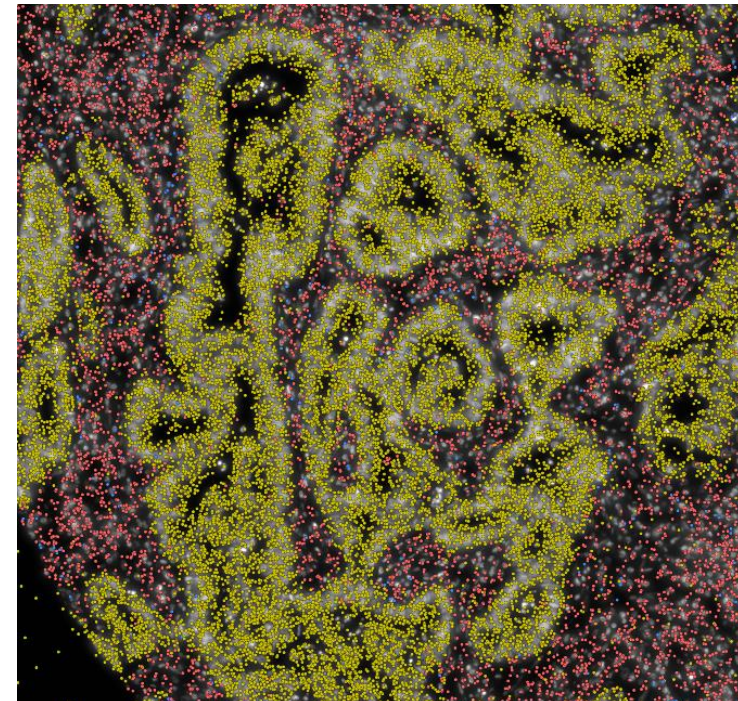
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Expression of
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neurons

Tissue domains



Expression of
genes associated
with tumor domain

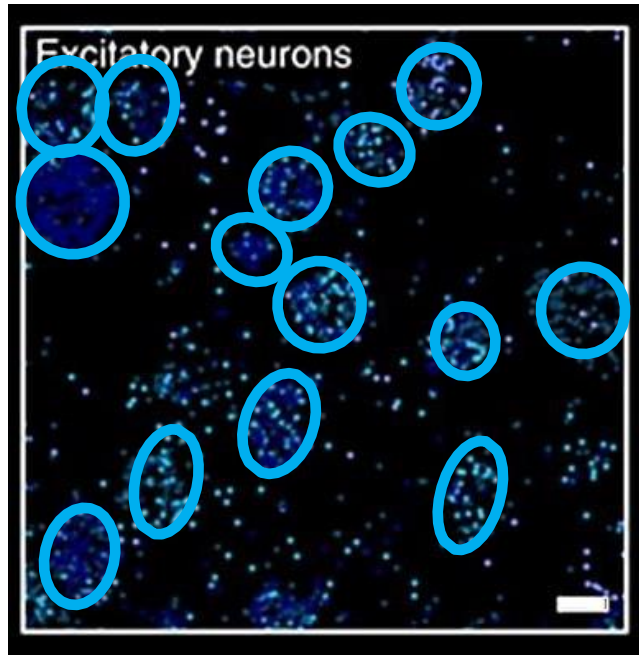
Expression of
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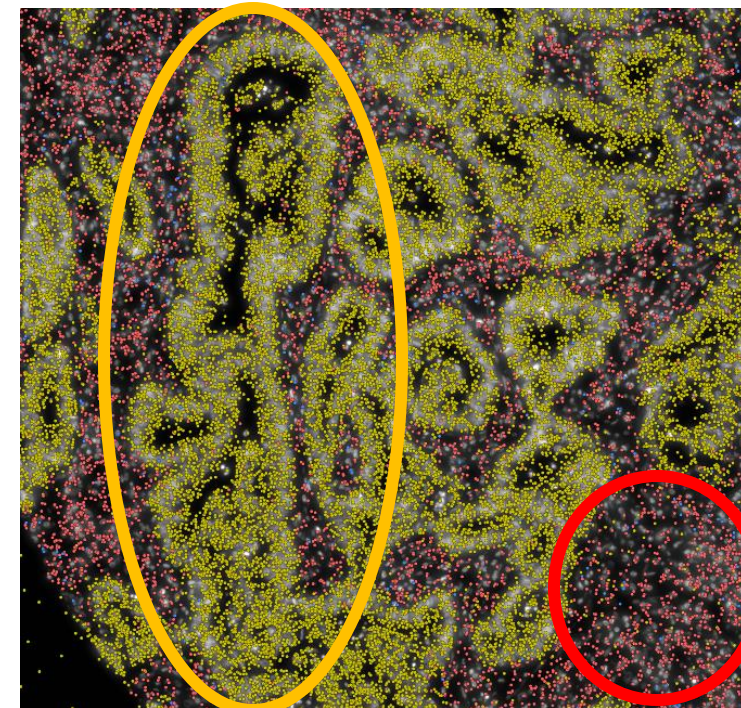
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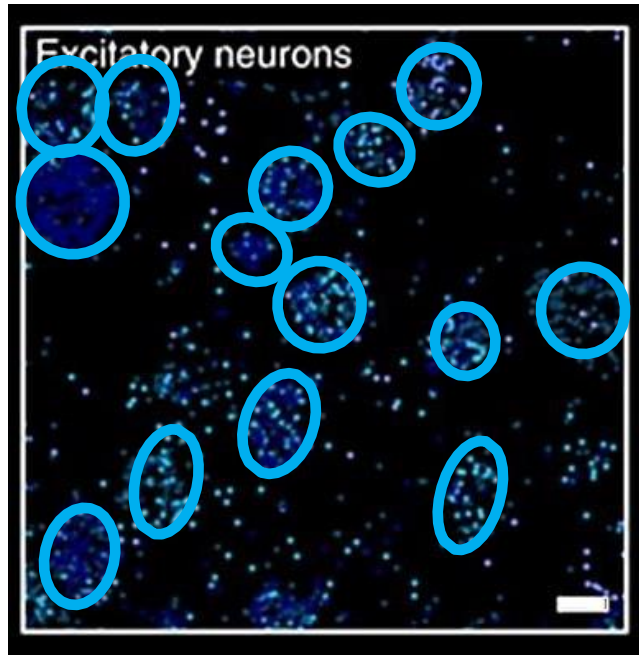
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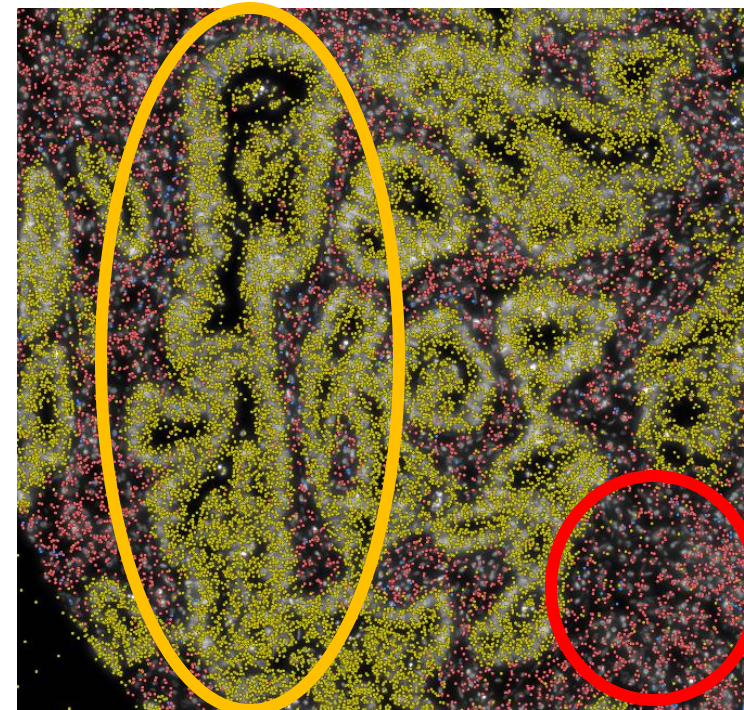
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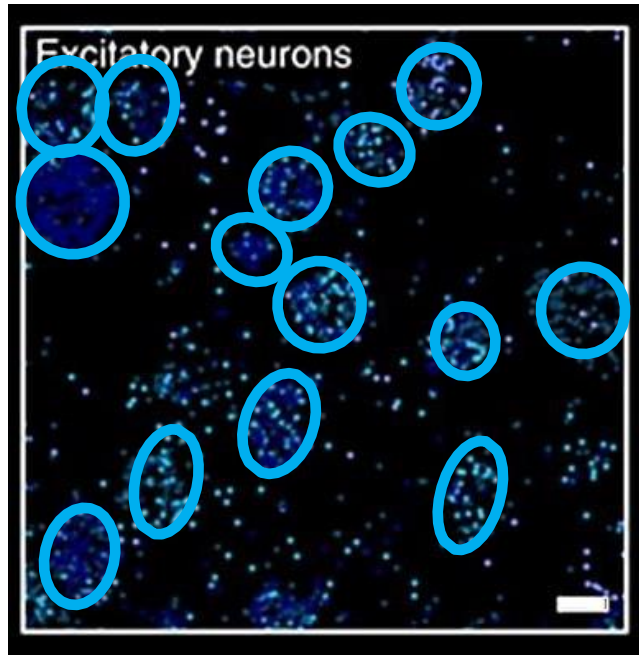
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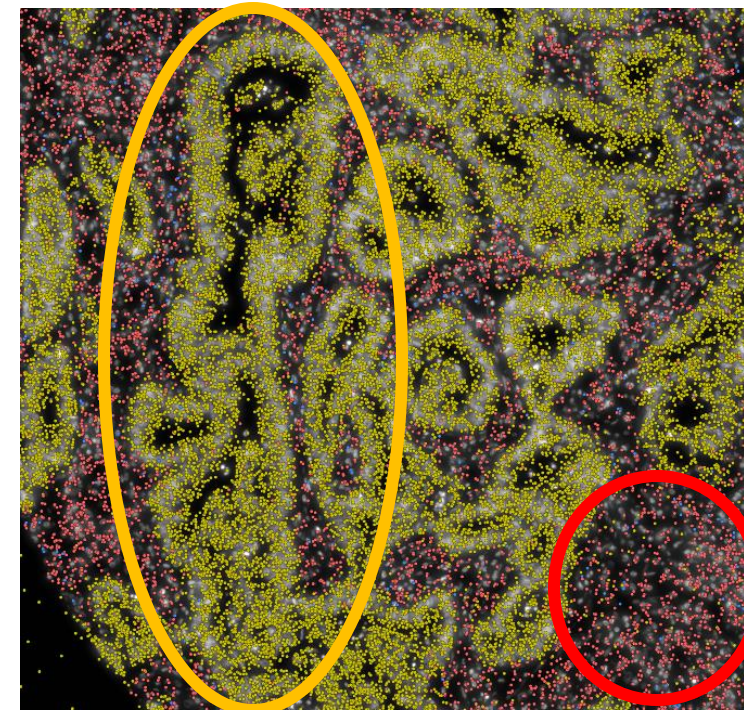
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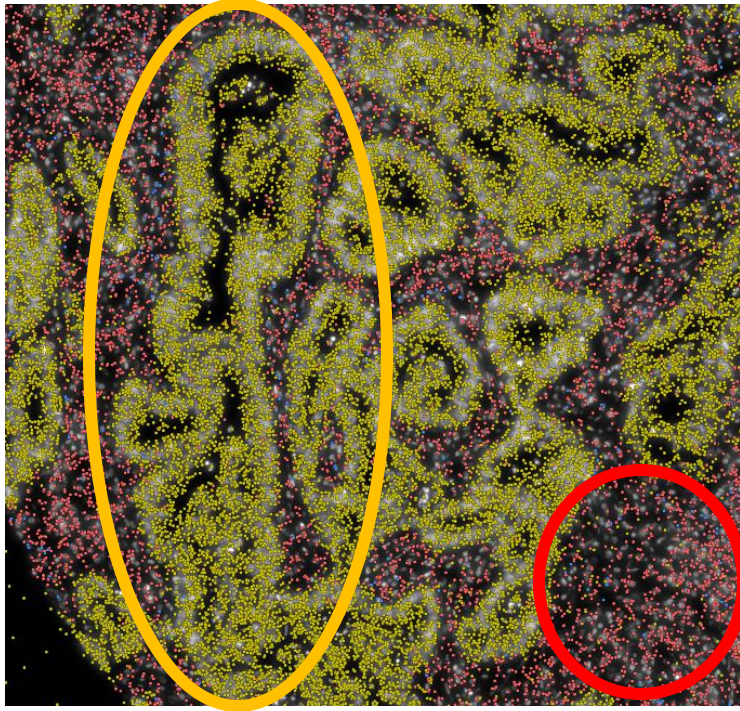


Expression of
genes associated
with tumor domain

Expression of
genes associated
with stromal domain

Tumors into two major biological compartments:

1. Epithelium. 2. Stroma.



Expression of genes associated with epithelium domain

Expression of genes associated with stromal domain

1. Epithelium regions

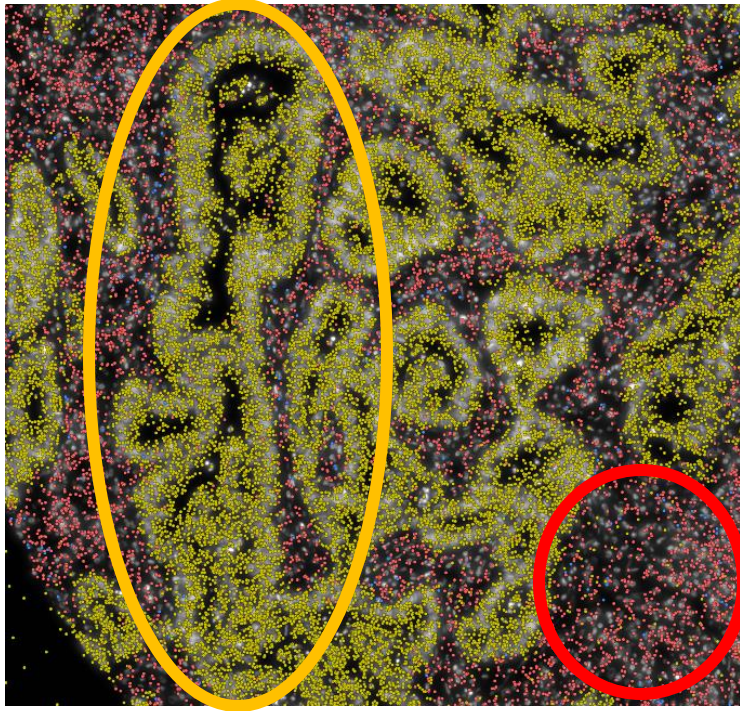
- Cancer cells

2. Stroma regions

- Immune and stromal cells

Tumors into two major biological compartments:

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Expression of genes associated with epithelium domain

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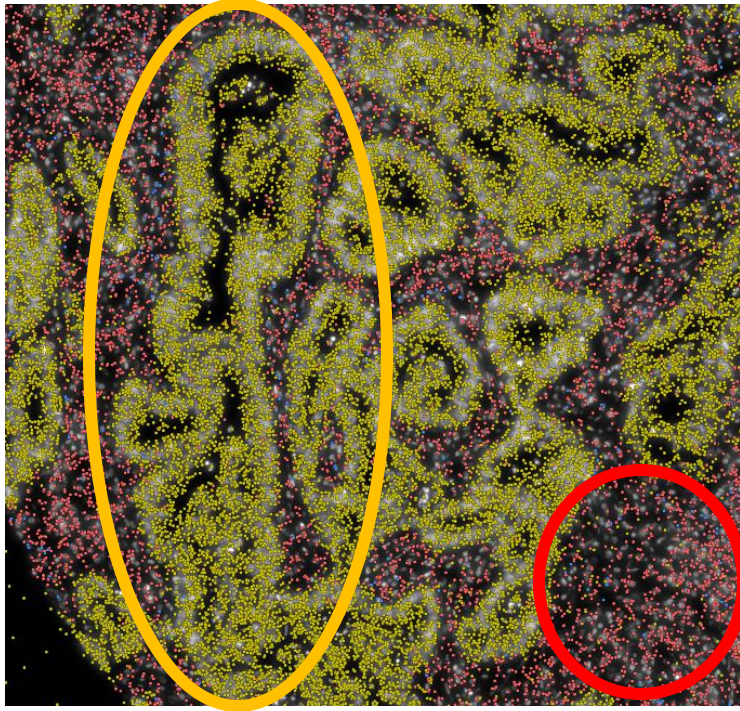
2. Stroma regions

- Immune and stromal cells

With GRIDGENE, these regions can be delineated based on areas **with high expression of defined gene sets**, bypassing the need for cell segmentation and facilitating downstream analyses targeting these specific compartments.

Tumors into two major biological compartments:

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Expression of genes associated with epithelium domain

Expression of genes associated with stromal domain

1. Epithelium regions

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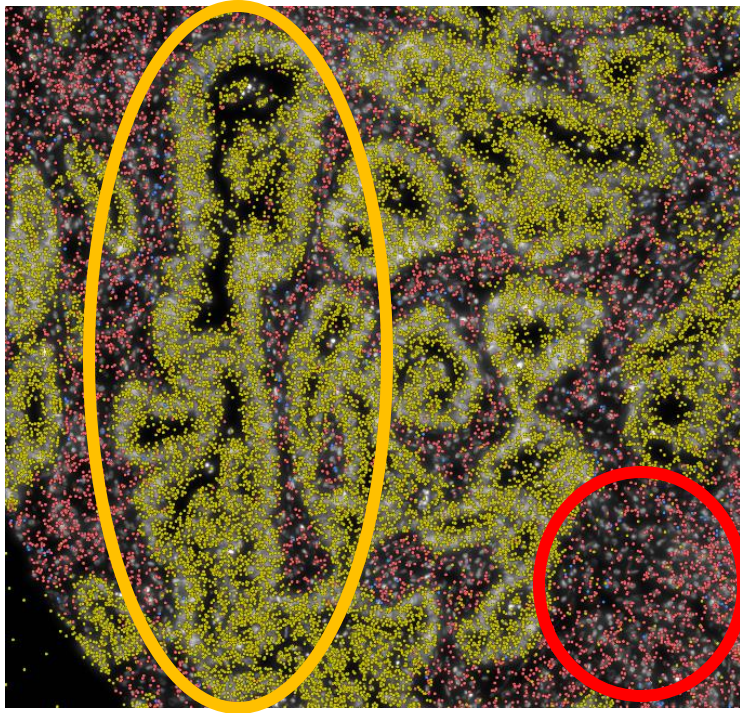
2. Stroma regions

- Immune and stromal cells

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Tumors into two major biological compartments:

1. Epithelium. 2. Stroma.



Expression of genes
associated with
epithelium domain

Expression of
genes associated
with stromal domain

1. Epithelium regions

- Cancer cells

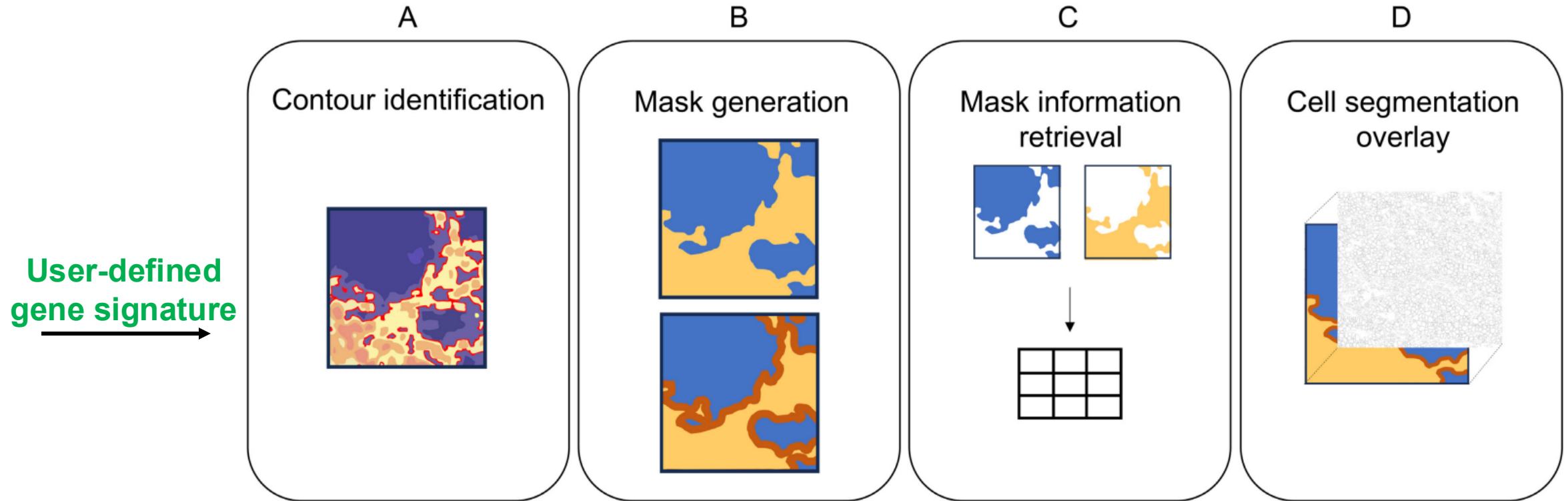
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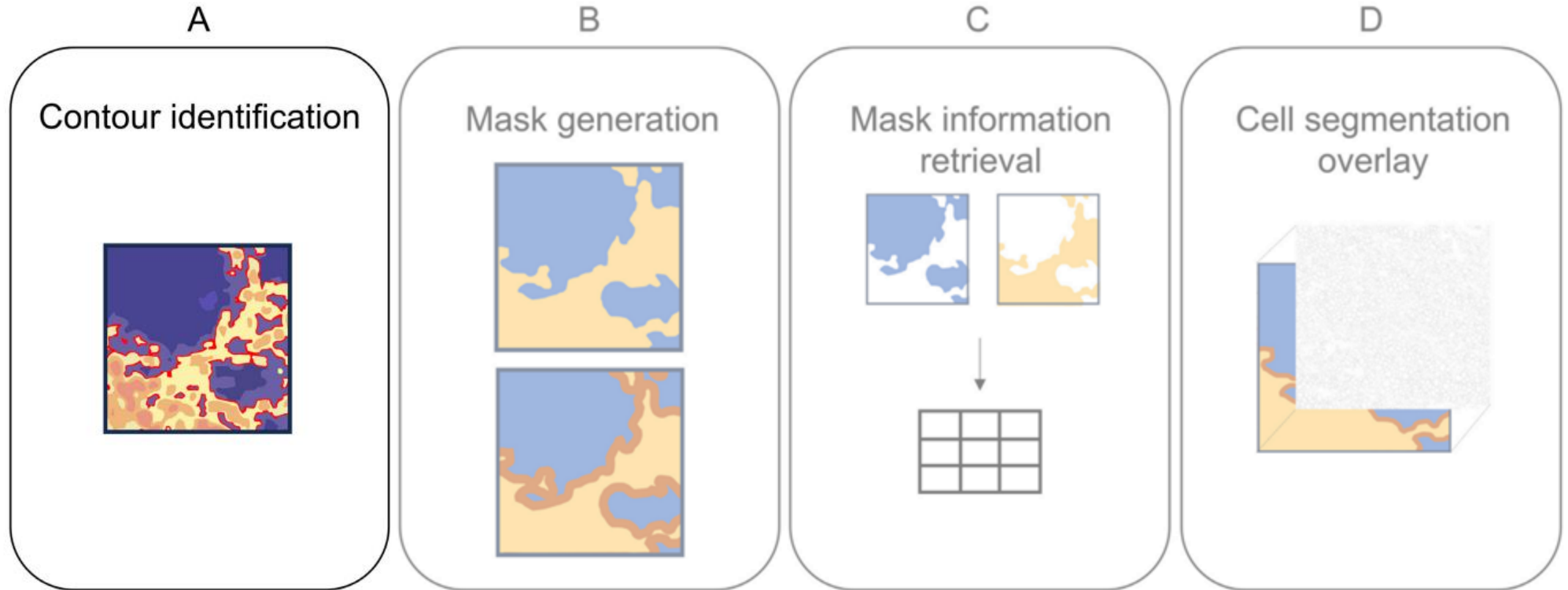
Let's use a **set of genes characteristic for epithelium** (cancer) regions as an example for a **gene set signature**

GRIDGENE workflow

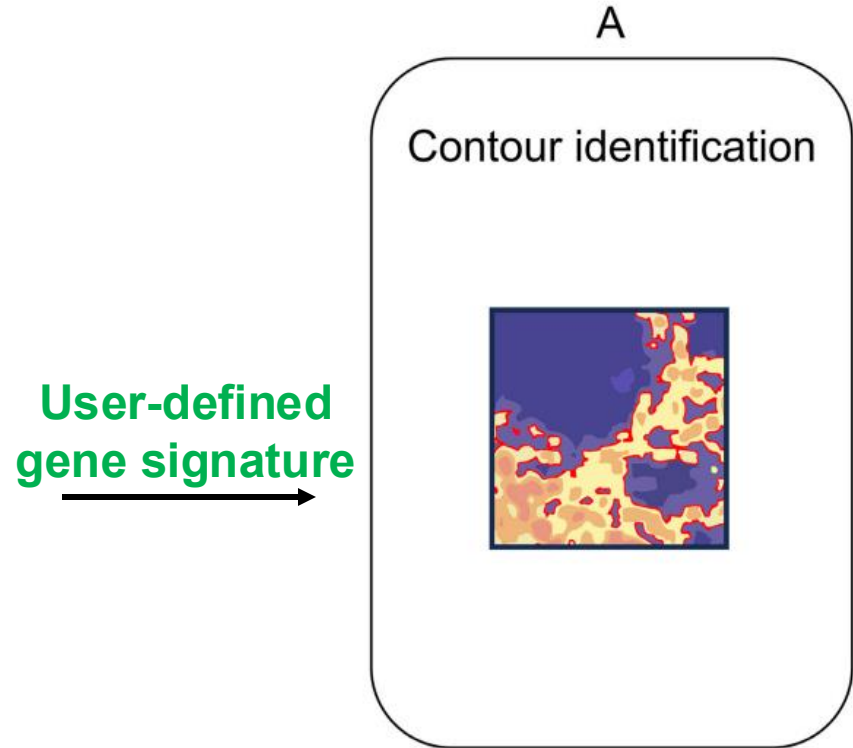


A. Contouring: identifying and outlining areas of interest

User-defined
gene signature



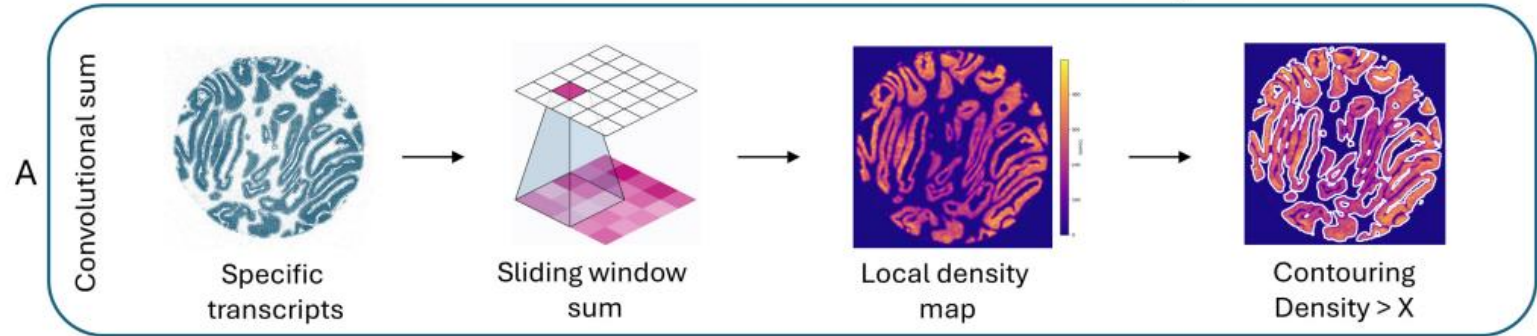
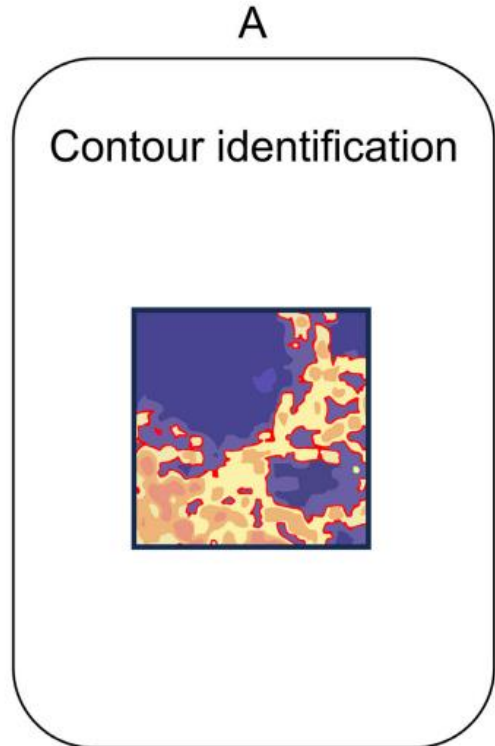
A. Contouring: identifying and outlining areas of interest



A. Contouring: identifying and outlining areas of interest

Contour identification - How to find and delimitate regions?

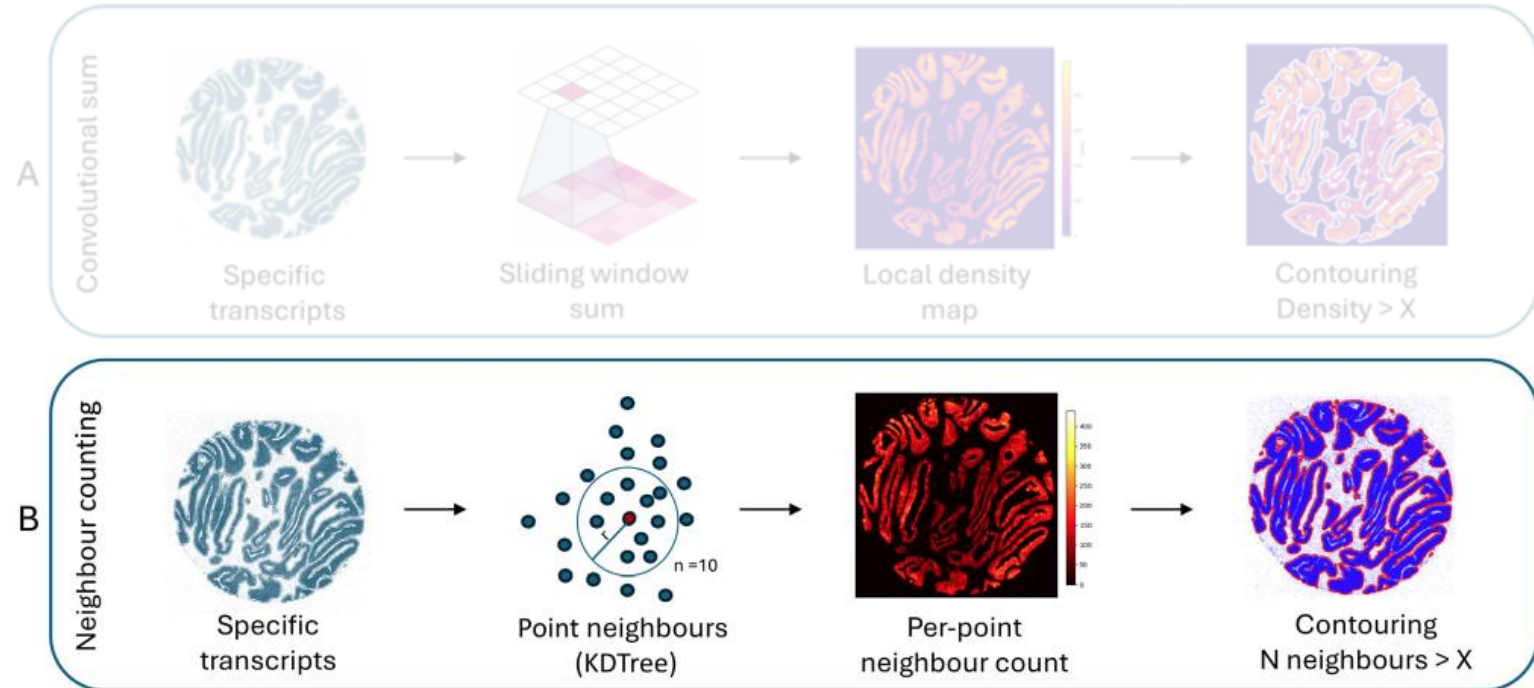
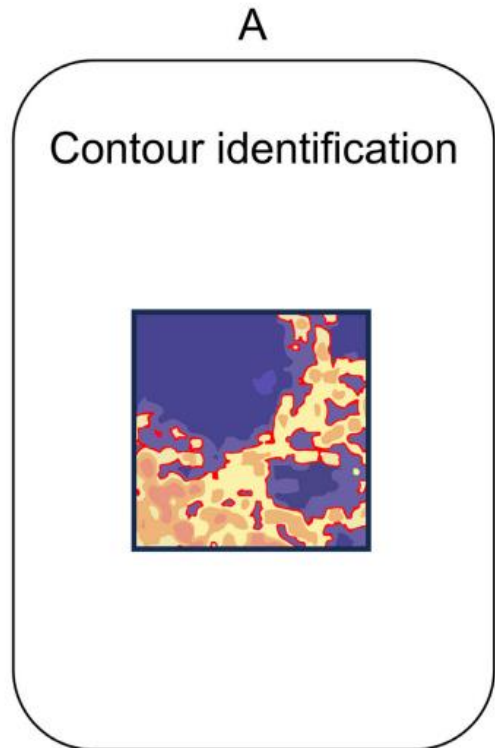
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Contour identification - How to find and delimitate regions?

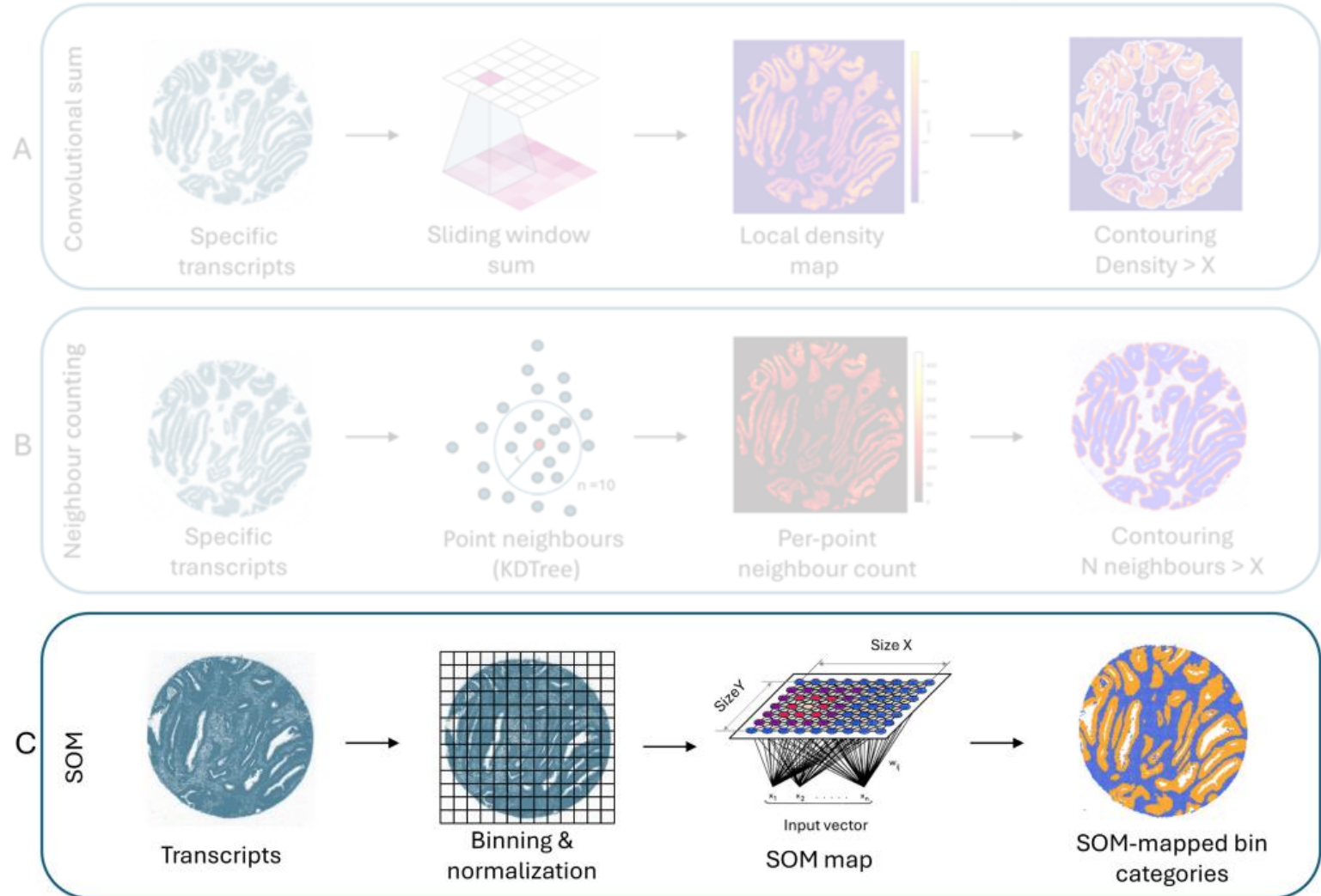
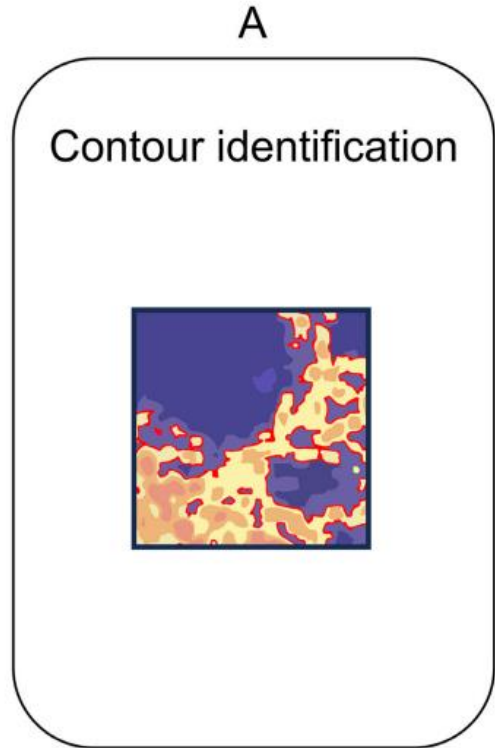
User-defined
gene signature



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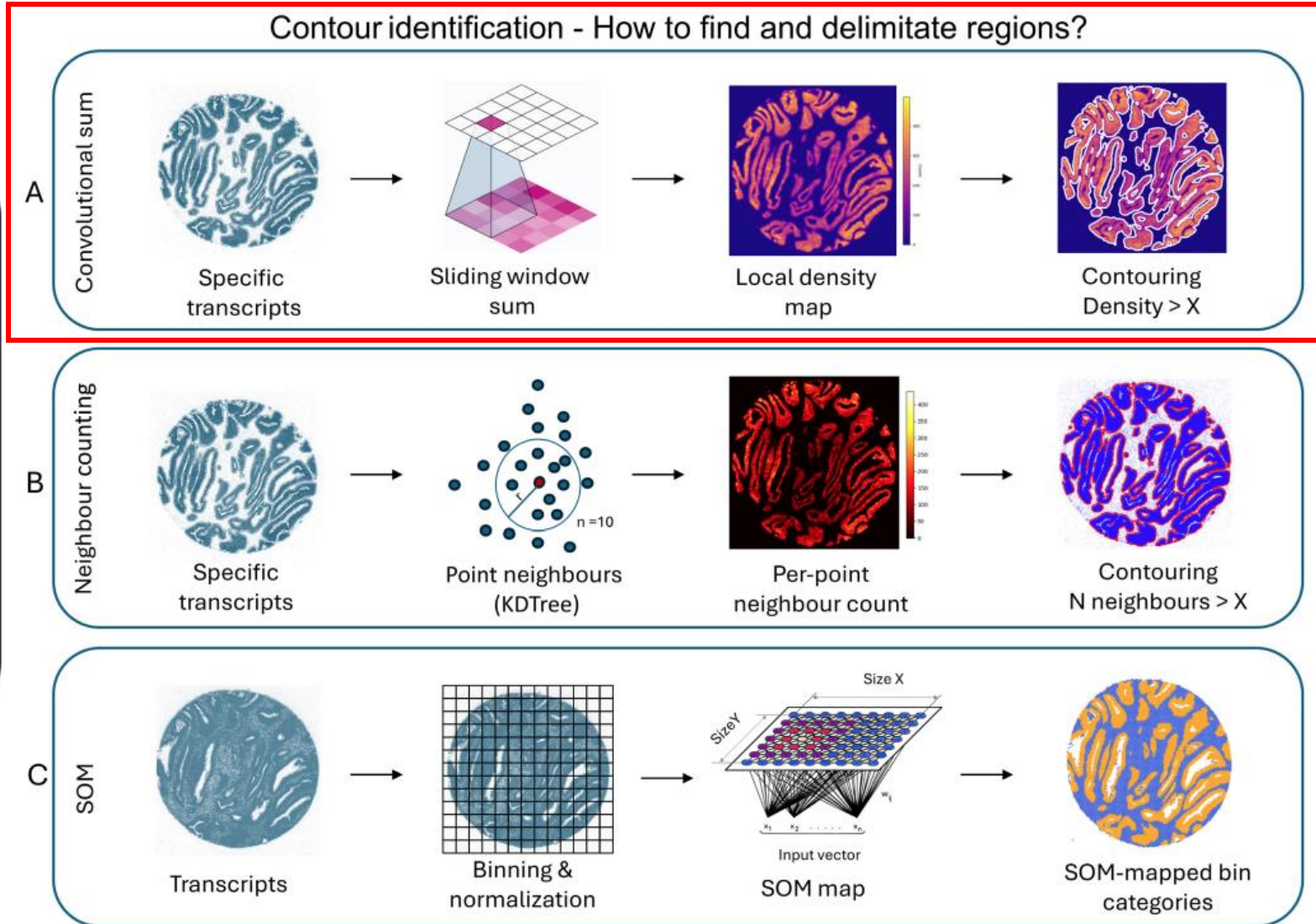
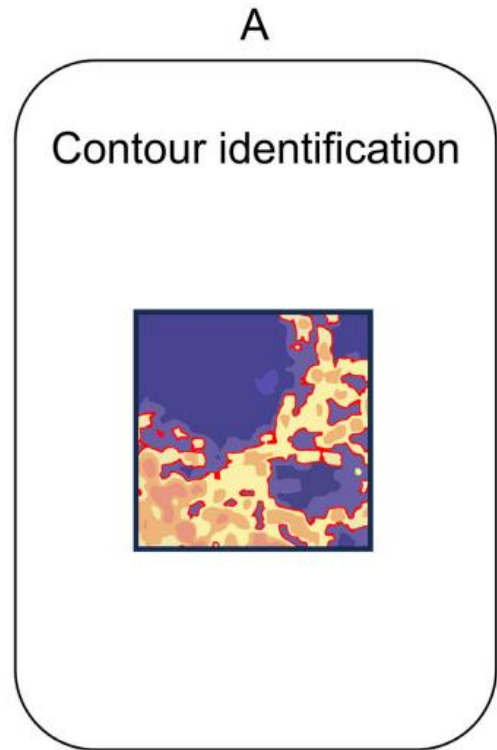
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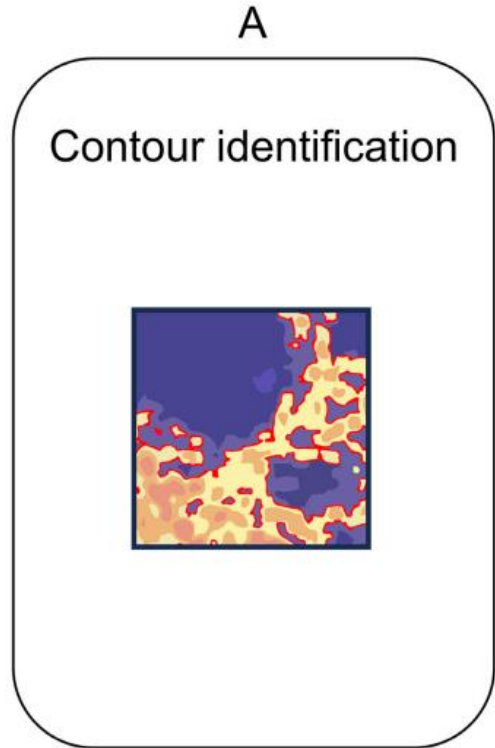
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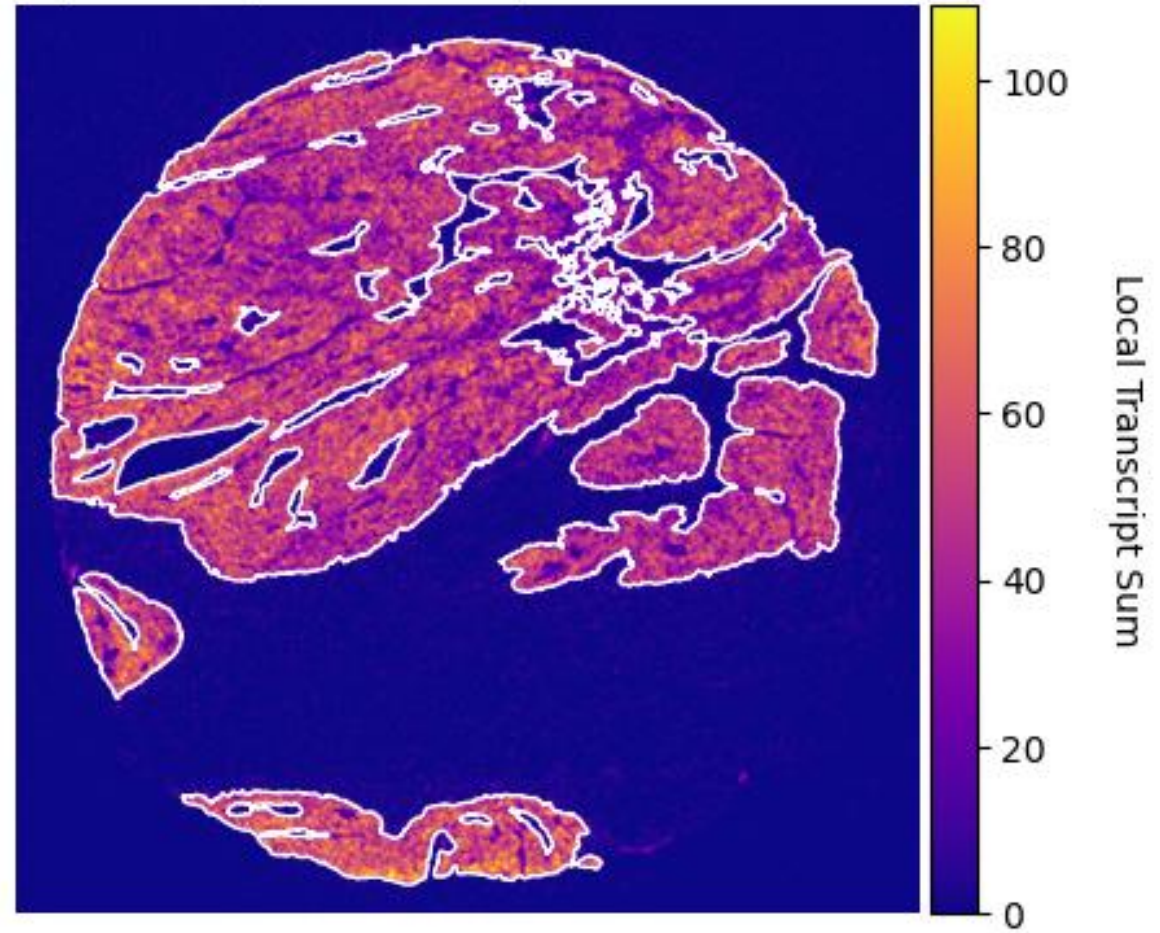


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User-defined
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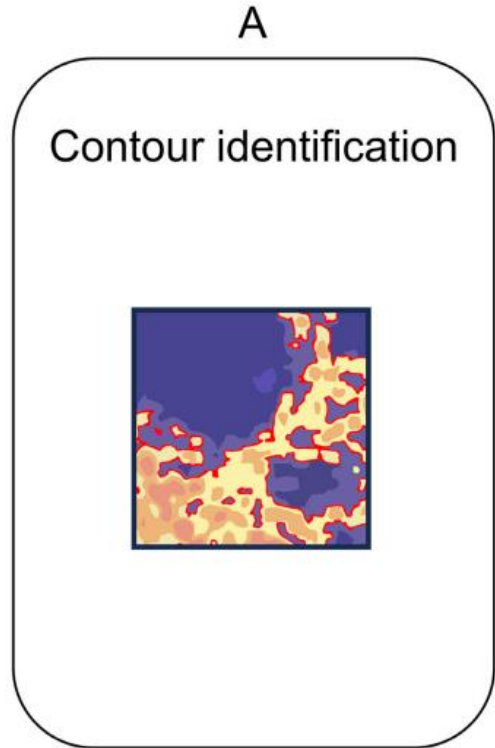


Epithelial points and epithelial contours

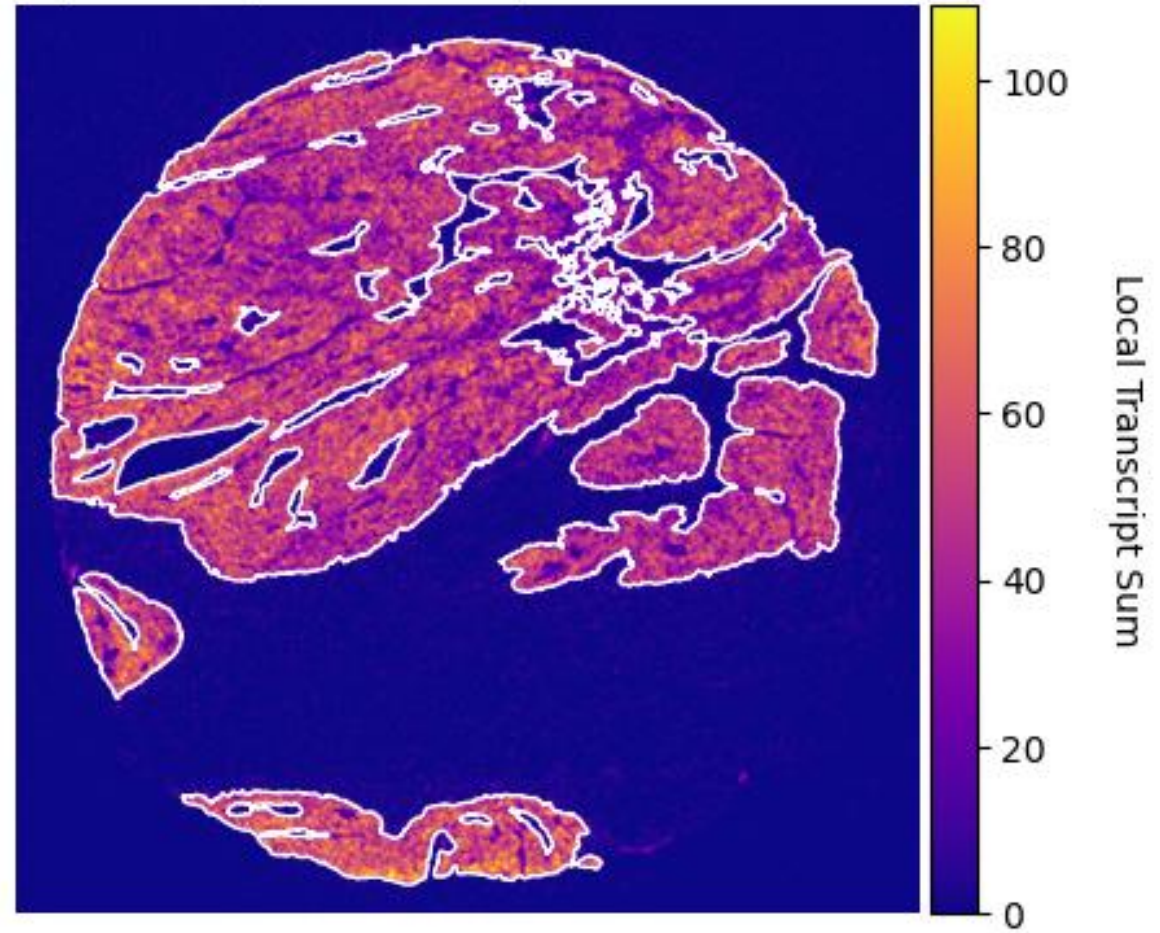


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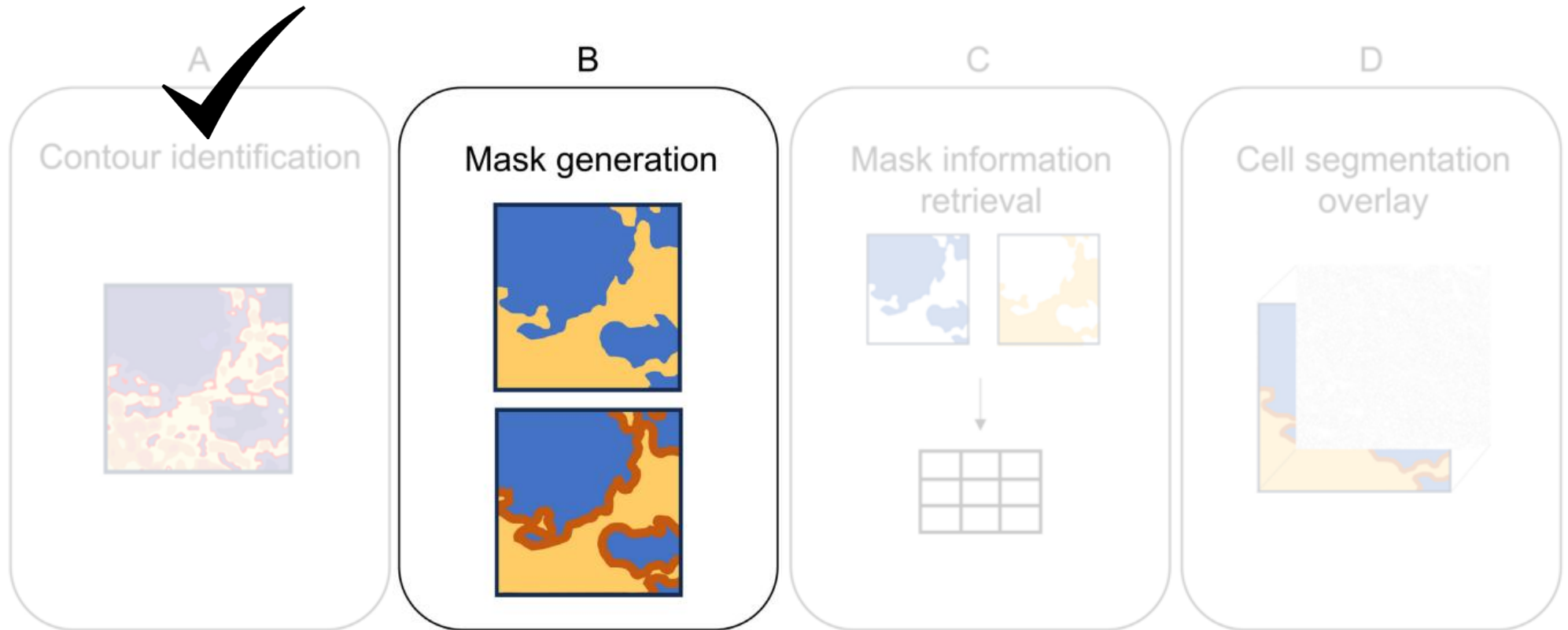
High expression of user-defined gene signature

Low expression of user-defined gene signature

GRIDGENE workflow



User-defined
gene signature



B. Mask generation: converting contours into binary masks

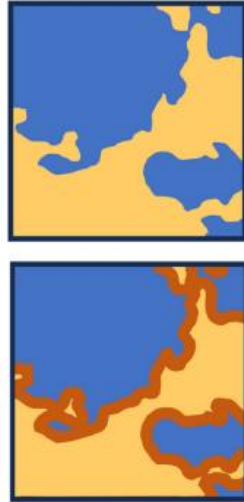
A

Contour identification

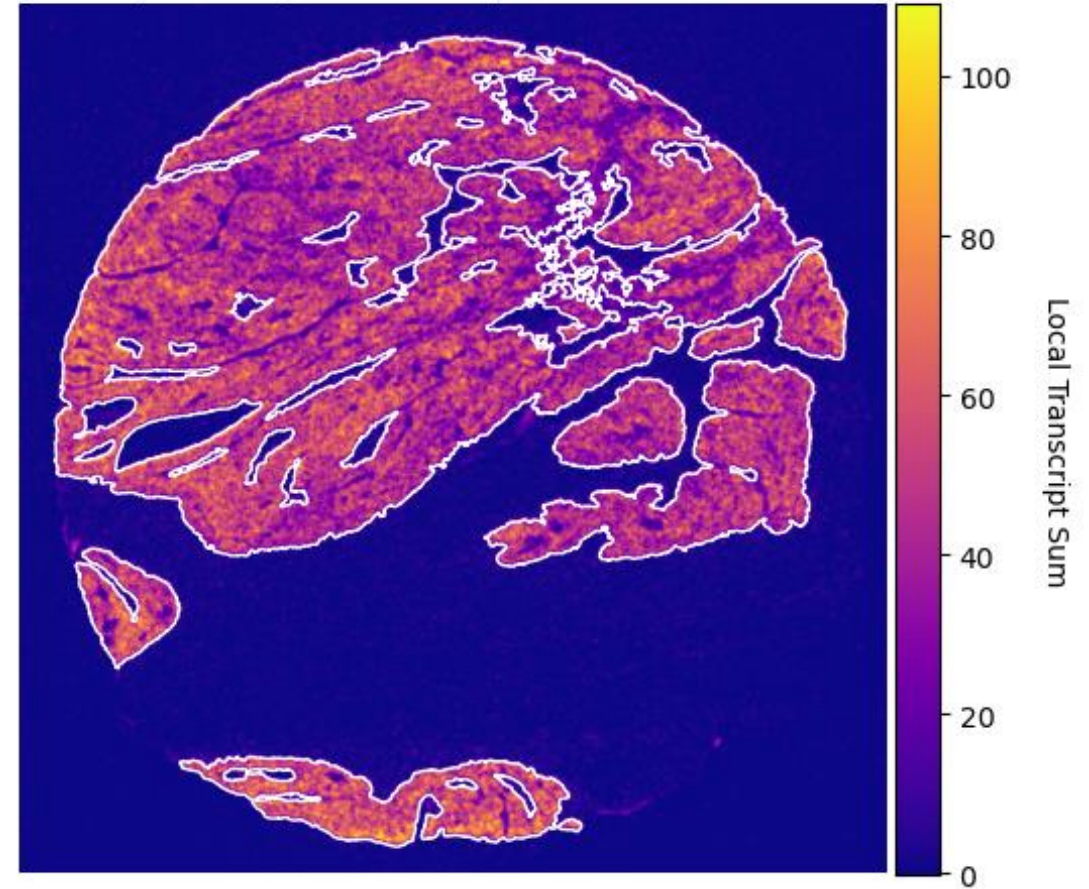


B

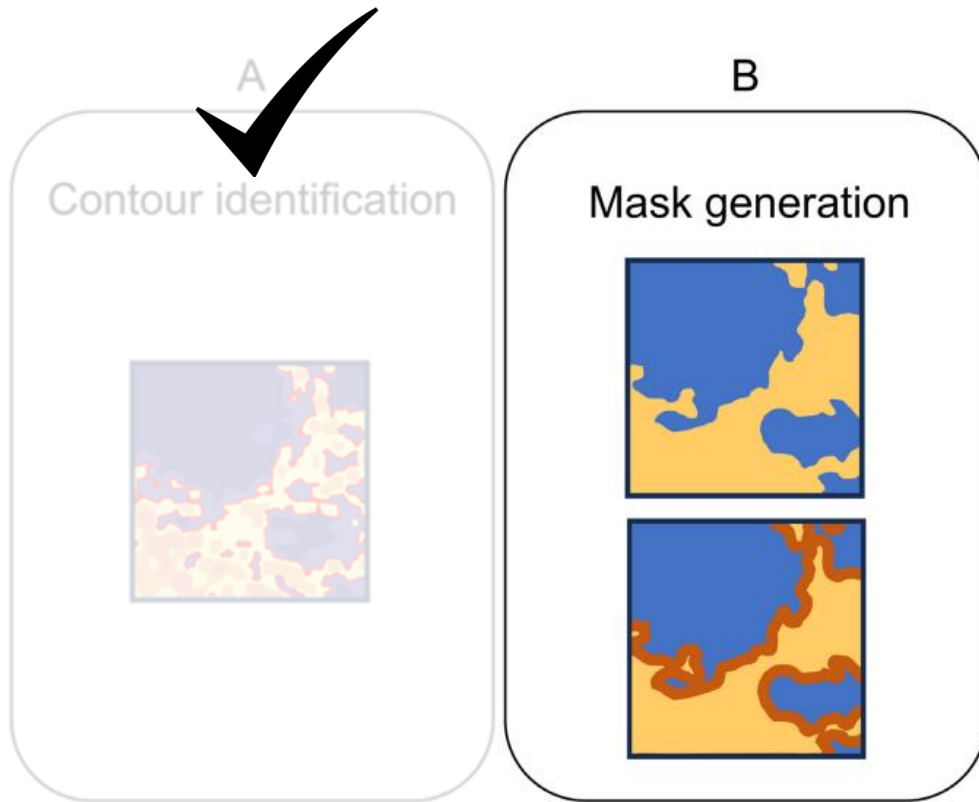
Mask generation



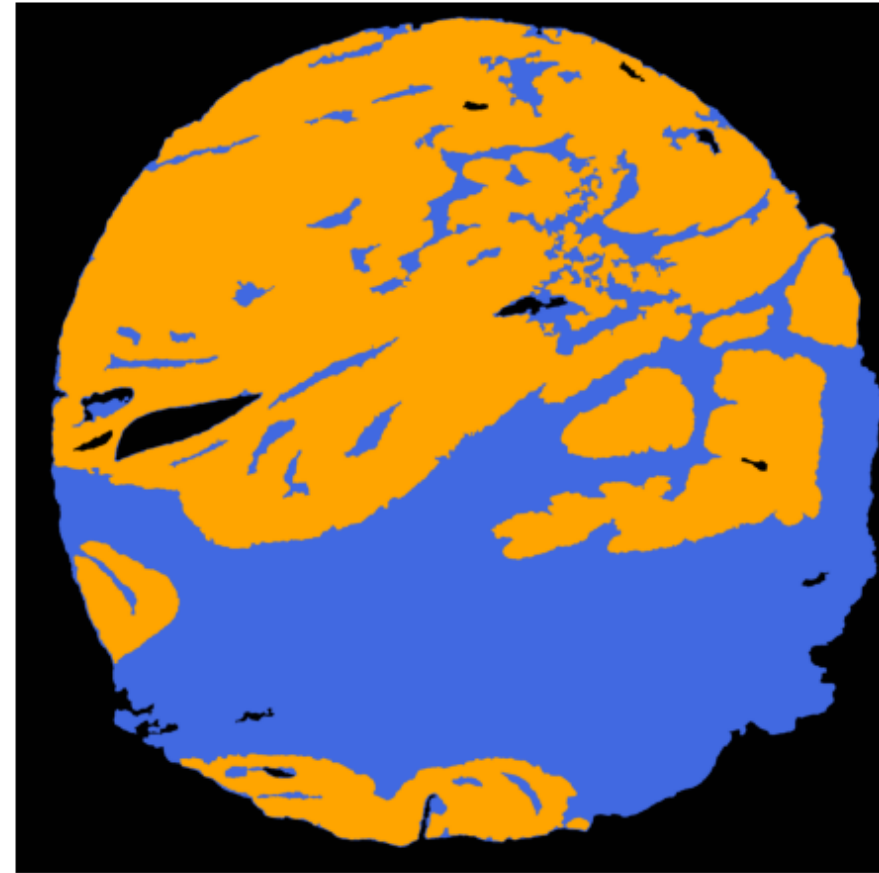
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B. Mask generation: converting contours into binary masks



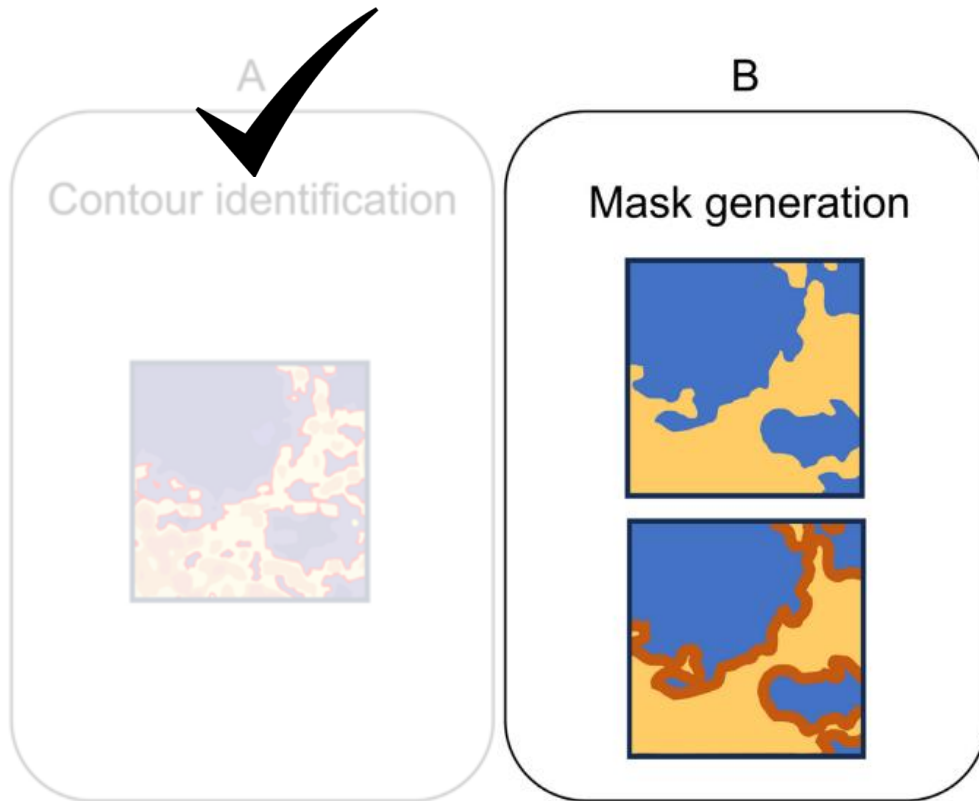
Mask generation



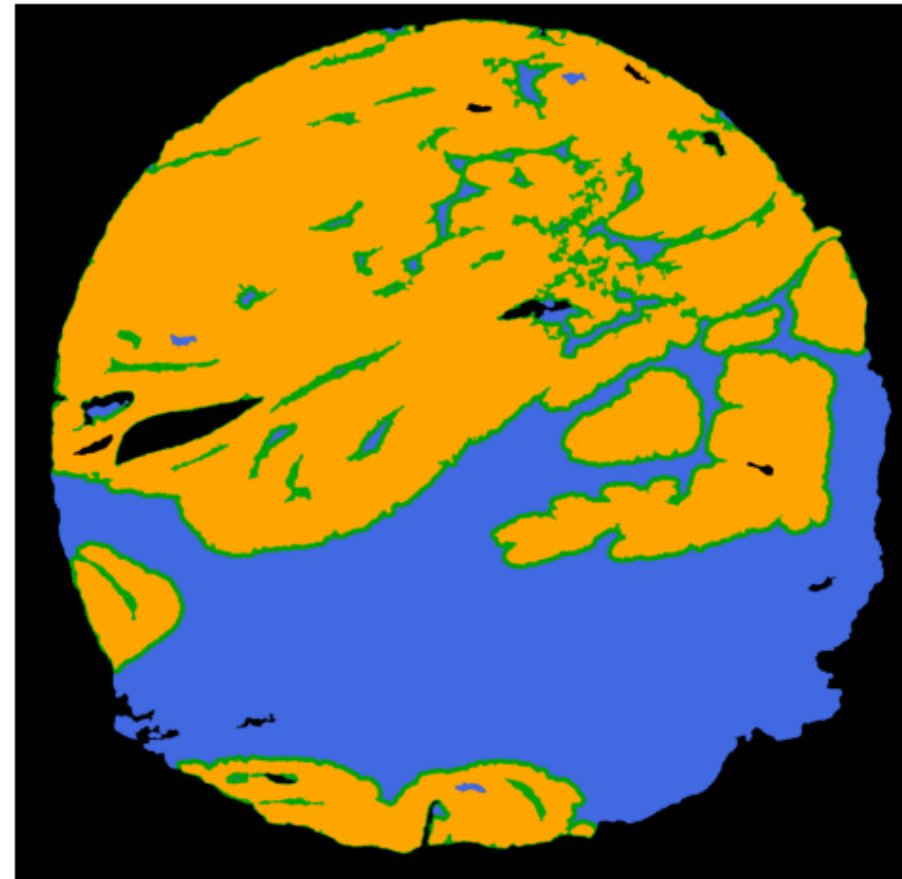
Tumor mask

Stroma mask

B. Mask generation: mask expansion



Mask generation

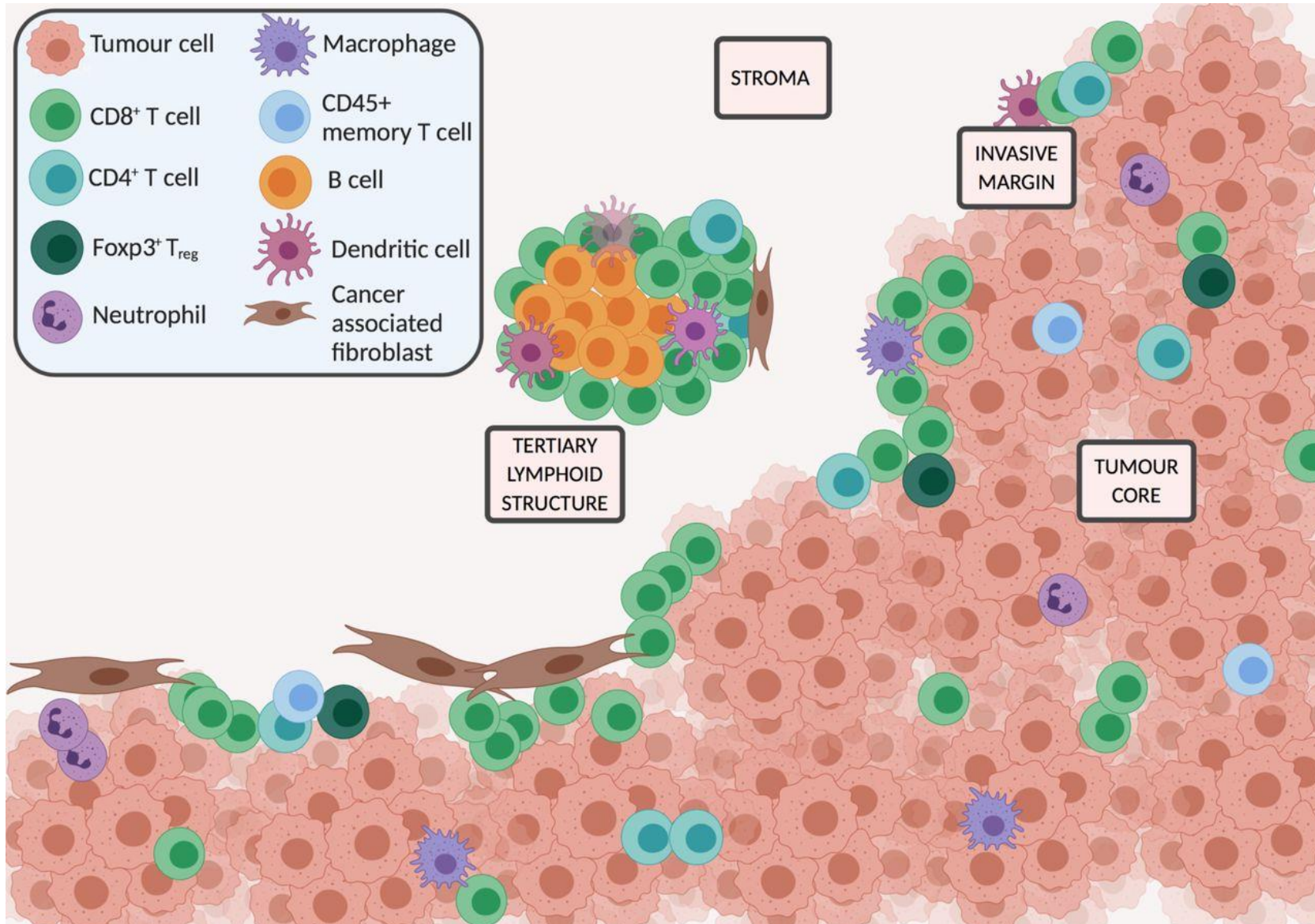


Tumor mask

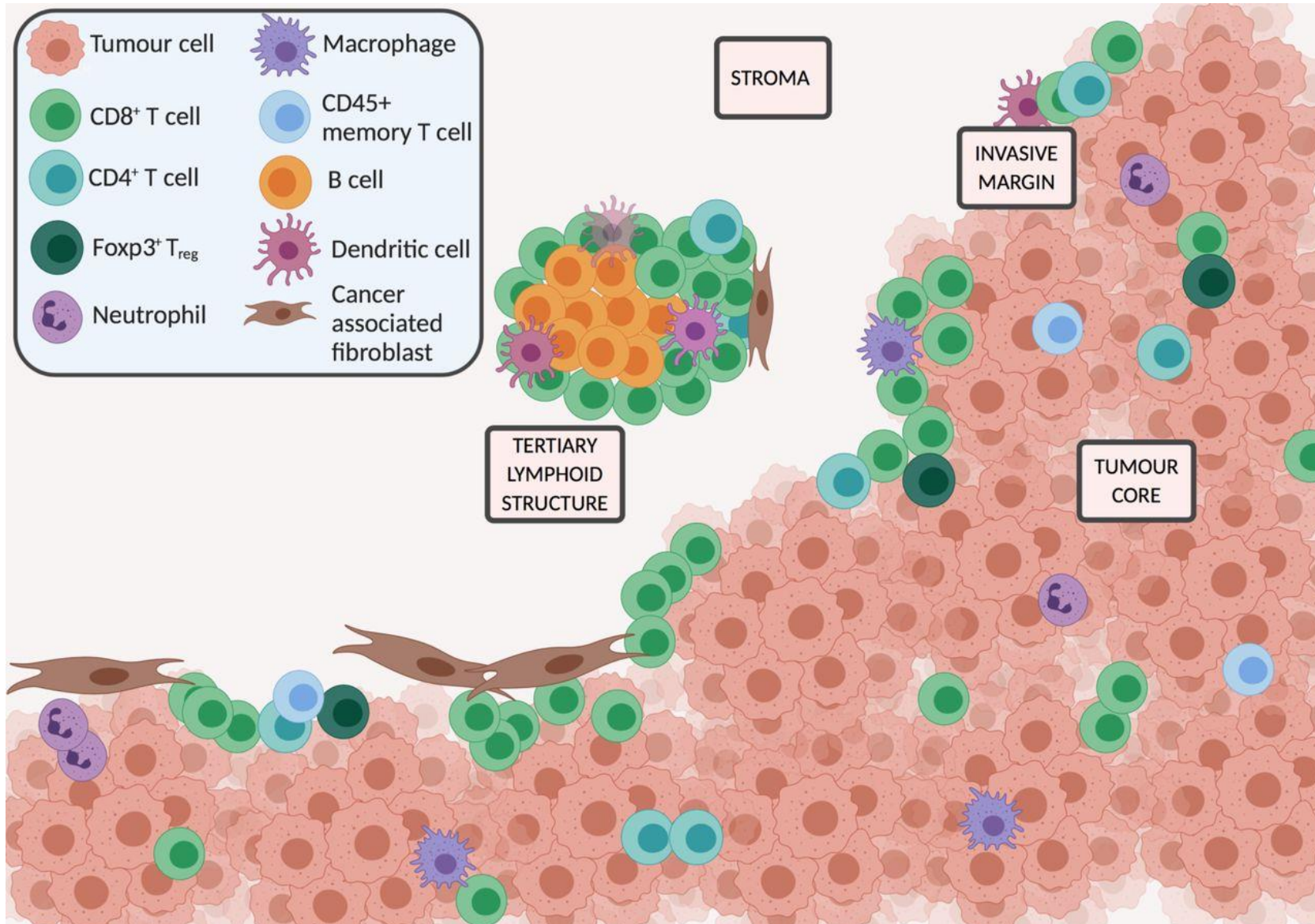
Stroma mask

Tumor-stroma border

The Tumor–Stroma Border as a Site of Tumor–Immune and Stromal Interactions

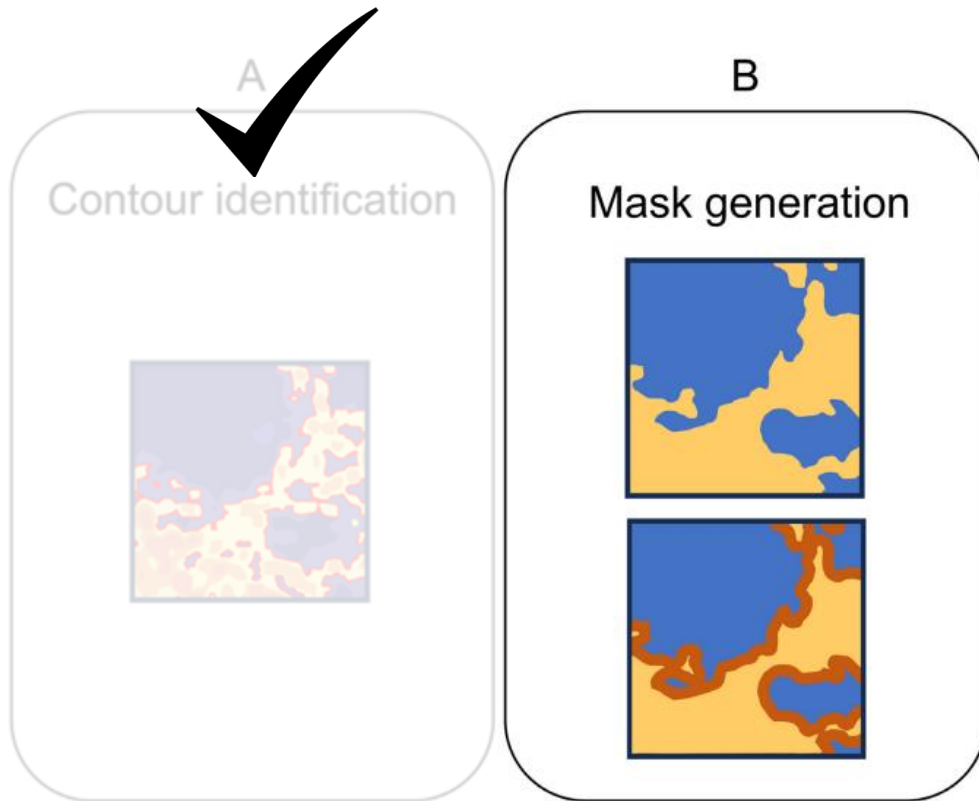


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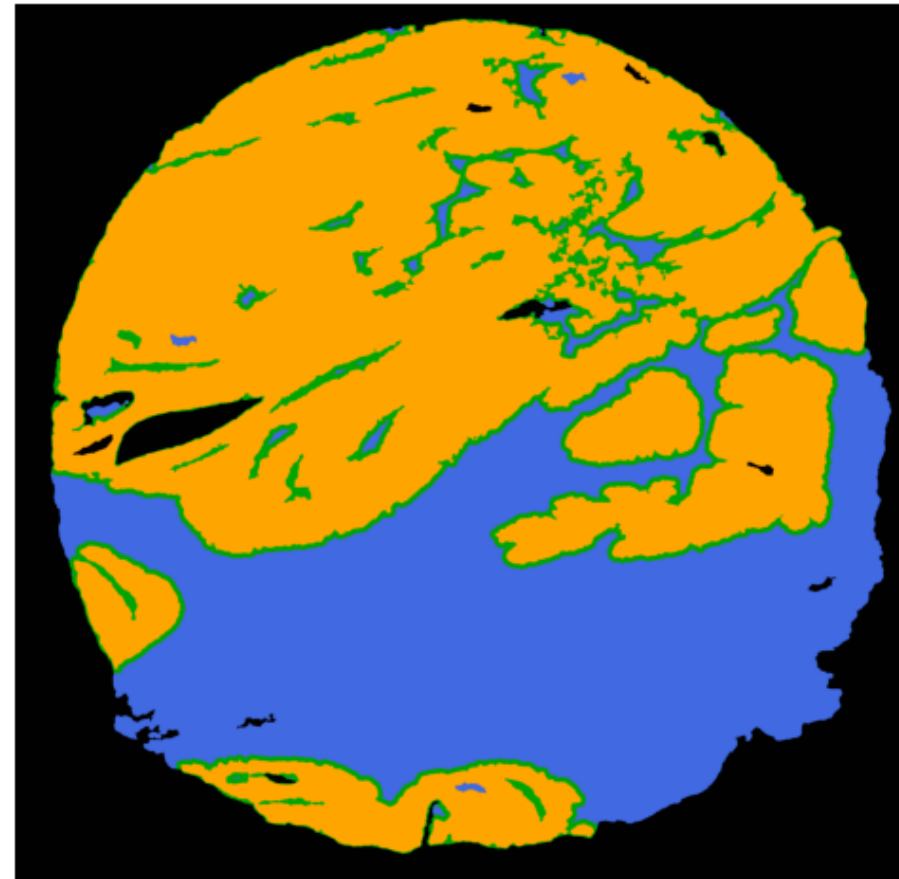


Important for tumor killing!

B. Mask generation: mask expansion



Mask generation

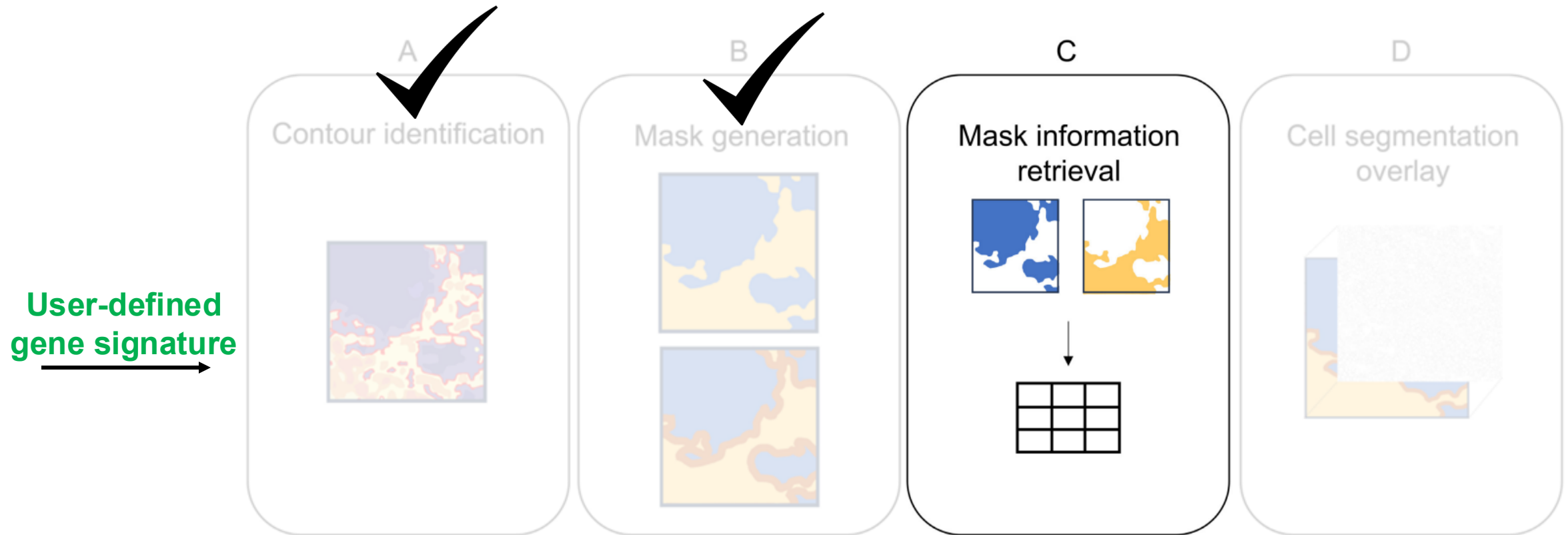


Tumor mask

Stroma mask

Tumor-stroma border

GRIDGENE workflow



GRIDGENE workflow



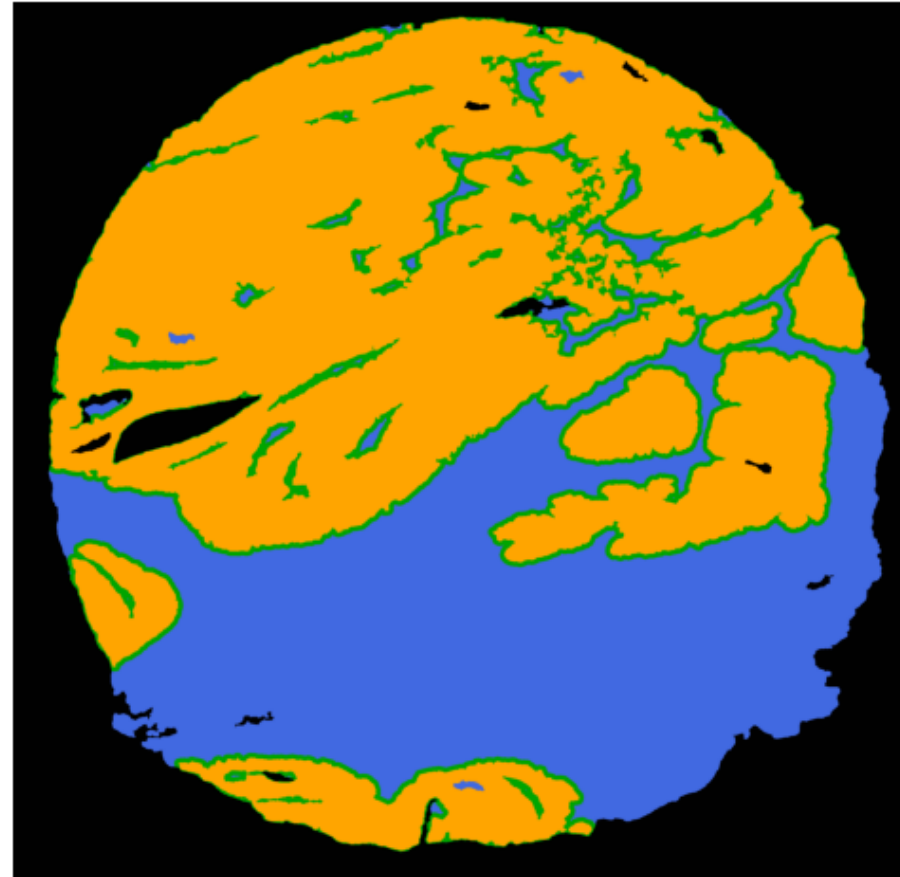
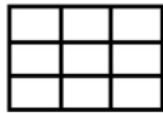
B

Mask generation



C

Mask information
retrieval



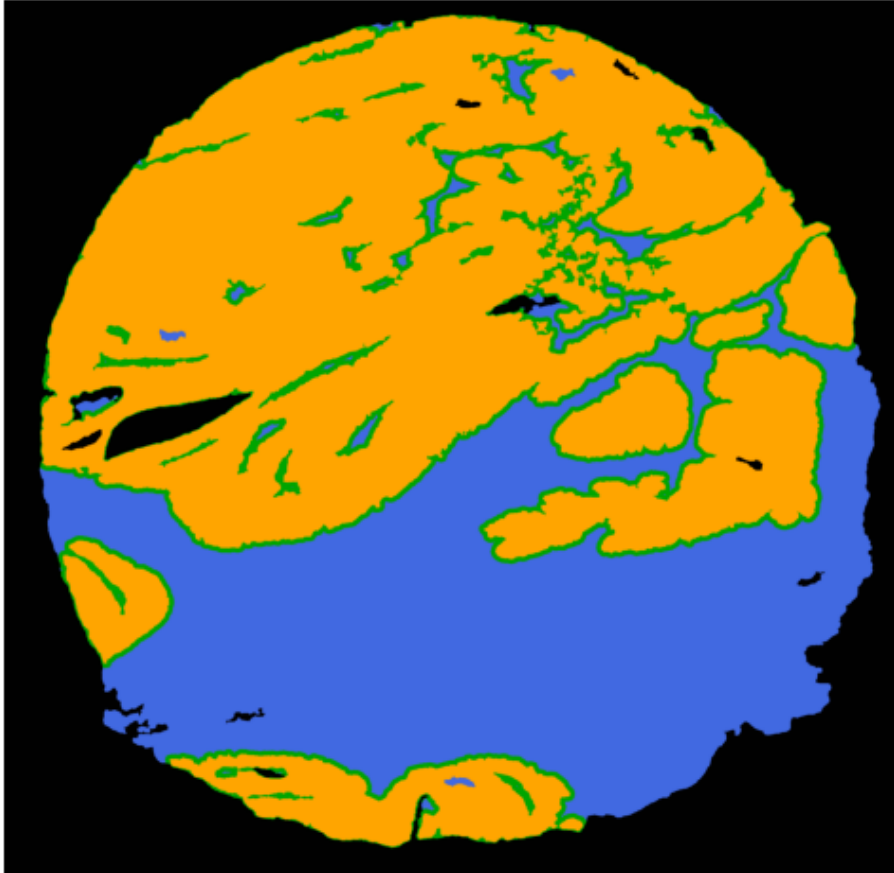
Tumor mask

Stroma mask

Tumor-stroma border

GRIDGENE workflow

C. Mask information retrieval



Tumor mask

Tumor-stroma border

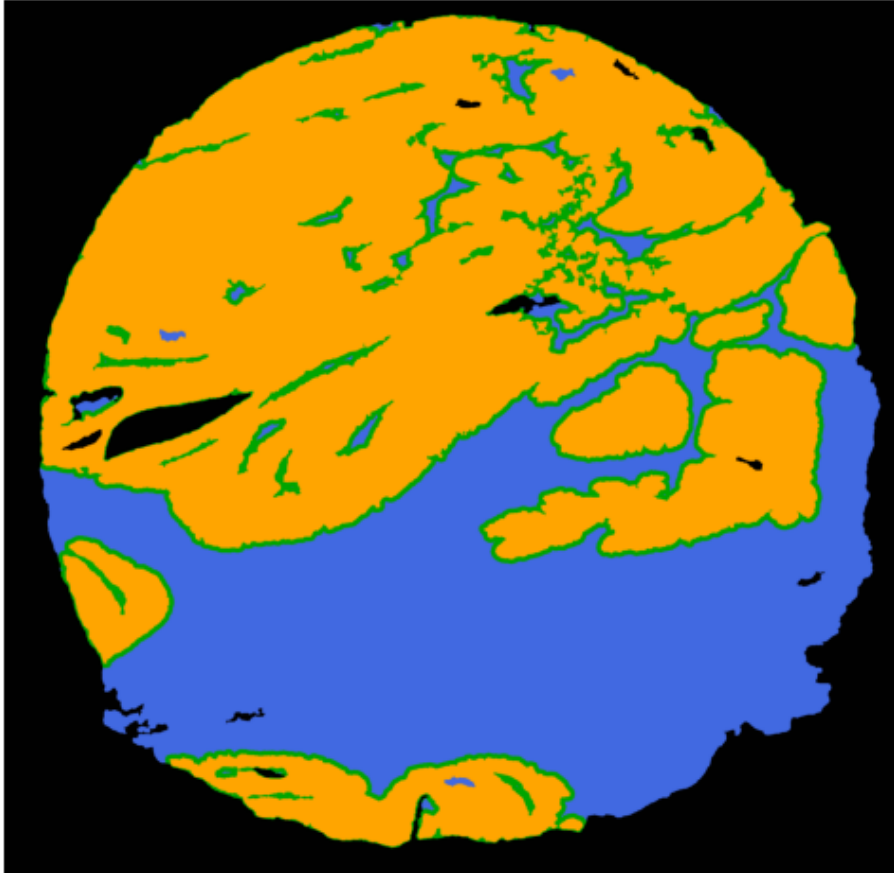
Stroma mask

Several types of information can be extracted from these masks:

- **Transcript data:** Obtain the transcript count per gene within each mask.
- **Morphological features:** Measure attributes such as area, perimeter, and shape descriptors.
- **Hierarchical annotations:** Capture hierarchical and topological relationships between original objects and their expansions, linking derived regions back to their parent contours.

GRIDGENE workflow

C. Mask information retrieval



Tumor mask

Tumor-stroma border

Stroma mask

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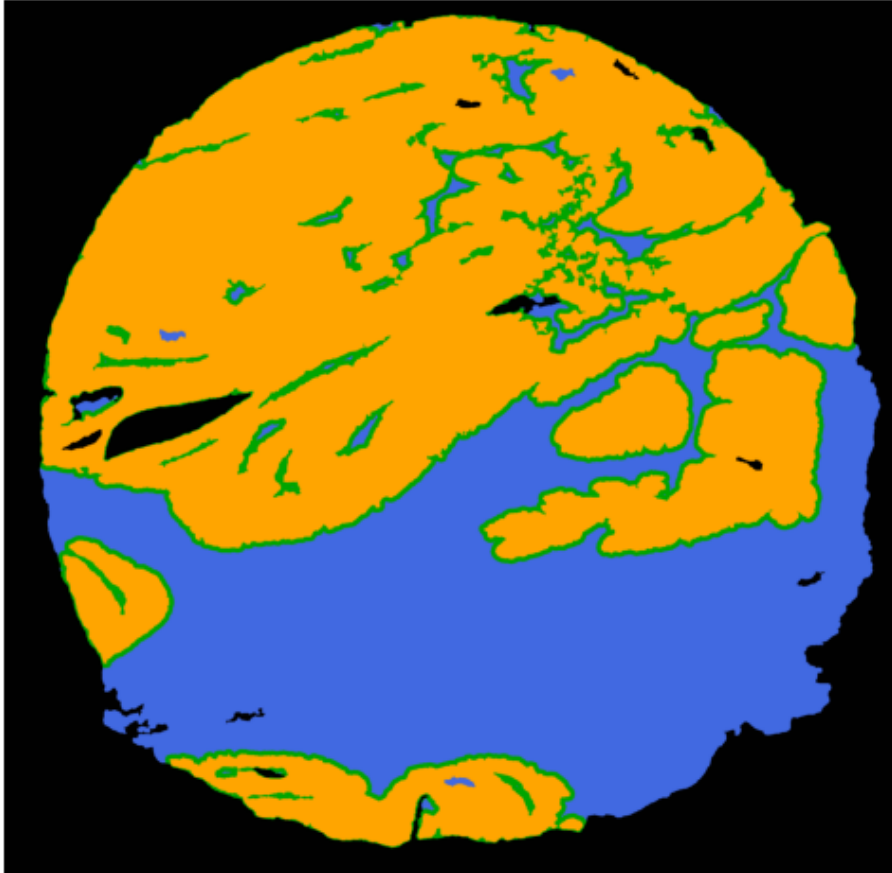
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This information can be retrieved at different levels:

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GRIDGENE workflow

C. Mask information retrieval



Tumor mask

Stroma mask

Tumor-stroma border

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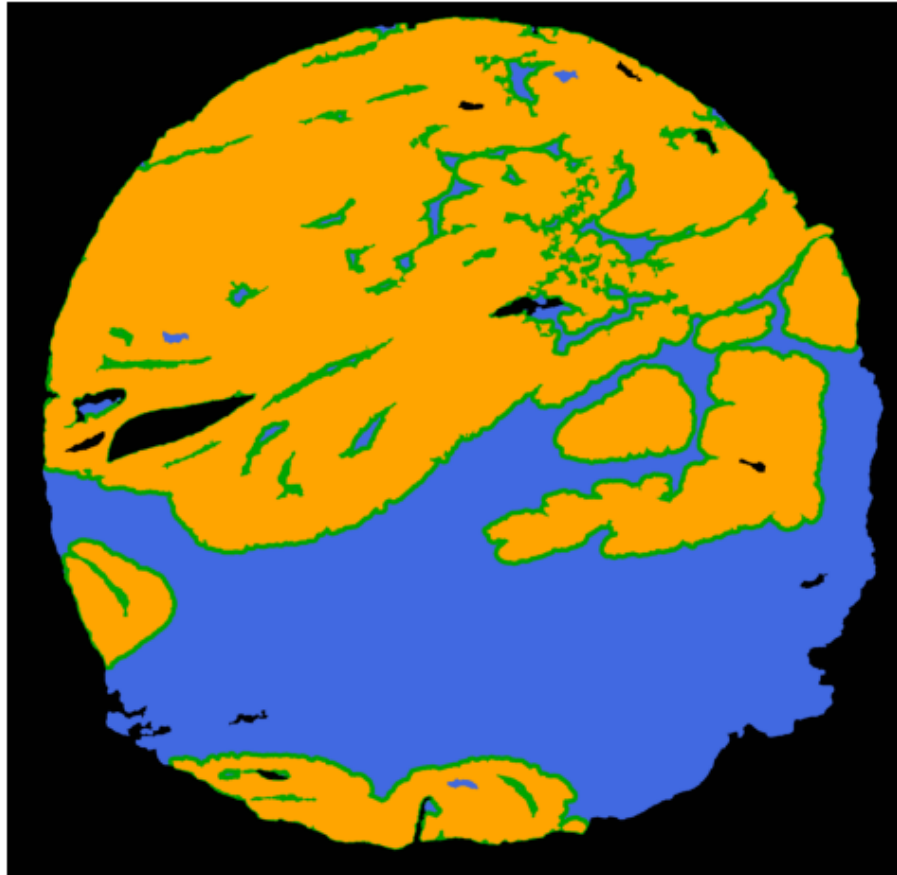
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GRIDGENE workflow

C. Mask information retrieval



Tumor mask

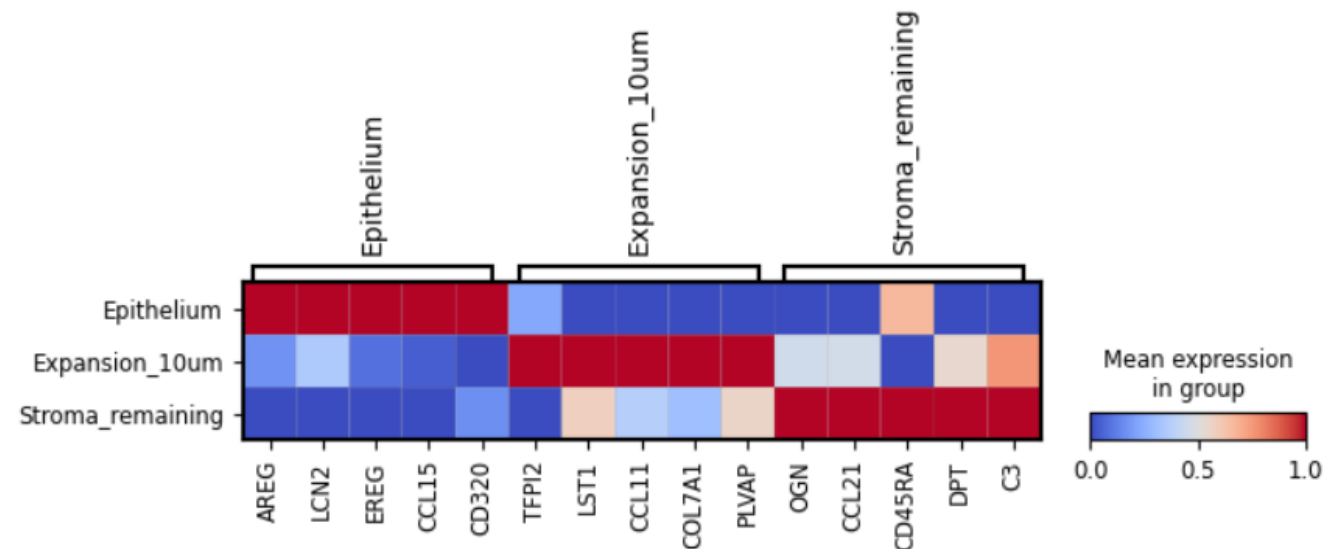
Tumor-stroma border

Stroma mask

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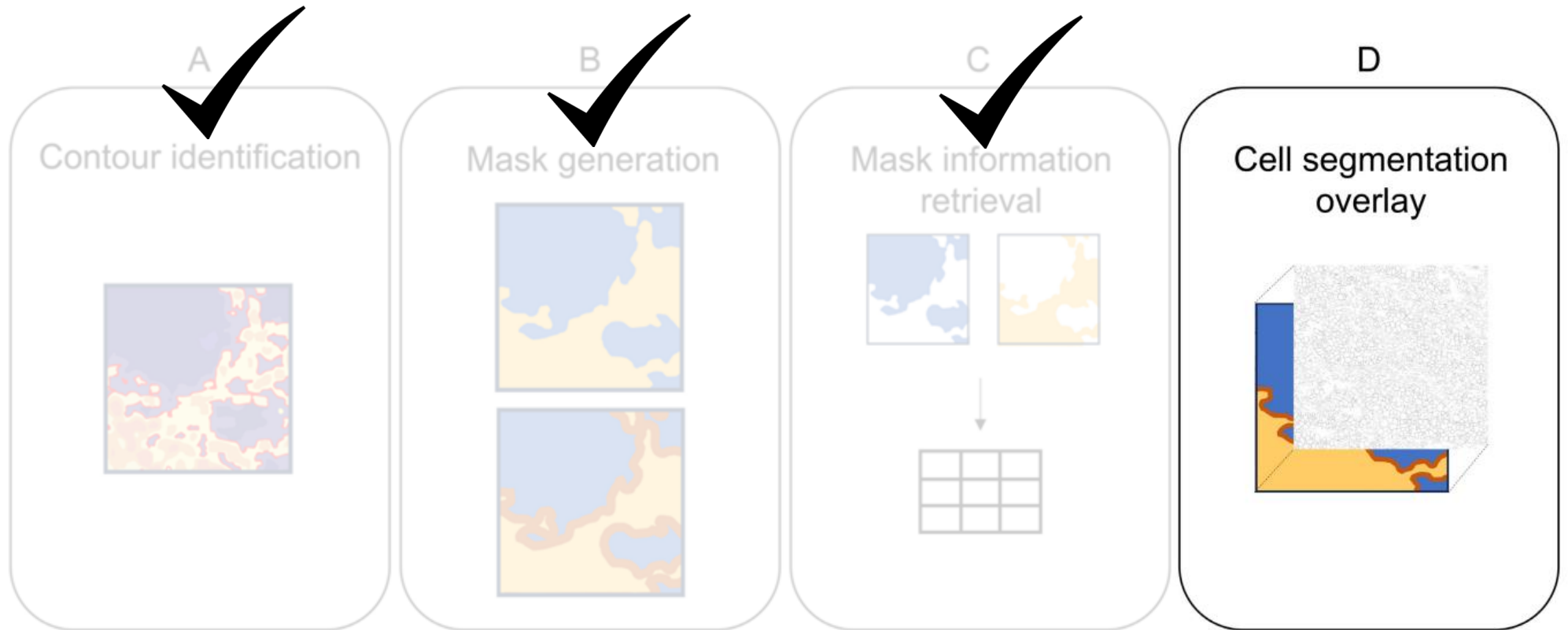
Differential gene expression between tissue masks



GRIDGENE workflow

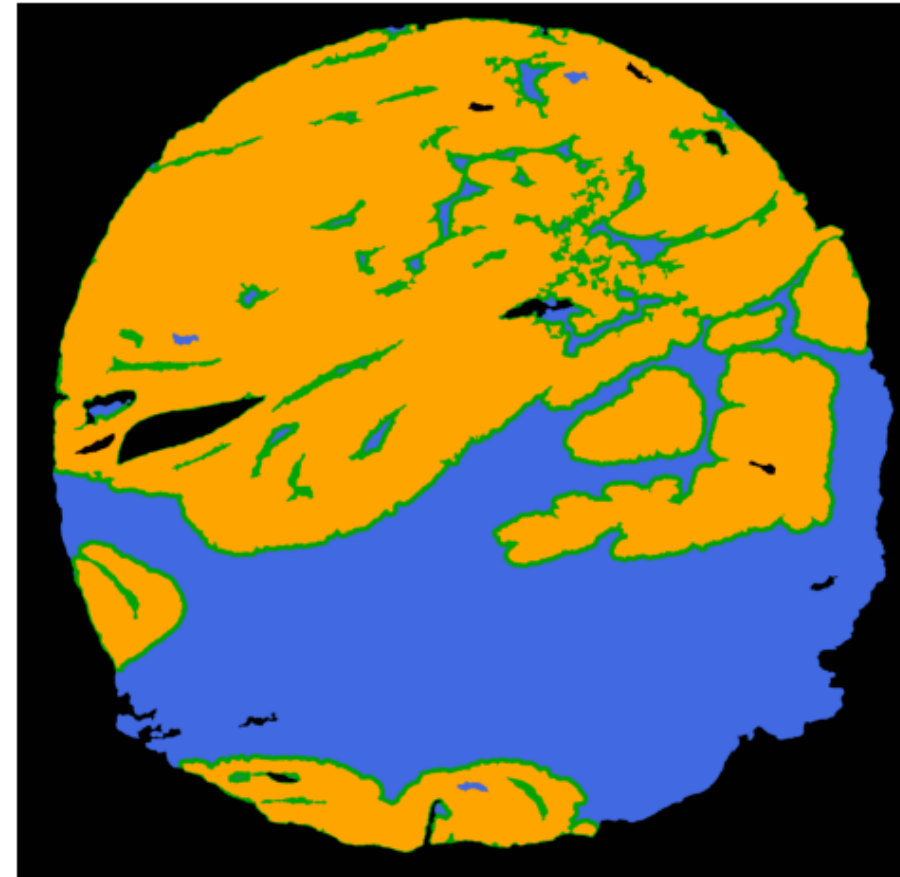
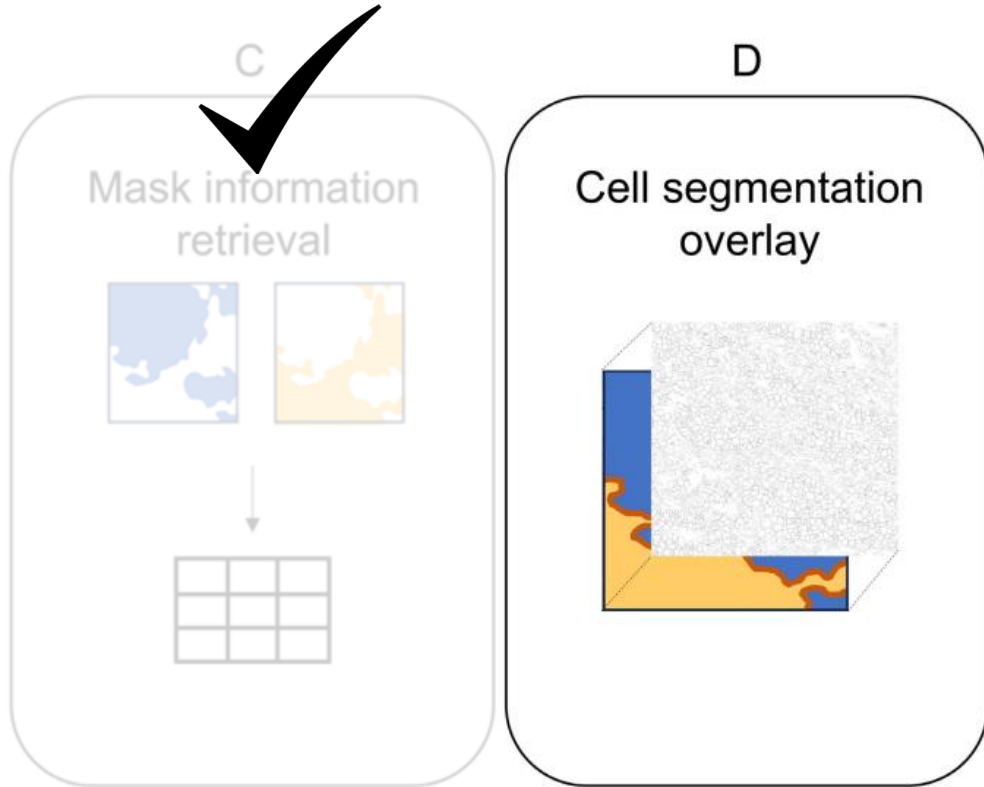


User-defined
gene signature
→



GRIDGENE workflow

D. Cell segmentation overlay

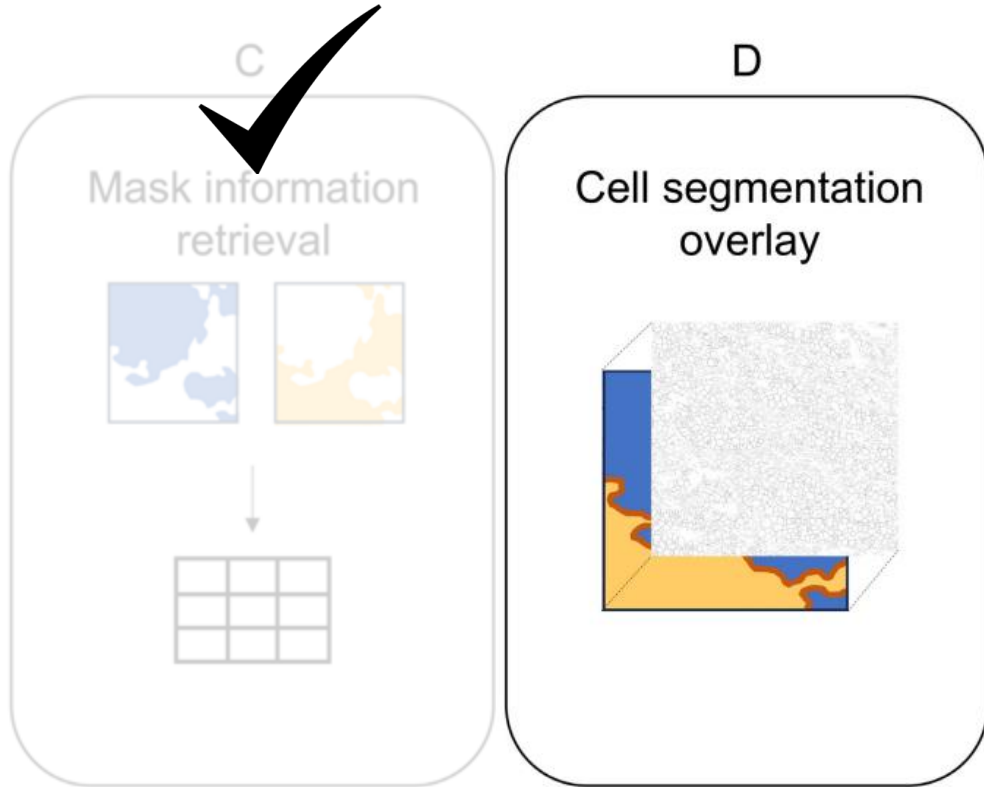


Tumor mask
Stroma mask

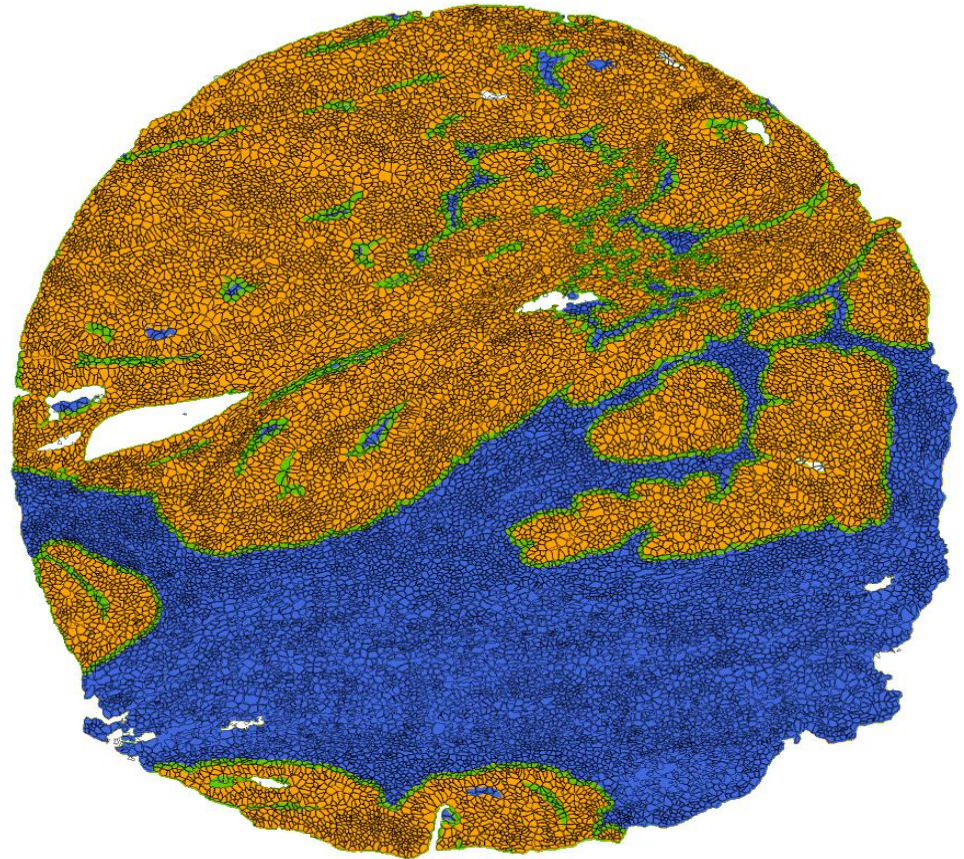
Tumor-stroma border

GRIDGENE workflow

D. Cell segmentation overlay



Overlay of Masks and Segmentation

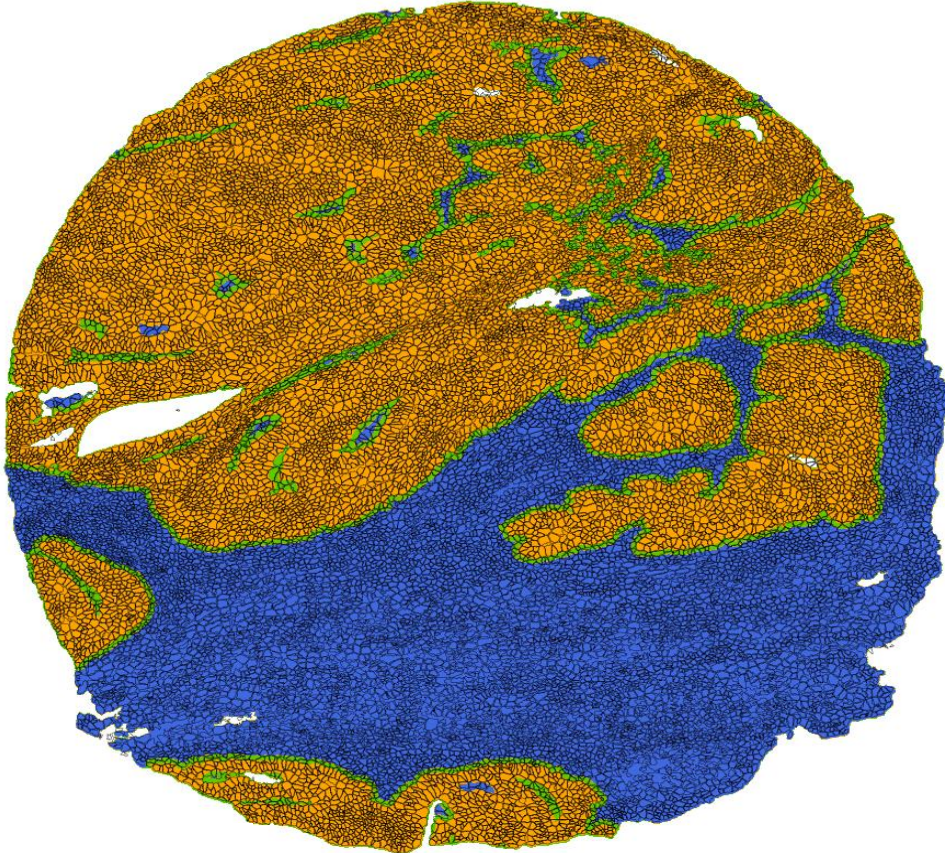


Tumor mask
Stroma mask

Tumor-stroma border

D. Cell segmentation overlay: cell type composition

Overlay of Masks and Segmentation



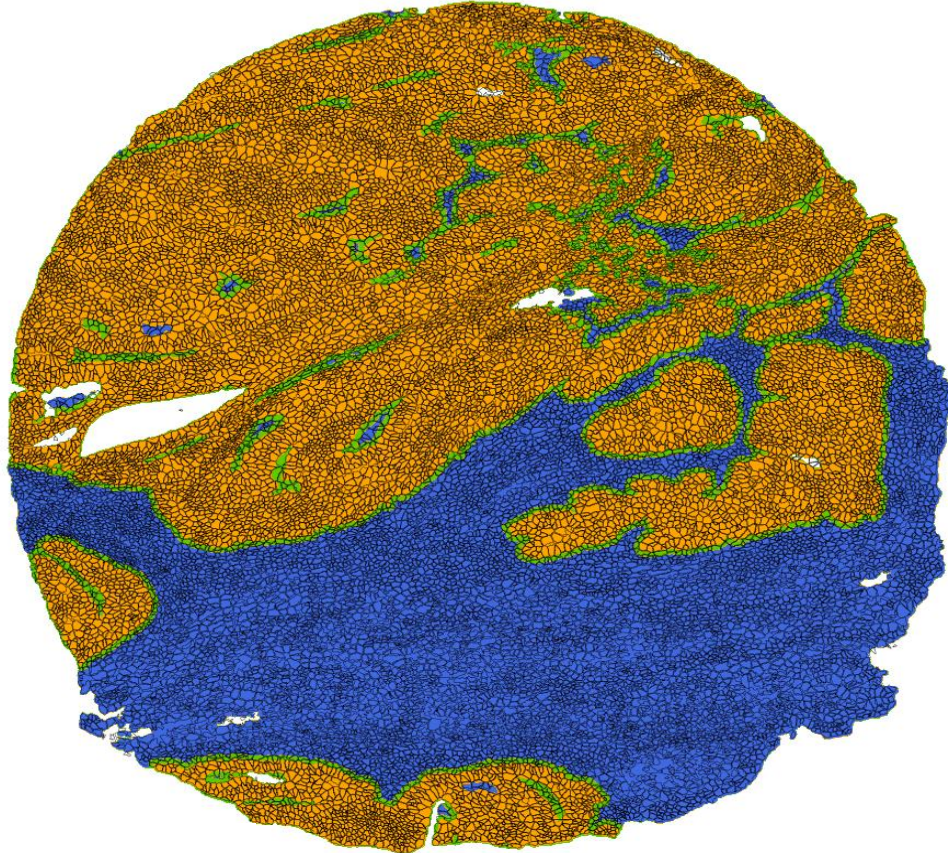
Tumor mask

Tumor-stroma border

Stroma mask

D. Cell segmentation overlay: cell type composition

Overlay of Masks and Segmentation

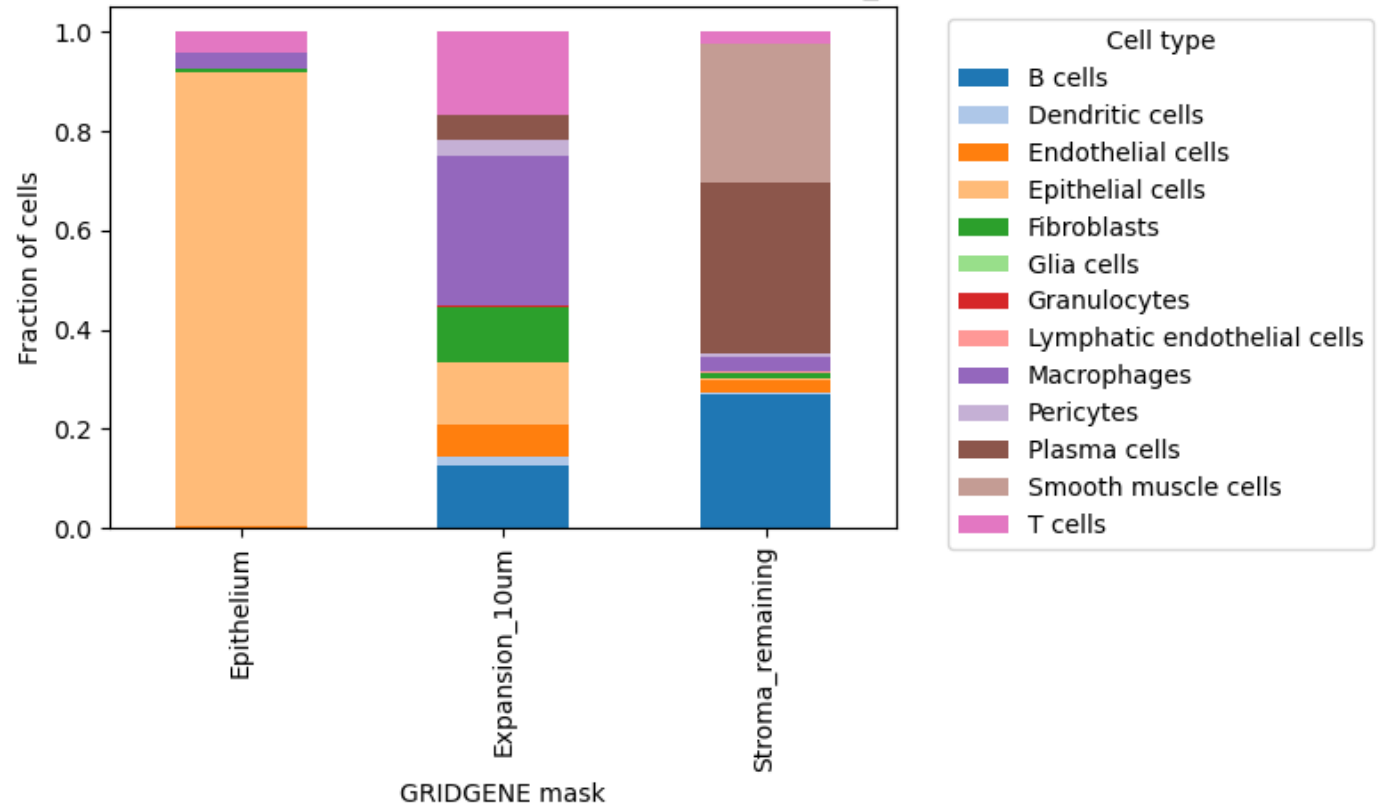


Tumor mask

Tumor-stroma border

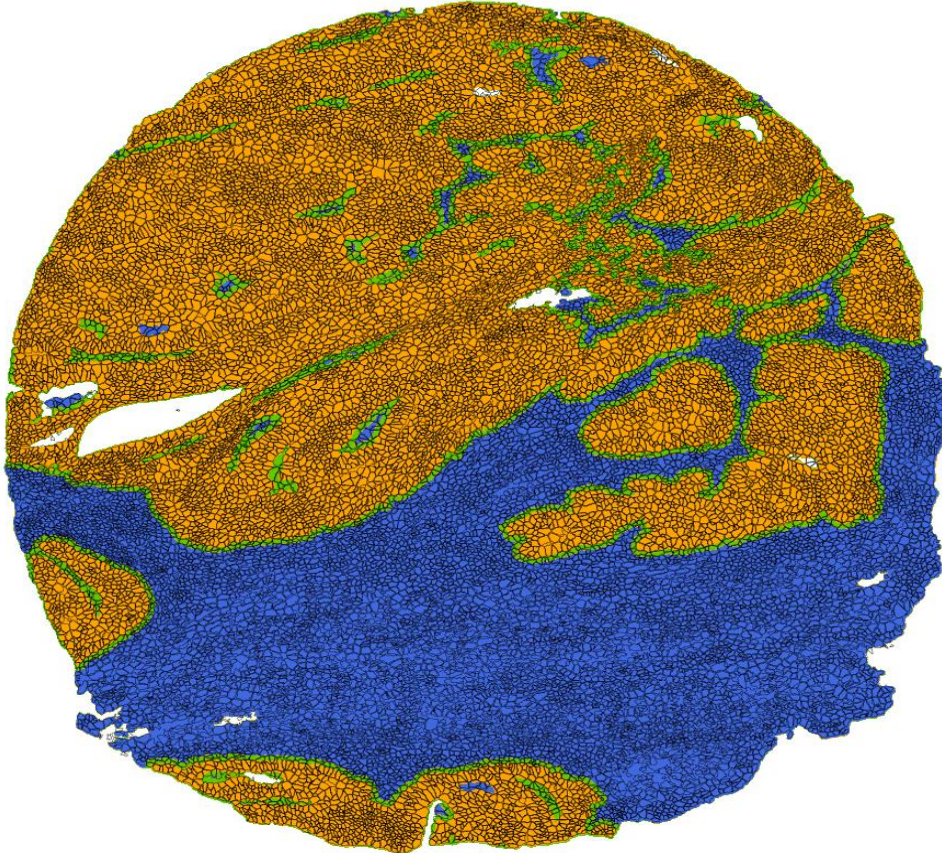
Stroma mask

Cell type composition per GRIDGENE mask (sample_1)



D. Cell segmentation overlay: differential gene expression

Overlay of Masks and Segmentation



Tumor mask

Tumor-stroma border

Stroma mask

How does a T cell in the tumor stroma-border functionally differ from a T cells that is located in the stroma?

Beyond scope of practical



Spatial transcriptomics goes beyond single cell segmentation

- Early (bad) cell segmentation can lead to inaccuracies and missing important signal
- Cell-segmentation free approaches are powerful parallel analysis avenues

Practical session

- Learn to use the GRIDGENE python package
- Analyze a Xenium dataset of two colorectal cancer samples
- Identify specific tissue domains
- Uncover unique spatial biology in these domains
- On Posit: Practical 3. Conda environment should load automatically, let me know if not.